

New Techniques for Preimage Sampling: Improved NIZKs and More from LWE

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Abstract

Recent constructions of vector commitments and non-interactive zero-knowledge (NIZK) proofs from LWE implicitly solve the following *shifted multi-preimage sampling problem*: given matrices $A_1, \dots, A_\ell \in \mathbb{Z}_q^{n \times m}$ and targets $t_1, \dots, t_\ell \in \mathbb{Z}_q^n$, sample a shift $c \in \mathbb{Z}_q^n$ and short preimages $\pi_1, \dots, \pi_\ell \in \mathbb{Z}_q^m$ such that $A_i \pi_i = t_i + c$ for all $i \in [\ell]$. In this work, we introduce a new technique for sampling $A_1, \dots, A_\ell$ together with a succinct public trapdoor for solving the multi-preimage sampling problem with respect to $A_1, \dots, A_\ell$. This enables the following applications:

- We provide a dual-mode instantiation of the hidden-bits model (and by correspondence, a dual-mode NIZK proof for NP) with (1) a linear-size common reference string (CRS); (2) a transparent setup in hiding mode (which yields statistical NIZK arguments); and (3) hardness from LWE with a polynomial modulus-to-noise ratio. This improves upon the work of Waters (STOC 2024) which required a quadratic-size structured reference string (in *both* modes) and LWE with a super-polynomial modulus-to-noise ratio.
- We give a statistically-hiding vector commitment with transparent setup and polylogarithmic-size CRS, commitments, and openings from SIS. This simultaneously improves upon the vector commitment schemes of de Castro and Peikert (EUROCRYPT 2023) as well as Wee and Wu (EUROCRYPT 2023).

At a conceptual level, our work provides a unified view of recent lattice-based vector commitments and hidden-bits model NIZKs through the lens of the shifted multi-preimage sampling problem.

1 Introduction

Starting from the seminal works of Ajtai [Ajt96] and of Gentry, Peikert, and Vaikuntanathan [GPV08], lattice trapdoors have played a critical role in building advanced cryptographic primitives from lattices. These include notions like hash-and-sign signatures [GPV08], identity-based and attribute-based encryption [GPV08, ABB10b, ABB10a, CHKP10, GVV13, BGG⁺14, GVV15a, BTW17], homomorphic signatures [GVW15b], functional commitments [dCP23, WW23b, BCFL23, WW23a], succinct non-interactive arguments [ACL⁺22, CLM23], and non-interactive zero-knowledge (NIZK) proofs [Wat24].¹

Lattice trapdoors. In this work, we focus on gadget trapdoors [MP12]. In this setting, a trapdoor for a matrix $A \in \mathbb{Z}_q^{n \times m}$ is a short matrix T where $AT = G$ and $G = I_n \otimes g^\top \in \mathbb{Z}_q^{n \times m}$ is the gadget matrix, I_n denotes the n -by- n identity matrix, and $g^\top = [2^0, 2^1, \dots, 2^{\lceil \log q \rceil - 1}]$. Given a trapdoor for a matrix A along with a target vector $c \in \mathbb{Z}_q^n$, we can efficiently compute a short preimage $\pi \in \mathbb{Z}_q^m$ satisfying $A \cdot \pi = c$. In fact, we can even sample *random* short discrete Gaussian preimages, whose distribution we denote by $A^{-1}(c)$.

Shifted multi-preimage sampling. Recent constructions of lattice-based vector commitments [PPS21, WW23b] and non-interactive zero-knowledge (NIZK) proofs [Wat24] implicitly considered variants of a “shifted multi-preimage sampling problem” which is parameterized by a collection of matrices $A_1, \dots, A_\ell \in \mathbb{Z}_q^{n \times m}$:

¹Earlier constructions of lattice-based NIZKs [CCH⁺19, PS19] did not require lattice trapdoors.

Shifted multi-preimage sampling: Given targets $\mathbf{t}_1, \dots, \mathbf{t}_\ell \in \mathbb{Z}_q^n$, sample a random vector $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ along with short discrete Gaussian preimages $\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell \in \mathbb{Z}_q^m$ satisfying $\mathbf{A}_i \boldsymbol{\pi}_i = \mathbf{t}_i + \mathbf{c}$ for all $i \in [\ell]$.

Solving this problem for arbitrary matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ requires knowing some hint (e.g. trapdoors) related to these matrices. The aforementioned applications also require “somewhere hardness:” namely, that the short integer solutions (SIS) or the learning with errors (LWE) problems is hard with respect to any individual $\mathbf{A}_i$ even given the hint. This rules out the trivial solution of taking the hint to be trapdoors for each matrix $\mathbf{A}_1, \dots, \mathbf{A}_\ell$. As a warm-up, observe that for $\ell = 1$ (and $\mathbf{A}_1$ being uniformly random), this problem is straightforward. One can sample a short Gaussian $\boldsymbol{\pi}_1 \in \mathbb{Z}_q^m$ and set $\mathbf{c} = \mathbf{A}_1 \boldsymbol{\pi}_1 - \mathbf{t}_1$. The work of [GPV08] shows that when $m \geq O(n \log q)$, the distribution of $\mathbf{c}$ is statistically close to uniform over $\mathbb{Z}_q^n$.

For the more general version with $\ell > 1$ targets, prior works [PPS21, WW23b, Wat24] required a hint of size $O(\ell^2)$, even for special cases of the problem where the target vectors $\mathbf{t}_1, \dots, \mathbf{t}_\ell$ are the all-zeroes vector $\mathbf{0}^n$. Among them, only the work of Wee and Wu [WW23b] solved the problem in full generality for arbitrary target vectors. They showed how to sample a random $\mathbf{c}$ together with *random* short discrete Gaussian preimages $\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell$ satisfying $\mathbf{A}_i \boldsymbol{\pi}_i = \mathbf{t}_i + \mathbf{c}$. In their construction, the hint corresponds to a gadget trapdoor for the matrix

$$\mathbf{D}_\ell := \left[\begin{array}{ccc|c} \mathbf{A}_1 & & & \mathbf{G} \\ & \ddots & & \vdots \\ & & \mathbf{A}_\ell & \mathbf{G} \end{array} \right] = [\text{diag}(\mathbf{A}_1, \dots, \mathbf{A}_\ell) \mid \mathbf{1}^\ell \otimes \mathbf{G}]. \quad (1.1)$$

The work of [WW23b] uses the gadget trapdoor to sample a random Gaussian preimage of $\mathbf{D}_\ell$ for the target vector $(\mathbf{t}_1, \dots, \mathbf{t}_\ell) \in \mathbb{Z}_q^{n\ell}$; namely, a vector $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \hat{\mathbf{c}})$ where

$$\mathbf{D}_\ell \begin{bmatrix} \boldsymbol{\pi}_1 \\ \vdots \\ \boldsymbol{\pi}_\ell \\ \hat{\mathbf{c}} \end{bmatrix} = \left[\begin{array}{ccc|c} \mathbf{A}_1 & & & \mathbf{G} \\ & \ddots & & \vdots \\ & & \mathbf{A}_\ell & \mathbf{G} \end{array} \right] \begin{bmatrix} \boldsymbol{\pi}_1 \\ \vdots \\ \boldsymbol{\pi}_\ell \\ \hat{\mathbf{c}} \end{bmatrix} = \begin{bmatrix} \mathbf{t}_1 \\ \vdots \\ \mathbf{t}_\ell \end{bmatrix}.$$

They then set $\mathbf{c} = -\mathbf{G}\hat{\mathbf{c}}$. In this case, for all $i \in [\ell]$, $\mathbf{A}_i \boldsymbol{\pi}_i = \mathbf{t}_i - \mathbf{G}\hat{\mathbf{c}} = \mathbf{t}_i + \mathbf{c}$, as required. Moreover, the ensuing distribution of $(\mathbf{c}, \boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell)$ is statistically close to that given by first sampling $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and then $\boldsymbol{\pi}_i \leftarrow \mathbf{A}_i^{-1}(\mathbf{t}_i + \mathbf{c})$.

Drawbacks of prior works. There are two major drawbacks of needing to include hints for $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ as part of the public parameters of the scheme:

- **Trusted setup.** First, sampling the hint typically requires *private* randomness. Existing constructions use private randomness to sample $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ along with their respective trapdoors. In the aforementioned applications (to vector commitments [PPS21, WW23b] and NIZKs [Wat24]), an adversary who knows the private randomness is able to break security of the associated scheme. Thus, the aforementioned constructions all rely on a *trusted* setup to sample the CRS.
- **Hint size.** The aforementioned approaches require a hint whose size is *quadratic* in the dimension ℓ . Here ℓ is the input dimension (in the case of vector commitments) or the length of the hidden-bits string (in the case of using a hidden-bits generator [FLS90, QRW19] to construct a NIZK). Thus, the existing schemes have large public parameters.

This work: *eliminating the hint.* In this work, we show how to construct a *shifted multi-preimage trapdoor sampler* that allows us to solve the shifted multi-preimage sampling problem with respect to a carefully-chosen set of matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ *without* hints. In our construction, the matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ will be correlated, but the marginal distribution of each individual $\mathbf{A}_i$ remains uniformly random, albeit with slightly larger dimensions: $n \times (m \cdot \lceil \log \ell \rceil)$. Moreover, both SIS and LWE are hard with respect to any individual $\mathbf{A}_i$. In fact, $\mathbf{A}_i$ is simply $\mathbf{B} - \mathbf{u}_i \otimes \mathbf{G}$, where $\mathbf{B} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times m \cdot \lceil \log \ell \rceil}$ and $\mathbf{u}_i \in \{0, 1\}^{\lceil \log \ell \rceil}$ is the binary representation of i . Given only the matrix $\mathbf{B}$, we show that we can *publicly* derive a gadget trapdoor for the matrix $\mathbf{D}_\ell$ in Eq. (1.1). Each entry in the public trapdoor of $\mathbf{D}_\ell$ lies in

$\{-1, 0, 1\}$. Here, we rely on the machinery from [GSW13, BGG⁺14, DHM⁺24] for homomorphic computation on matrix encodings. Having a publicly-computable gadget trapdoor means that in our applications, the public parameters only need to specify the single (uniformly) random matrix $\mathbf{B}$; we do not need to include the matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ or its trapdoor for $\mathbf{D}_\ell$ (say, as needed in [WW23b]). This immediately yields improvements to existing constructions of lattice-based vector commitments and dual-mode NIZKs which we discuss in more detail below.

Vector commitments. Recall that a vector commitment allows a user to succinctly commit to a vector $\mathbf{x}$ and also succinctly open to individual components x_i of the committed vector. The security properties are binding and hiding. Binding says that an adversary should not be able to open a commitment σ to two distinct values $x_i \neq x'_i$ at any index i . Hiding says that the openings for any set of indices should not reveal anything about the values at unopened indices.

By integrating our shifted multi-preimage trapdoor sampler with the framework of [WW23b], we obtain a statistically-hiding and computationally-binding vector commitment scheme (Corollary 6.15) from the SIS assumption with a *transparent* (i.e., public-coin) setup. For committing to ℓ -dimensional inputs over $\mathbb{Z}_q^\ell$, the size of the (uniformly-random) common reference string (CRS), the commitment, and the openings are all $\text{poly}(\lambda, \log \ell)$. Our construction simultaneously improves upon and inherits the properties of prior vector commitments from the SIS assumption [PSTY13, LLNW16, PPS21, dCP23, WW23b]. Here, we focus on comparing with the most recent schemes [dCP23, WW23b]:

- (1) Like [dCP23, WW23b], our scheme is linearly homomorphic and supports stateless updates (i.e., given a commitment σ to a vector $\mathbf{x}$ it is possible to transform it into a commitment to $\mathbf{x}'$ using only knowledge of σ and the difference $\mathbf{x}' - \mathbf{x}$);
- (2) We achieve a *transparent* and polylogarithmic-size CRS, matching [dCP23] and improving upon the quadratic-size CRS in [WW23b];
- (3) We achieve statistical hiding and can directly commit to vectors over $\mathbb{Z}_q^\ell$ (while preserving linear homomorphism), as achieved in [WW23b] but not in [dCP23].

We refer to Section 2.3 for further discussion and comparison of our approach with prior work.

Dual-mode NIZKs. Our second application is to (dual-mode) hidden-bits generators [FLS90, QRW19, LPWW20], which imply dual-mode NIZKs. In a dual-mode NIZK [GOS06, GOS12], the CRS can be sampled from one of two computationally-indistinguishable distributions: one distribution yields computational NIZK proofs while the other yields statistical NIZK arguments. Previously, Peikert and Shiehian [PS19] showed how to construct dual-mode NIZKs for NP from LWE by constructing a correlation-intractable hash function [CGH04, KRR17, HL18, CCR18, CCH⁺19]. For many years, the correlation-intractability approach was the only way of realizing NIZKs for NP from lattices.

Very recently, Waters [Wat24] showed a new path for constructing NIZKs from lattices by constructing a hidden-bits generator from the LWE assumption. A hidden-bits generator [FLS90, QRW19] is a cryptographic primitive that generates a succinct commitment to a pseudorandom sequence of hidden bits (and relative to a common reference string). Unlike the case of vector commitments, we require the commitment to *statistically* bind to the sequence of hidden bits (relative to the *long* CRS). Since the commitments are succinct and the hidden-bits generator is statistically binding, the number of possible hidden-bit strings that can be associated with a commitment is small.

In this work, we present a new hidden-bits generator by combining our shifted multi-preimage trapdoor sampler with ideas and techniques from [WW23b] and [Wat24]. Our hidden-bits generator improves upon the [Wat24] hidden-bits generator in three key aspects (Corollary 5.19):

- (1) We achieve a shorter CRS whose size scales *linearly* with the number of hidden bits ℓ . This improves upon the quadratic dependency in [Wat24].
- (2) In hiding mode, our scheme has a transparent setup (i.e., a *uniform* random CRS). This yields statistical (multi-theorem) NIZK arguments with a transparent setup. The [Wat24] construction required a structured CRS in both modes.

- (3) Security relies on LWE with a *polynomial* modulus-to-noise ratio, improving upon the *super-polynomial* modulus-to-noise ratio in [Wat24].

Waters [Wat24] implicitly constructed a hidden-bits generator starting from a shifted multi-preimage trapdoor sampler for the special case where the ℓ target vectors are set to the all-zeroes vector (the vector $\mathbf{c}$ corresponds to the succinct commitment and the preimage $\boldsymbol{\pi}_i$ corresponds to an opening for the i^{th} bit). The sampler in [Wat24] essentially outputs random linear combinations of quantities in the CRS.

Substituting our shifted multi-preimage trapdoor sampler into the [Wat24] approach yields our improved hidden-bits generator. In particular, Properties (1) and (2) follow immediately from the improved parameters for our shifted multi-preimage trapdoor sampler. The binding analysis for our hidden-bits generator is the same as that in [Wat24], whereas the hiding analysis follows the proof of statistical hiding for the vector commitment from [WW23b]. The latter eliminates the use of noise flooding used in [Wat24], which allows us to achieve Property (3). We refer to Section 5 for a more detailed technical comparison of our approach with that of [Wat24].

Taken together, we obtain a dual-mode NIZK for NP from LWE via the hidden-bits model approach that achieves all of the properties of the Peikert-Shiehian construction [PS19] based on correlation-intractable hash functions. Our scheme has the additional appealing feature that it does not need to make non-black-box use of cryptographic primitives or lattice-sampling algorithms. But more broadly, our results show that the hidden-bits model approach is just as versatile for realizing NIZKs for NP from the LWE assumption as the correlation-intractability framework.

A new abstraction. At a conceptual level, our work provides a unified and more modular view of recent lattice-based vector commitments [PPS21, WW23b] and dual-mode NIZKs [Wat24] through the lens of the shifted multi-preimage sampling problem. By focusing on and giving an improved construction of this key cryptographic object (the shifted multi-preimage trapdoor sampler), we immediately obtain improvements to both vector commitments and dual-mode NIZKs. We believe that the notion of shifted multi-preimage sampling, as well as our new techniques for solving this preimage sampling problem, will find additional applications to other lattice-based primitives in the future.

2 Technical Overview

The main technical building block in this work is a shifted multi-preimage trapdoor sampler for solving the shifted multi-preimage sampling problem. Namely, starting from a seed (i.e., a random matrix) $\mathbf{B} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times t}$ where $t = \lceil \log \ell \rceil \cdot m$, we construct a structured matrix of the form in Eq. (1.1):

$$\mathbf{D}_\ell := \left[\begin{array}{ccc|c} \mathbf{A}_1 & & & \mathbf{G} \\ & \ddots & & \vdots \\ & & \mathbf{A}_\ell & \mathbf{G} \end{array} \right] = [\text{diag}(\mathbf{A}_1, \dots, \mathbf{A}_\ell) \mid \mathbf{1}^\ell \otimes \mathbf{G}].$$

The description of $\mathbf{D}_\ell$ is extremely simple: let $\mathbf{u}_i \in \{0, 1\}^{\lceil \log \ell \rceil}$ be the binary representation of i , and set

$$\mathbf{A}_i := \mathbf{B} - \mathbf{u}_i^\top \otimes \mathbf{G} \in \mathbb{Z}_q^{n \times t} \quad (2.1)$$

Moreover, $\mathbf{D}_\ell$ has the following remarkable property: given only the seed $\mathbf{B}$, we can derive a gadget trapdoor for $\mathbf{D}_\ell$ where the entries of the trapdoor are in the set $\{-1, 0, 1\}$. Before we describe how to construct the trapdoor, we describe three properties of $\mathbf{D}_\ell$ that are useful for our applications:

- **Shifted multi-preimage sampling:** Given $\mathbf{D}_\ell$, its trapdoor, and target vectors $\mathbf{t}_1, \dots, \mathbf{t}_\ell \in \mathbb{Z}_q^n$, we can sample a uniform randomly $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ together with short vectors $\boldsymbol{\pi}_i \in \mathbb{Z}_q^m$ such that $\mathbf{A}_i \boldsymbol{\pi}_i = \mathbf{t}_i + \mathbf{c} \in \mathbb{Z}_q^n$ for all $i \in [\ell]$.
- **Hardness:** For all $i \in [\ell]$, both SIS and LWE are hard with respect to any $\mathbf{A}_i$ even given $\mathbf{D}_\ell$ and its trapdoor.
- **Preimage distribution:** We require that the joint distribution of $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c})$ sampled using $\mathbf{D}_\ell$ and its trapdoor to be statistically close to sampling $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and $\boldsymbol{\pi}_i \leftarrow \mathbf{A}_i^{-1}(\mathbf{t}_i + \mathbf{c})$. In other words, the distribution of each $\boldsymbol{\pi}_i$ should be statistically close to a random discrete Gaussian $\boldsymbol{\pi}_i$ conditioned on $\mathbf{A}_i \boldsymbol{\pi}_i = \mathbf{t}_i + \mathbf{c}$.

In the applications to vector commitments (resp., hidden-bits generators), the first property will be used to sample the commitment and the openings, the second will be used to argue binding (resp., mode indistinguishability), and the third will be used to argue hiding.

A gadget trapdoor for D_ℓ . To derive a gadget trapdoor T for D_ℓ , we rely on the machinery from [GSW13, BGG⁺14, DHM⁺24] for homomorphic computation on matrix encodings. We consider the specific application to the family of indicator functions $\{\delta_u : \{0, 1\}^k \rightarrow \{0, 1\}\}_{u \in \{0, 1\}^k}$ given by

$$\delta_u(x) := \begin{cases} 1 & x = u \\ 0 & x \neq u. \end{cases}$$

Next, let $B \in \mathbb{Z}_q^{n \times km}$ be a matrix. Then, there is an efficient (and explicit) algorithm that takes as input $u, x \in \{0, 1\}^k$ and outputs a short matrix $H_{B,u,x}$ (with entries in $\{-1, 0, 1\}$) where

$$(B - x^\top \otimes G) \cdot H_{B,u,x} = \begin{cases} B_u - G & \text{if } u = x \\ B_u & \text{if } u \neq x, \end{cases} \quad (2.2)$$

and moreover, we can efficiently compute $B_u \in \mathbb{Z}_q^{n \times m}$ given just B and u .² Setting $k = \lceil \log \ell \rceil$, we have:

$$\underbrace{\begin{bmatrix} B - u_1^\top \otimes G & & \\ & \ddots & \\ & & B - u_\ell^\top \otimes G \end{bmatrix}}_{D_\ell} \cdot \underbrace{\begin{bmatrix} G & & \\ \vdots & \ddots & \\ G & & \end{bmatrix}}_T \cdot \begin{bmatrix} -H_{B,u_1,u_1} & \cdots & -H_{B,u_\ell,u_1} \\ \vdots & \ddots & \vdots \\ -H_{B,u_1,u_\ell} & \cdots & -H_{B,u_\ell,u_\ell} \\ G^{-1}(B_{u_1}) & \cdots & G^{-1}(B_{u_\ell}) \end{bmatrix} = \begin{bmatrix} G & & \\ & \ddots & \\ & & G \end{bmatrix}. \quad (2.3)$$

Shifted multi-preimage sampling using T . Given the trapdoor T for $D_\ell = [\text{diag}(A_1, \dots, A_\ell) \mid \mathbf{1}^\ell \otimes G]$, we can use the [WW23b] approach to solve the shifted multi-preimage sampling problem with respect to the matrices $A_1, \dots, A_\ell$. Specifically, given target vectors $t_1, \dots, t_\ell \in \mathbb{Z}_q^n$, we use T to sample a Gaussian preimage $(\pi_1, \dots, \pi_\ell, \hat{c})$ to the linear system

$$\underbrace{\begin{bmatrix} A_1 & & \\ & \ddots & \\ & & A_\ell \end{bmatrix}}_{D_\ell} \cdot \begin{bmatrix} G \\ \vdots \\ G \end{bmatrix} \cdot \begin{bmatrix} \pi_1 \\ \vdots \\ \pi_\ell \\ \hat{c} \end{bmatrix} = \begin{bmatrix} t_1 \\ \vdots \\ t_\ell \end{bmatrix}. \quad (2.4)$$

By construction, for all $i \in [\ell]$, we have $A_i \pi_i + G \hat{c} = t_i$, or equivalently, $A_i \pi_i = t_i - G \hat{c}$. Defining $c = -G \hat{c}$, we obtain a solution $(\pi_1, \dots, \pi_\ell, c)$ to the shifted multi-preimage sampling problem.

Hardness. The second property we require is that the SIS and LWE problems are hard with respect to any A_i given $D_\ell = [\text{diag}(A_1, \dots, A_\ell) \mid \mathbf{1}^\ell \otimes G]$ together with the trapdoor for D_ℓ . It suffices to show that given any index $i \in [\ell]$ and any matrix $A^* \in \mathbb{Z}_q^{n \times t}$, we can simulate a seed $B \in \mathbb{Z}_q^{n \times km}$ that expands into matrices $A_1, \dots, A_\ell$ where $A_i = A^*$. We refer to this as a “somewhere programmability” property on our shifted multi-preimage trapdoor sampler. Recall from Eq. (2.1) that $A_i = B - u_i^\top \otimes G$. To simulate a seed B that expands to A^* in position i , we can set

$$B := A^* + u_i^\top \otimes G.$$

Under this definition, $A_i = B - u_i^\top \otimes G = A^*$, as required. Moreover, when A^* is uniformly random (i.e., A^* is an SIS or LWE challenge), then the simulated seed is distributed identically to the real seed. This shows that given an SIS or LWE challenge matrix $A^* \in \mathbb{Z}_q^{n \times t}$, we can simulate an identically-distributed seed B that expands to A^* in position i . This suffices to demonstrate hardness of the SIS or LWE problems with respect to any of the matrices A_i associated with a seed.

² B_u is the homomorphic evaluation of δ_u on B .

Preimage distribution. The preimage distribution property requires that the *joint* distribution of $(\pi_1, \dots, \pi_\ell, \mathbf{c})$ sampled using $\mathbf{D}_\ell, \mathbf{T}$ to be statistically close to sampling $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and $\pi_i \leftarrow \mathbf{A}_i^{-1}(\mathbf{t}_i + \mathbf{c})$. This basic requirement follows directly by properties of the discrete Gaussian distribution (see [Lemma 3.6](#)). For the application to statistically-hiding vector commitments (and for proving a stronger simulation property on our dual-mode hidden-bits generator), we rely on a stronger simulatability property that stipulates that there is an efficient algorithm that samples a seed $\mathbf{B}$ together with trapdoors for the *individual* matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell$. These trapdoors are used to efficiently *simulate* the distribution $\pi_i \leftarrow \mathbf{A}_i^{-1}(\mathbf{c})$. To satisfy this stronger hiding requirement, we append to the seed a random $\mathbf{A} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times m}$ and derive $\mathbf{D}_\ell$ by setting

$$\mathbf{A}_i := [\mathbf{A} \mid \mathbf{B} - \mathbf{u}_i \otimes \mathbf{G}]$$

This way, we can derive gadget trapdoors for $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ starting from a gadget trapdoor for $\mathbf{A}$. It is straightforward to check that we can still derive a gadget trapdoor for the modified $\mathbf{D}_\ell$ given just $\mathbf{A}, \mathbf{B}$, and moreover, that the required hardness properties continue to hold. We provide the formal definition and construction details of our shifted multi-preimage trapdoor sampler in [Section 4](#).

2.1 Application to Vector Commitments

Our shifted multi-preimage trapdoor sampler can be directly applied to the Wee-Wu vector commitment [\[WW23b\]](#) to obtain a statistically-hiding (and computationally-binding) scheme with a short transparent setup. We first recall their construction (rephrased in the language of shifted multi-preimage sampling). In the following description, let ℓ be the vector dimension.

- The common reference string contains matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell \in \mathbb{Z}_q^{n \times t}$ together with a hint for solving the shifted multi-preimage sampling problem with respect to $\mathbf{A}_1, \dots, \mathbf{A}_\ell$.
- The commitment to an input $\mathbf{x} \in \mathbb{Z}_q^\ell$ is a vector $\mathbf{c} \in \mathbb{Z}_q^n$ and an opening to value $x_i \in \mathbb{Z}_q$ at index $i \in [\ell]$ is a short preimage π_i where $\mathbf{A}_i \pi_i = x_i \mathbf{e}_1 + \mathbf{c}$ and $\mathbf{e}_1 = [1, 0, \dots, 0]^\top$ is the first canonical basis vector.

In other words, a commitment $\mathbf{c}$ and the openings $(\pi_1, \dots, \pi_\ell)$ form a solution to the shifted multi-preimage sampling problem with respect to matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ and target vectors $x_1 \mathbf{e}_1, \dots, x_\ell \mathbf{e}_1$. The work of [\[WW23b\]](#) solve the shifted multi-preimage sampling problem by publishing random matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ in the CRS together with a gadget trapdoor for the matrix $\mathbf{D}_\ell = [\text{diag}(\mathbf{A}_1, \dots, \mathbf{A}_\ell) \mid \mathbf{1}^\ell \otimes \mathbf{G}]$. Thus, their construction requires a structured CRS whose size scales quadratically with the input dimension ℓ .

Vector commitments with short transparent setup. To obtain a vector commitment scheme with a succinct transparent setup, we replace the structured matrix $\mathbf{D}_\ell$ and its trapdoor $\mathbf{T}$ in the [\[WW23b\]](#) CRS with the seed $[\mathbf{A} \mid \mathbf{B}]$ of our shifted multi-preimage trapdoor sampler. This is sufficient for correctness. Security then follows by the properties of the shifted multi-preimage trapdoor sampler:

- **Binding:** The [\[WW23b\]](#) scheme is computationally binding if SIS is hard with respect to $\mathbf{A}_i$ (even given $\mathbf{D}_\ell$ and $\mathbf{T}$).³ This is the same hardness property satisfied by our shifted multi-preimage trapdoor sampler.
- **Hiding:** The [\[WW23b\]](#) scheme is statistically hiding if there is an alternative (and statistically indistinguishable) way to sample $\mathbf{D}_\ell, \mathbf{T}$ together with knowledge of a gadget trapdoor for each $\mathbf{A}_1, \dots, \mathbf{A}_\ell$. This is the simulatability property of our shifted multi-preimage trapdoor sampler.

Taken together, we obtain a statistically-hiding vector commitment scheme with a transparent setup. Moreover, the size of the common random string is polylogarithmic in the vector dimension (in contrast to [\[WW23b\]](#) which required a structured quadratic-size CRS). We compare with other vector commitment schemes in [Section 2.3](#) and give the formal description in [Section 6](#).

³Technically, binding holds if the SIS assumption holds with respect to $\mathbf{A}_i$ *without* the first row, but we elide this detail in this overview.

Another view: a randomized version of [dCP23]. We can also view our vector commitment scheme as a *randomized* variant of the de Castro-Peikert functional commitment scheme [dCP23], specialized to the setting of vector commitments. Recall that a [dCP23] commitment to a function f is the matrix $\mathbf{B}_f \in \mathbb{Z}_q^{n \times m}$ associated with homomorphically evaluating f on $\mathbf{B}$ à la [GSW13, BGG⁺14]. To commit to a vector $\mathbf{x} \in \{0, 1\}^\ell$, the [dCP23] approach is to commit to the function $f_{\mathbf{x}}(\mathbf{y}) = \sum_{i \in [\ell]} x_i \cdot \delta_{\mathbf{u}_i}(\mathbf{y})$, where $\mathbf{u}_i$ is the vector whose components are the binary representation of the index $i \in [\ell]$. In this case, the commitment to the function $f_{\mathbf{x}}$ is $\mathbf{C} = \mathbf{B}_{f_{\mathbf{x}}} = \sum_{i \in [\ell]} x_i \mathbf{B}_{\mathbf{u}_i}$ and the opening Π_i for index $i \in [\ell]$ is the homomorphic evaluation matrix $\Pi_i = -\mathbf{H}_{\mathbf{B}, f_{\mathbf{x}}, \mathbf{u}_i} = -\sum_{j \in [\ell]} x_j \mathbf{H}_{\mathbf{B}, \mathbf{u}_j, \mathbf{u}_i}$. Here, $\mathbf{B}_{\mathbf{u}_i}$ and $\mathbf{H}_{\mathbf{B}, \mathbf{u}_j, \mathbf{u}_i}$ are the homomorphic evaluation matrices from Eq. (2.2). The verification relation (for index i and value $x_i \in \{0, 1\}$) checks that Π_i has low norm and that,

$$(\mathbf{B} - \mathbf{u}_i^\top \otimes \mathbf{G}) \cdot \Pi_i = x_i \mathbf{G} - \mathbf{C}.$$

Our vector commitment scheme relies on the same verification relation but instead of computing $\mathbf{C}$ and the Π_i 's deterministically, we jointly sample random $\mathbf{C}, \Pi_1, \dots, \Pi_\ell$ that satisfy the above verification relation. Roughly speaking, we would compute

$$\mathbf{C} = \sum_{j \in [\ell]} \mathbf{B}_{\mathbf{u}_j} \mathbf{G}_{\text{rand}}^{-1}(x_j \mathbf{G}), \quad \Pi_i = - \sum_{j \in [\ell]} \mathbf{H}_{\mathbf{B}, \mathbf{u}_j, \mathbf{u}_i} \mathbf{G}_{\text{rand}}^{-1}(x_j \mathbf{G})$$

where $\mathbf{G}_{\text{rand}}^{-1}(\mathbf{Y})$ denotes sampling $\mathbf{X} \in \mathbb{Z}^{m \times m}$ from a discrete Gaussian distribution conditioned on $\mathbf{G}\mathbf{X} = \mathbf{Y}$ (formally, we also need to introduce an additional perturbation matrix following [MP12]). We omit further details since our formal analysis does not rely on this viewpoint. We note that extending the de Castro-Peikert scheme to achieve statistical hiding when opening to general circuits (i.e., a statistically-hiding *functional* commitment) remains an open problem.

2.2 Application to Dual-Mode NIZKs for NP

The second application of our shifted multi-preimage trapdoor sampler is to dual-mode hidden-bits generators, which in turn, implies a dual-mode NIZK for NP. We start with a more modular view of the dual-mode hidden-bits generator by Waters [Wat24] in the language of the shifted multi-preimage sampling and then show how our shifted multi-preimage trapdoor sampler can simultaneously reduce the CRS size (from quadratic to linear), achieve transparent setup in hiding mode, and only rely on LWE with a polynomial modulus-to-noise ratio. Thus, we simultaneously improve on functionality and security.

Hidden-bits generators. At a high level, a hidden-bits generator [FLS90, QRW19] is a cryptographic primitive that generates a pseudorandom sequence of hidden bits from a short seed. Moreover, the user can provide local openings to any subset of the bits (with respect to a commitment of the seed). The unopened bits should remain pseudorandom. While both vector commitments and hidden-bits generators have the flavor of committing to a long input with a short commitment, there is a key distinction between the two notions:

- In a vector commitment, the user can choose *any* string of bits $\mathbf{r} \in \{0, 1\}^\ell$ and derive a succinct commitment to $\mathbf{r}$. The scheme is *computationally* binding (i.e., it is hard for an adversary to open a commitment to two distinct values at any single index).
- In a hidden-bits generator, the user samples a succinct commitment σ . The commitment together with a *long* CRS determines an associated hidden-bits string $\mathbf{r} \in \{0, 1\}^\ell$. In this setting, the binding property is *statistical*; each commitment can only be opened to at most one bit-string $\mathbf{r}$ with respect to the CRS. Since the commitment is succinct, the number of potential bit-strings $\mathbf{r} \in \{0, 1\}^\ell$ that has an associated commitment σ is small.

More formally, a hidden-bits generator consists of three main algorithms:

- The $\text{Setup}(1^\lambda, 1^\ell)$ algorithm takes the security parameter λ and the output length ℓ and outputs a common reference string crs .
- The $\text{GenBits}(\text{crs})$ algorithm takes the common reference string and outputs a *short* commitment σ , a bit-string $\mathbf{r} \in \{0, 1\}^\ell$, and openings $\pi_1, \dots, \pi_\ell$.

- The $\text{Verify}(\text{crs}, \sigma, i, r_i, \pi_i)$ algorithm takes the common reference string crs , the commitment σ , an index $i \in [\ell]$, a bit $r_i \in \{0, 1\}$, and a proof π_i and decides whether to accept or reject.

The security requirements are (1) binding which says that the adversary should not be able to open a commitment σ to both a 0 and a 1 at any index i ; and (2) hiding, which says that given the commitment σ , the bits r_i together with their openings π_i for all $i \in S$, the unopened bits r_i for $i \notin S$ should be pseudorandom. In conjunction with information-theoretic NIZKs for NP in the hidden-bits model [FLS90], a hidden-bits generator gives a NIZK for NP [QRW19].

In a dual-mode hidden-bits generator [LPWW20], we impose an additional property where the crs output by Setup is in one of two computationally indistinguishable modes. In binding mode, the hidden-bits generator should be *statistically* binding while in hiding mode, the hidden-bits generator should be *statistically* hiding. Dual-mode hidden-bits generators imply dual-mode NIZKs for NP (i.e., NIZKs where the CRS can be sampled in one of two computationally-indistinguishable modes, with one mode yielding computational NIZK proofs and the other yielding statistical NIZK arguments). Recently, Waters [Wat24] constructed the first dual-mode hidden-bits generator from LWE.

The [Wat24] hidden-bits generator. We now describe the recent dual-mode hidden-bits generator by Waters [Wat24] in the language of the shifted multi-preimage sampling problem. In the following description, let ℓ be the length of the hidden-bits string.

- The common reference string consists of matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell \in \mathbb{Z}_q^{n \times t}$.
- To generate a hidden-bits string, the prover invokes the shifted multi-preimage sampling procedure for $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ to obtain a random $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ together with short preimages $\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell \in \mathbb{Z}_q^t$ satisfying $\mathbf{A}_i \boldsymbol{\pi}_i = \mathbf{c}$ for all $i \in [\ell]$. In this setting, all of the target vectors $\mathbf{t}_1, \dots, \mathbf{t}_\ell$ in the shifted multi-preimage sampling problem are the all-zeroes vector $\mathbf{0}^n$. The commitment is $\mathbf{c}$ and an opening at index i is $\boldsymbol{\pi}_i$.
- The CRS also contains vectors $\mathbf{v}_1, \dots, \mathbf{v}_\ell \in \mathbb{Z}_q^t$ which in conjunction with the openings, determine the hidden-bits string: the i^{th} bit b_i is given by $\lfloor \mathbf{v}_i^\top \boldsymbol{\pi}_i \rfloor$, where we write $\lfloor \cdot \rfloor$ to denote the rounding operation (i.e., $\lfloor x \rfloor$ outputs 0 if $|x| < q/4$ and 1 if $|x - q/2| < q/4$).

The distribution of $\mathbf{v}_i$ determines whether the bit b_i is binding or hiding:

- When $\mathbf{v}_i^\top = \mathbf{s}_i^\top \mathbf{A}_i + \mathbf{e}_i^\top$ is an LWE sample, then the bit b_i is completely determined by the CRS component $\mathbf{v}_i$ and the commitment $\mathbf{c}$. To see this, consider *any* valid opening $(\boldsymbol{\pi}_i, b_i)$ for any commitment $\mathbf{c} \in \mathbb{Z}_q^n$ on index i . Then, it must be the case that $\boldsymbol{\pi}_i$ is short and $\mathbf{A}_i \boldsymbol{\pi}_i = \mathbf{c}$. Moreover,

$$b_i = \lfloor \mathbf{v}_i^\top \boldsymbol{\pi}_i \rfloor = \lfloor \mathbf{v}_i^\top \boldsymbol{\pi}_i - \mathbf{e}_i^\top \boldsymbol{\pi}_i \rfloor = \lfloor (\mathbf{s}_i^\top \mathbf{A}_i + \mathbf{e}_i^\top) \boldsymbol{\pi}_i - \mathbf{e}_i^\top \boldsymbol{\pi}_i \rfloor = \lfloor \mathbf{s}_i^\top \mathbf{c} \rfloor,$$

where the second equality holds as long as $\mathbf{v}_i^\top \boldsymbol{\pi}_i$ is far from a rounding boundary (enforced by the verification relation) and the fact that if $\boldsymbol{\pi}_i$ is small, then so is $\mathbf{e}_i^\top \boldsymbol{\pi}_i$. Thus, in this case, the commitment $\mathbf{c}$ completely determines the associated bit b_i . Importantly, this analysis holds for *any* (even adversarial) choice of the commitment $\mathbf{c}$.

- When $\mathbf{v}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$ is a uniform random vector, then $\mathbf{v}_i$ and $\mathbf{c}$ leaks no information about b_i ; in particular, the bit b_i is statistically close to uniform given the commitment $\mathbf{c}$ and the openings $\boldsymbol{\pi}_j$ for all $j \neq i$. In our work, we argue this by relying on the hiding property of our shifted multi-preimage trapdoor sampler. Namely, when $\boldsymbol{\pi}_i \leftarrow \mathbf{A}_i^{-1}(\mathbf{c})$ and $\mathbf{v}_i$ is uniform, the distribution of $\mathbf{v}_i^\top \boldsymbol{\pi}_i \in \mathbb{Z}_q$ is statistically close to uniform over $\mathbb{Z}_q$; this holds by appealing to the leftover hash lemma and the fact that the discrete Gaussian distribution has high min-entropy. Moreover, the preimage distribution property of shifted multi-preimage trapdoor sampler also ensures that each $\boldsymbol{\pi}_i$ is independent of $\boldsymbol{\pi}_j$ for $j \neq i$ (indeed, the preimage distribution property stipulates that each $\boldsymbol{\pi}_i$ is sampled independently from $\mathbf{A}_i^{-1}(\mathbf{c})$, independently of all other $\boldsymbol{\pi}_j$). The work of [Wat24] relied on a noise smudging argument to argue hiding. We refer to Section 5 for a more detailed comparison between the two approaches for analyzing hiding.

Finally, the distribution of $\mathbf{v}_i$ in the two modes is computationally indistinguishable under the LWE assumption. In the context of the shifted multi-preimage trapdoor sampler, we require that LWE hold with respect to the matrix $\mathbf{A}_i$.

Instantiating [Wat24] with the shifted multi-preimage trapdoor sampler. The work of [Wat24] solve the shifted multi-preimage sampling problem by publishing a collection of short preimages in the common reference string. This leads to a structured CRS in both binding and hiding modes, and moreover, the size of the CRS scales quadratically with the output length of the hidden-bits generator. The [Wat24] scheme also requires a super-polynomial modulus to implement the noise smudging argument needed for hiding.

In this work, we replace the CRS with the seed for our shifted multi-preimage trapdoor sampler. This provides an efficient way to solve the shifted multi-preimage sampling problem and thus, suffices to instantiate the general blueprint of [Wat24]. In this case, the CRS for the hidden-bits generator consists of the seed for the sampler together with the vectors $\mathbf{v}_1, \dots, \mathbf{v}_\ell \in \mathbb{Z}_q^t$. This yields a hidden-bits generator with a CRS that is linear in the output length. Moreover, in statistically-hiding mode, each $\mathbf{v}_i$ is uniformly random, and we obtain a scheme in the common *random* string model (equivalently, a scheme with a *transparent* setup). We provide the full details in Section 5.

Dual-mode NIZKs for NP. Taken together, we obtain a dual-mode hidden-bits generator from plain LWE (Corollary 5.19). In conjunction with existing compilers [FLS90, QRW19, LPWW20, Wat24], this yields a dual-mode (multi-theorem) NIZK for NP from LWE with a polynomial modulus-to-noise ratio (Corollary 5.20). Our construction achieves the same set of properties as the Peikert-Shiehian construction [PS19] based on correlation-intractable hash functions. Our approach thus yields an alternative route to constructing NIZKs for NP that does not rely on non-black-box use of cryptographic primitives or lattice sampling algorithms.

2.3 Related Work

Vector commitments. Starting from Merkle’s construction of vector commitments from any collision-resistant hash function [Mer87], many works have studied constructions of vector commitments from algebraic assumptions over groups with bilinear maps [LY10, KZG10, CF13, LRY16, LM19, TAB⁺20, GRWZ20], groups of unknown order [CF13, LM19, CFG⁺20, AR20, TXN20], and lattice-based assumptions [PSTY13, LLNW16, PPS21, dCP23, WW23b]. Compared to schemes like Merkle [Mer87] based on collision-resistant hash functions, the advantage of the algebraic approach we take is the support for properties like linear homomorphism (e.g., given commitments to $\mathbf{x}, \mathbf{x}'$, we can compute a commitment to the sum $\mathbf{x} + \mathbf{x}'$) or the support for stateless updates. Moreover, the basic approach based on collision-resistant hash functions does not satisfy hiding. It is possible to augment Merkle commitments to be (statistically) hiding using (lossy) public-key encryption or (statistical) NIZK arguments.

Prior to this work, the state-of-the-art in lattice-based vector commitments are [dCP23, WW23b]; both works give constructions from the SIS assumption. The work of de Castro and Peikert [dCP23] has a transparent polylogarithmic-size CRS (similar to our scheme). However, their scheme does not natively support hiding. While their work describes a way to achieve statistical hiding, the transformation comes at the price of relaxing binding to the weaker notion of target binding (i.e., where an adversary cannot open an *honestly-generated* commitment to two different values). Alternatively, one could compose the [dCP23] commitment scheme with a statistical NIZK argument [PS19, Wat24] (or a lossy public-key encryption scheme [PW08, HLOV11]) to obtain a statistically-hiding scheme. This approach would additionally bring in the LWE assumption. The advantage of our approach is we achieve statistical hiding directly without needing additional tools. Conversely, the scheme of Wee and Wu [WW23b] is statistically hiding. However, their scheme requires a structured CRS whose size scales *quadratically* with the vector dimension. Our construction has a transparent and polylogarithmic-size CRS.

Non-interactive zero-knowledge. NIZKs have been extensively studied and we have constructions from most standard algebraic assumptions, such as factoring [FLS90], pairing-based assumptions [CHK03, GOS06], (sub-exponential) decisional Diffie-Hellman [JJ21], learning with errors [CCH⁺19, PS19, Wat24], and the combination of learning parity with noise in conjunction with multivariate quadratic equations [DJJ24]. Among the lattice-based constructions, the initial constructions [CCH⁺19, PS19] leveraged correlation-intractable hash functions to provably instantiate the Fiat-Shamir heuristic, while the recent work of [Wat24] use the LWE assumption to implement the classic hidden-bits model.

Concurrent work. In a concurrent and independent work, Branco et al. [BCD⁺25] show how to construct NIZKs from vector trapdoor hashing. Similar to our work, they improve upon the dual-mode hidden-bits generator [Wat24]

to obtain a construction with a transparent setup in hiding mode and where security relies on LWE with a polynomial modulus-to-noise ratio. Like [Wat24], the size of the CRS in their construction remains *quadratic* in the output length ℓ of the hidden-bits generator, while our construction has a linear-size CRS. The size of and time to verify the openings in their scheme also scale linearly with ℓ whereas in our scheme (and in [Wat24]), the sizes of the openings scale polylogarithmically with ℓ . At a technical level, the two works take different and incomparable approaches. The work of [BCD⁺25] show how to avoid lattice trapdoors altogether, whereas our approach relies on building a trapdoor for solving the shifted multi-preimage sampling problem that has a succinct description. Our work also differs in how we handle inputs that land within a “rounding boundary;” we provide more discussion in Remark 5.21. Finally, our shifted multi-preimage trapdoor sampler allows us to directly show our hidden-bits generator satisfies a stronger simulation-based notion of security (see Remark 5.3). Then, using the results from [LPWW20], we can directly argue that the resulting NIZKs obtained via our hidden-bits generator satisfies *multi-theorem* statistical zero-knowledge in the *uniform* random string model. In contrast, the work of [BCD⁺25] needs to invoke an additional [FLS90]-style compiler based on “or-proofs.” This step requires non-black-box use of a cryptographic language (i.e., the single-theorem NIZK is used to prove membership in a cryptographic language). Our approach avoids this step entirely.

3 Preliminaries

Throughout this work, we write λ to denote the security parameter. For a positive integer $n \in \mathbb{N}$, we define the set $[n] := \{1, \dots, n\}$. For a positive integer $q \in \mathbb{N}$, we write $\mathbb{Z}_q$ to denote the ring of integers modulo q . We write $\text{poly}(\lambda)$ to denote a fixed polynomial in λ . We write $\text{negl}(\lambda)$ to denote a function that is negligible in λ (i.e., a function that is $o(\lambda^{-c})$ for all $c \in \mathbb{N}$). We say an event occurs with overwhelming probability if the probability of its complement occurring is negligible. We say an algorithm is efficient if it runs in probabilistic polynomial time in the length of its input. For two ensembles of distributions $\mathcal{D}_1 = \{\mathcal{D}_{1,\lambda}\}_{\lambda \in \mathbb{N}}$ and $\mathcal{D}_2 = \{\mathcal{D}_{2,\lambda}\}_{\lambda \in \mathbb{N}}$ indexed by a security parameter, we say they are computationally indistinguishable if for all efficient adversaries $\mathcal{A}$, there exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$,

$$\left| \Pr[\mathcal{A}(1^\lambda, x) = 1 : x \leftarrow \mathcal{D}_{1,\lambda}] - \Pr[\mathcal{A}(1^\lambda, x) = 1 : x \leftarrow \mathcal{D}_{2,\lambda}] \right| = \text{negl}(\lambda).$$

We say they are statistically indistinguishable if there exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$, the statistical distance between them is $\text{negl}(\lambda)$. We write $\mathcal{D}_1 \stackrel{c}{\approx} \mathcal{D}_2$ (resp., $\mathcal{D}_1 \stackrel{s}{\approx} \mathcal{D}_2$) if $\mathcal{D}_1$ and $\mathcal{D}_2$ are computationally (resp., statistically) indistinguishable. We write $\mathcal{D}_1 \equiv \mathcal{D}_2$ if the distributions are identical.

Vectors and matrices. Throughout, we use bold uppercase letters (e.g., $\mathbf{A}, \mathbf{B}$) to denote matrices, bold lowercase letters (e.g., $\mathbf{u}, \mathbf{v}$) to denote vectors, and non-boldface letters to refer to their components (e.g., $\mathbf{v} = [v_1, \dots, v_n]$). For matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell$, we write $\text{diag}(\mathbf{A}_1, \dots, \mathbf{A}_\ell)$ to denote the block diagonal matrix where the blocks are the matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell$. For a vector $\mathbf{v} \in \mathbb{Z}^n$, we write $\|\mathbf{v}\|$ to denote the ℓ_∞ -norm of $\mathbf{v}$. When $\mathbf{v} \in \mathbb{Z}_q^n$, we write $\|\mathbf{v}\|$ to denote the ℓ_∞ -norm of the vector (over $\mathbb{Z}^n$) obtained by first associating each component $v_i \in \mathbb{Z}_q$ with its unique representative in the set $(-q/2, q/2] \cap \mathbb{Z}$. For a matrix $\mathbf{A}$, we write $\|\mathbf{A}\|$ to denote the ℓ_∞ -norm of the vector obtained by concatenating together the columns of $\mathbf{A}$ (i.e., $\|\mathbf{A}\| = \max_{i,j} |A_{i,j}|$).

Lemma 3.1 (Full Rank Matrices [GPV08, Lemma 5.1]). *Let n, m, q be lattice parameters where q is prime and $m \geq 2n \log q$. Then, all but a $\text{negl}(n)$ fraction of matrices $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ are full rank.*

Discrete Gaussians and gadget matrices. We write $D_{\mathbb{Z},s}$ to denote the discrete Gaussian distribution over $\mathbb{Z}$ with width parameter $s > 0$. For a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and a target vector $\mathbf{y} \in \mathbb{Z}_q^n$ in the column-span of $\mathbf{A}$, we write $\mathbf{A}^{-1}(\mathbf{y})$ to denote the random variable $\mathbf{x} \leftarrow D_{\mathbb{Z},s}^m$ conditioned on $\mathbf{A}\mathbf{x} = \mathbf{y} \bmod q$. Note that if $\mathbf{y}$ is not in the column-span of $\mathbf{A}$, then the distribution $\mathbf{A}^{-1}(\mathbf{y})$ simply outputs $\perp$ with probability 1. We extend $\mathbf{A}^{-1}$ to operate on matrices by applying $\mathbf{A}^{-1}$ column-wise. For positive integers $n, q \in \mathbb{N}$, we write $\mathbf{G} = \mathbf{I}_n \otimes \mathbf{g}^\top \in \mathbb{Z}_q^{n \times m'}$ to be the gadget matrix [MP12] where $\mathbf{I}_n$ is the identity matrix of dimension n , $\mathbf{g}^\top = [1, 2, \dots, 2^{\lceil \log q \rceil - 1}]$, and $m' = n \lceil \log q \rceil$. For dimensions $m \geq m'$, we overload the notation and write $\mathbf{G} \in \mathbb{Z}_q^{n \times m}$ to denote the “padded gadget matrix” $[\mathbf{I}_n \otimes \mathbf{g}^\top \mid \mathbf{0}^{n \times (m-m')}]$. The inverse function $\mathbf{G}^{-1}: \mathbb{Z}_q^n \rightarrow \mathbb{Z}_q^{m'}$ expands each entry $x \in \mathbb{Z}_q$ into a column of size $\lceil \log q \rceil$ corresponding to the bits in the binary

representation of x . Similarly, when $G \in \mathbb{Z}_q^{n \times m}$ is a padded gadget matrix with dimension $m \geq m'$, we extend the output of $G^{-1} : \mathbb{Z}_q^n \rightarrow \mathbb{Z}_q^m$ by zero-padding each column. By construction, for all $\mathbf{t} \in \mathbb{Z}_q^n$, it follows that $G \cdot G^{-1}(\mathbf{t}) = \mathbf{t} \bmod q$.

Lemma 3.2 (Gaussian Tail Bound [MP12, Lemma 2.6, adapted]). *Let n, m, q be lattice parameters where $m \geq 2n \log q$. Then, for all but a $\text{negl}(n)$ -fraction of matrices $A \in \mathbb{Z}_q^{n \times m}$, all width parameters $s > \log m$ and all vectors $\mathbf{y} \in \mathbb{Z}_q^n$ in the column-span of A ,*

$$\Pr[\|\mathbf{x}\| > \sqrt{ms} : \mathbf{x} \leftarrow A_s^{-1}(\mathbf{y})] \leq O(2^{-m}).$$

For the particular case of the discrete Gaussian over the integers and any $\lambda \in \mathbb{N}$,

$$\Pr[|x| > \sqrt{\lambda}s : x \leftarrow D_{\mathbb{Z},s}] \leq 2^{-\lambda}.$$

Lemma 3.3 (Discrete Gaussian Preimages [GPV08, adapted]). *Let n, m, q, s be lattice parameters where $m \geq n \lceil \log q \rceil$ and $s \geq \log m$. Then the statistical distance between the following distributions is at most $\text{negl}(n)$:*

$$\{G\mathbf{x} : \mathbf{x} \leftarrow D_{\mathbb{Z},s}^m\} \quad \text{and} \quad \left\{ \mathbf{u} : \mathbf{u} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n \right\}.$$

Discrete Gaussian preimages. We will need to reason about the distribution of $[\text{diag}(A_1, \dots, A_\ell) \mid B]_s^{-1}(\mathbf{t})$ where $A_i \in \mathbb{Z}_q^{n \times m}$ and $B \in \mathbb{Z}_q^{n \ell \times k}$.

Definition 3.4 (Minimum Distance of $\Lambda(A_i)$). For a matrix $A \in \mathbb{Z}_q^{n \times m}$, we write $\Lambda(A)$ to denote the q -ary lattice $\Lambda(A) := \{\mathbf{y} \in \mathbb{Z}^m : \mathbf{y} = A^T \mathbf{x} \bmod q \text{ for some } \mathbf{x} \in \mathbb{Z}^m\}$. For a lattice $\Lambda \subset \mathbb{R}^m$, we write $\lambda_1^\infty(\Lambda)$ to denote the minimum distance $\lambda_1^\infty(\Lambda) := \min_{\mathbf{0} \neq \mathbf{v} \in \Lambda} \|\mathbf{v}\|$.

Lemma 3.5 (Minimum Distance of Random Matrix [GPV08, Lemma 5.3]). *Let n, m, q be lattice parameters where q is prime and $m \geq 2n \log q$. Then, for all but a $q^{-n} = \text{negl}(n)$ fraction of matrices $A \in \mathbb{Z}_q^{n \times m}$, $\lambda_1^\infty(\Lambda(A)) \geq q/4$.*

Lemma 3.6 (Discrete Gaussian Preimages [WW23b, Corollary 2.11]). *Let n, m, q, t be parameters where $m \geq n$. Take any $\ell, k = \text{poly}(n, \log q)$, any collection of matrices $A_1, \dots, A_\ell \in \mathbb{Z}_q^{n \times m}$ where A_i is full rank and $\lambda_1^\infty(\Lambda(A_i)) \geq t$ for all $i \in [\ell]$, any collection of matrices $B_1, \dots, B_\ell \in \mathbb{Z}_q^{n \times k}$, and any target vector $\mathbf{t} \in \mathbb{Z}_q^{n\ell}$. Define the following matrices*

$$C = \left[\begin{array}{ccc|c} A_1 & & & B_1 \\ & \ddots & & \vdots \\ & & A_\ell & B_\ell \end{array} \right] \quad \text{and} \quad \mathbf{t} = \begin{bmatrix} \mathbf{t}_1 \\ \vdots \\ \mathbf{t}_\ell \end{bmatrix}.$$

Then, for all width parameters $s \geq q/t \cdot \log(\ell m)$, the statistical distance between the following distributions is $\text{negl}(m)$:

$$\left\{ \mathbf{v} : \mathbf{v} \leftarrow C_s^{-1}(\mathbf{t}) \right\} \quad \text{and} \quad \left\{ \begin{bmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_\ell \\ \mathbf{v}_{\ell+1} \end{bmatrix} : \begin{array}{l} \mathbf{v}_{\ell+1} \xleftarrow{D_{\mathbb{Z},s}^k} \\ \mathbf{v}_i \leftarrow (A_i)_s^{-1}(\mathbf{t}_i - B_i \mathbf{u}). \end{array} \right\}. \quad (3.1)$$

Min-entropy. Let $\mathcal{D}$ be a distribution with finite support $\mathcal{X}$. We define the min-entropy of $\mathcal{D}$ to be $H_\infty(\mathcal{D}) := -\log \max_{x \in \mathcal{X}} \Pr[X = x : X \leftarrow \mathcal{D}]$. We now state the following corollary of the leftover hash lemma [HILL99]:

Lemma 3.7 (Leftover Hash Lemma). *Let $m, q \in \mathbb{N}$ be positive integers where q is prime. Let $\mathcal{D}$ be a distribution over $\mathbb{Z}_q^m$ where $H_\infty(\mathcal{D}) \geq 2\lambda + \log q$. Then, for all $\ell = \ell(\lambda)$, the statistical distance between the following two distributions is at most $\ell(\lambda) \cdot 2^{-\lambda}$:*

$$\left\{ (\mathbf{v}, \mathbf{v}^T \mathbf{x}_1, \dots, \mathbf{v}^T \mathbf{x}_\ell) : \mathbf{v} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^m, \forall i \in [\ell] : \mathbf{x}_i \leftarrow \mathcal{D} \right\} \quad \text{and} \quad \left\{ (\mathbf{v}, r_1, \dots, r_\ell) : \mathbf{v} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^m, \forall i \in [\ell] : r_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q \right\}.$$

Lemma 3.8 (Min-Entropy of Discrete Gaussian [PR06, Lemma 2.11, adapted]). *Let n, q be lattice parameters and suppose $m \geq 2n \log q$. Then, for all but a $\text{negl}(n)$ -fraction of matrices $A \in \mathbb{Z}_q^{n \times m}$, all width parameters $s \geq \log m$, all target vectors $\mathbf{y} \in \mathbb{Z}_q^n$, the random variable $\mathbf{x} \leftarrow A_s^{-1}(\mathbf{y})$ has min-entropy $H_\infty(\mathbf{x}) \geq n/2$.*

Gadget trapdoors. Our constructions will use the gadget trapdoors from [MP12], which builds on a long sequence of works on constructing lattice trapdoors [Ajt96, GPV08, AP09, ABB10a, ABB10b, CHKP10].

Theorem 3.9 (Gadget Trapdoor [MP12, adapted]). *Let n, m, q be lattice parameters with $m \geq 3n \lceil \log q \rceil$. Then there exist efficient algorithms (TrapGen, SamplePre) with the following syntax:*

- $\text{TrapGen}(1^n, q, m) \rightarrow (\mathbf{A}, \mathbf{T})$: On input the lattice dimension n , the modulus q , and the number of samples m , the trapdoor-generation algorithm outputs a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ together with a trapdoor $\mathbf{T}$.
- $\text{SamplePre}(\mathbf{A}, \mathbf{T}, \mathbf{y}, s) \rightarrow \mathbf{x}$: On input a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, a trapdoor $\mathbf{T}$, a target vector $\mathbf{y} \in \mathbb{Z}_q^n$, and a Gaussian width parameter s , the preimage-sampling algorithm outputs a vector $\mathbf{x} \in \mathbb{Z}_q^m$.

Moreover, the above algorithms satisfy the following properties:

- **Trapdoor distribution:** If $(\mathbf{A}, \mathbf{T}) \leftarrow \text{TrapGen}(1^n, q, m)$, then the distribution of $\mathbf{A}$ is 2^{-n} -close to the uniform distribution over $\mathbb{Z}_q^{n \times m}$. Moreover, $\mathbf{AT} = \mathbf{G}$ and $\|\mathbf{T}\| = 1$.
- **Preimage sampling:** For all matrices $\mathbf{T}$, width parameters $s > 0$, and all target vectors $\mathbf{y} \in \mathbb{Z}_q^n$ in the column span of $\mathbf{A}$, if we sample $\mathbf{x} \leftarrow \text{SamplePre}(\mathbf{A}, \mathbf{T}, \mathbf{y}, s)$, then $\mathbf{Ax} = \mathbf{y}$.
- **Preimage distribution:** Suppose $\mathbf{T}$ is a gadget trapdoor for $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ (i.e., $\mathbf{AT} = \mathbf{G}$). Then, for all $s \geq m \|\mathbf{T}\| \log n$, and all target vectors $\mathbf{y} \in \mathbb{Z}_q^n$, the statistical distance between the following distributions is at most 2^{-n} :

$$\{\mathbf{x} \leftarrow \text{SamplePre}(\mathbf{A}, \mathbf{T}, \mathbf{y}, s)\} \quad \text{and} \quad \{\mathbf{x} \leftarrow \mathbf{A}_s^{-1}(\mathbf{y})\}.$$

Homomorphic evaluation. Our construction of succinct functional commitments will rely on the lattice homomorphic evaluation procedure developed in [GSW13, BGG⁺14, DHM⁺24]. In this work, we consider a specialization to indicator functions $\delta_{\mathbf{u}}: \{0, 1\}^\ell \rightarrow \{0, 1\}$ where

$$\delta_{\mathbf{u}}(\mathbf{x}) := \begin{cases} 1 & \mathbf{x} = \mathbf{u} \\ 0 & \mathbf{x} \neq \mathbf{u}. \end{cases}$$

Specifically, we focus on the version for “database reads” from [DHM⁺24, §4.1].

Theorem 3.10 (Homomorphic Encodings [DHM⁺24, Theorem 4.5, adapted]). *Let λ be a security parameter and $n = n(\lambda)$, $q = q(\lambda)$ be lattice parameters. Take any $m \geq n \lceil \log q \rceil$ and let $k = k(\lambda)$ be an input length. Then there exist a pair of efficient algorithms (EvalF, EvalFX) with the following properties:*

- $\text{EvalF}(\mathbf{A}, \delta_{\mathbf{u}}) \rightarrow \mathbf{A}_{\mathbf{u}}$: On input a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times km}$ and the indicator function $\delta_{\mathbf{u}}$ (where $\mathbf{u} \in \{0, 1\}^k$), the input-independent evaluation algorithm outputs a matrix $\mathbf{A}_{\mathbf{u}} \in \mathbb{Z}_q^{n \times m}$.
- $\text{EvalFX}(\mathbf{A}, \delta_{\mathbf{u}}, \mathbf{x}) \rightarrow \mathbf{H}_{\mathbf{A}, \mathbf{u}, \mathbf{x}}$: On input a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times km}$, an indicator function $\delta_{\mathbf{u}}$ (where $\mathbf{u} \in \{0, 1\}^k$), and an input $\mathbf{x} \in \{0, 1\}^k$, the input-dependent evaluation algorithm outputs a matrix $\mathbf{H}_{\mathbf{A}, \mathbf{u}, \mathbf{x}} \in \mathbb{Z}_q^{km \times m}$.

Moreover, for all security parameters $\lambda \in \mathbb{N}$, matrices $\mathbf{A} \in \mathbb{Z}_q^{n \times km}$, all indicator functions $\delta_{\mathbf{u}}$, and all inputs $\mathbf{x} \in \{0, 1\}^k$, the matrices $\mathbf{A}_{\mathbf{u}} \leftarrow \text{EvalF}(\mathbf{A}, \delta_{\mathbf{u}})$ and $\mathbf{H}_{\mathbf{A}, \mathbf{u}, \mathbf{x}} \leftarrow \text{EvalFX}(\mathbf{A}, \delta_{\mathbf{u}}, \mathbf{x})$ satisfy the following properties:

- $\mathbf{H}_{\mathbf{A}, \mathbf{u}, \mathbf{x}} \in \{-1, 0, 1\}^{km \times m}$.
- $(\mathbf{A} - \mathbf{x}^\top \otimes \mathbf{G}) \cdot \mathbf{H}_{\mathbf{A}, \mathbf{u}, \mathbf{x}} = \mathbf{A}_{\mathbf{u}} - \delta_{\mathbf{u}}(\mathbf{x}) \cdot \mathbf{G}$.

The running time of $\text{EvalF}(\mathbf{A}, \delta_{\mathbf{u}})$ and $\text{EvalFX}(\mathbf{A}, \delta_{\mathbf{u}}, \mathbf{x})$ is bounded by $2^k \cdot \text{poly}(n, m, k, \log q)$.

Lattice assumptions. We recall the short integer solutions (SIS) [Ajt96] and learning with errors (LWE) [Reg05] problems.

Assumption 3.11 (Short Integer Solutions [Ajt96]). Let λ be a security parameter and $n = n(\lambda)$, $m = m(\lambda)$, $q = q(\lambda)$, and $\beta = \beta(\lambda)$ be lattice parameters. We say the short integer solution problem $\text{SIS}_{n,m,q,\beta}$ holds if for all efficient adversaries $\mathcal{A}$,

$$\Pr \left[\mathbf{Ax} = \mathbf{0} \text{ and } 0 < \|\mathbf{x}\| \leq \beta : \begin{array}{l} \mathbf{A} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times m}; \\ \mathbf{x} \leftarrow \mathcal{A}(1^\lambda, \mathbf{A}) \end{array} \right] = \text{negl}(\lambda).$$

Assumption 3.12 (Learning with Errors [Reg05]). Let λ be a security parameter and $n = n(\lambda)$, $m = m(\lambda)$, $q = q(\lambda)$, and $s = s(\lambda)$ be lattice parameters. We say the learning with errors problem $\text{LWE}_{n,m,q,s}$ holds if for all efficient adversaries $\mathcal{A}$,

$$\left| \Pr \left[\mathcal{A}(\mathbf{A}, \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top) = 1 : \begin{array}{l} \mathbf{A} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times m} \\ \mathbf{s} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n, \mathbf{e} \leftarrow D_{\mathbb{Z}_s}^m \end{array} \right] - \Pr \left[\mathcal{A}(\mathbf{A}, \mathbf{u}^\top) = 1 : \begin{array}{l} \mathbf{A} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times m} \\ \mathbf{u} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^m \end{array} \right] \right| = \text{negl}(\lambda).$$

4 Shifted Multi-Preimage Trapdoor Sampler

In this section, we describe our technique for deriving a collection of matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ together with a trapdoor for solving the shifted multi-preimage sampling problem from a short common *random* string. Our construction relies on the following key lemma asserting that a certain structured matrix has a *public* trapdoor:

Lemma 4.1 (Structured Lattice with a Public Trapdoor). *Let λ be a security parameter and $n = n(\lambda)$, $m = m(\lambda)$ and $q = q(\lambda)$ be lattice parameters. Suppose $m \geq n \lceil \log q \rceil$. Then, for all $\ell \in \mathbb{N}$, and $t = m \lceil \log \ell \rceil$, there exists an explicit polynomial-time algorithm StructTrapGen that takes as input $(\mathbf{B}, \mathbf{u}_1, \dots, \mathbf{u}_\ell)$ where $\mathbf{B} \in \mathbb{Z}_q^{n \times t}$, and $\mathbf{u}_1, \dots, \mathbf{u}_\ell \in \{0, 1\}^{\lceil \log \ell \rceil}$ are distinct vectors, and outputs a gadget trapdoor $\mathbf{T} \in \mathbb{Z}_q^{(t+m) \times \ell m}$ where $\|\mathbf{T}\| = 1$ for the matrix*

$$\mathbf{D}'_\ell = \left[\begin{array}{ccc|c} \mathbf{B} - \mathbf{u}_1^\top \otimes \mathbf{G} & & & \mathbf{G} \\ & \ddots & & \vdots \\ & & \mathbf{B} - \mathbf{u}_\ell^\top \otimes \mathbf{G} & \mathbf{G} \end{array} \right] \in \mathbb{Z}_q^{\ell n \times (\ell t + m)}.$$

Proof. Algorithm StructTrapGen($\mathbf{B}, \mathbf{u}_1, \dots, \mathbf{u}_\ell$) works as follows:

1. For $i \in [\ell]$, let $\delta_{\mathbf{u}_i} : \{0, 1\}^{\lceil \log \ell \rceil} \rightarrow \{0, 1\}$ be the indicator function where $\delta_{\mathbf{u}_i}(\mathbf{v})$ outputs 1 if $\mathbf{v} = \mathbf{u}_i$ and 0 otherwise. For each $i, j \in [\ell]$, compute $\mathbf{H}_{\mathbf{B}, \mathbf{u}_i, \mathbf{u}_j} \leftarrow \text{EvalFX}(\mathbf{B}, \delta_{\mathbf{u}_i}, \mathbf{u}_j)$ and $\mathbf{B}_{\mathbf{u}_i} \leftarrow \text{EvalF}(\mathbf{B}, \delta_{\mathbf{u}_i})$.
2. Output the trapdoor

$$\mathbf{T} = \left[\begin{array}{ccc|c} -\mathbf{H}_{\mathbf{B}, \mathbf{u}_1, \mathbf{u}_1} & \cdots & -\mathbf{H}_{\mathbf{B}, \mathbf{u}_\ell, \mathbf{u}_1} & \vdots \\ \vdots & \ddots & \vdots & \vdots \\ -\mathbf{H}_{\mathbf{B}, \mathbf{u}_1, \mathbf{u}_\ell} & \cdots & -\mathbf{H}_{\mathbf{B}, \mathbf{u}_\ell, \mathbf{u}_\ell} & \vdots \\ \mathbf{G}^{-1}(\mathbf{B}_{\mathbf{u}_1}) & \cdots & \mathbf{G}^{-1}(\mathbf{B}_{\mathbf{u}_\ell}) & \mathbf{G} \end{array} \right] \in \mathbb{Z}_q^{(\ell t + m) \times \ell m}. \quad (4.1)$$

To complete the proof, we show that the trapdoor $\mathbf{T}$ output by StructTrapGen satisfies the required properties. Namely, we need to show that

$$\mathbf{D}'_\ell \mathbf{T} = \left[\begin{array}{ccc|c} \mathbf{B} - \mathbf{u}_1^\top \otimes \mathbf{G} & & & \mathbf{G} \\ & \ddots & & \vdots \\ & & \mathbf{B} - \mathbf{u}_\ell^\top \otimes \mathbf{G} & \mathbf{G} \end{array} \right] \cdot \left[\begin{array}{ccc|c} -\mathbf{H}_{\mathbf{B}, \mathbf{u}_1, \mathbf{u}_1} & \cdots & -\mathbf{H}_{\mathbf{B}, \mathbf{u}_\ell, \mathbf{u}_1} & \vdots \\ \vdots & \ddots & \vdots & \vdots \\ -\mathbf{H}_{\mathbf{B}, \mathbf{u}_1, \mathbf{u}_\ell} & \cdots & -\mathbf{H}_{\mathbf{B}, \mathbf{u}_\ell, \mathbf{u}_\ell} & \vdots \\ \mathbf{G}^{-1}(\mathbf{B}_{\mathbf{u}_1}) & \cdots & \mathbf{G}^{-1}(\mathbf{B}_{\mathbf{u}_\ell}) & \mathbf{G} \end{array} \right] = \left[\begin{array}{ccc|c} \mathbf{G} & & & \\ & \ddots & & \\ & & \mathbf{G} & \end{array} \right]. \quad (4.2)$$

Since $t = m \cdot \lceil \log \ell \rceil \geq n \lceil \log q \rceil \lceil \log \ell \rceil$, by Theorem 3.10 (with the input length $k = \lceil \log \ell \rceil$), we have

$$(\mathbf{B} - \mathbf{u}_i^\top \otimes \mathbf{G}) \cdot \mathbf{H}_{\mathbf{B}, \mathbf{u}_j, \mathbf{u}_i} = \mathbf{B}_{\mathbf{u}_j} - \delta_{\mathbf{u}_j}(\mathbf{u}_i) \cdot \mathbf{G} = \begin{cases} \mathbf{B}_{\mathbf{u}_j} - \mathbf{G} & i = j \\ \mathbf{B}_{\mathbf{u}_j} & i \neq j. \end{cases}$$

Correspondingly, Eq. (4.2) holds. Again by Theorem 3.10, for all $i, j \in [\ell]$, we have that $\mathbf{H}_{\mathbf{B}, \mathbf{u}_i, \mathbf{u}_j} \in \{-1, 0, 1\}^{t \times m}$. Moreover $\mathbf{G}^{-1}(\mathbf{B}_{\mathbf{u}_i}) \in \{0, 1\}^{m \times m}$, so it follows that $\|\mathbf{T}\| = 1$. $\square$

Shifted multi-preimage trapdoor syntax. At a high-level, a shifted multi-preimage trapdoor sampler provides a way to sample a set of matrices $A_1, \dots, A_\ell$ together with a trapdoor td that allows us to efficiently solve the shifted multi-preimage sampling problem with respect to $A_1, \dots, A_\ell$. Formally, the sampler consists of a Gen algorithm that samples a common reference string crs , an Expand algorithm that expands crs into the matrices $A_1, \dots, A_\ell$ and the trapdoor td , and a shifted multi-preimage sampler algorithm SampleMultPre . The main properties we require are as follows:

- **Correctness:** Given the trapdoor td and any set of target vectors $\mathbf{t}_1, \dots, \mathbf{t}_\ell$, the shifted multi-preimage sampler $\text{SampleMultPre}(td, \mathbf{t}_1, \dots, \mathbf{t}_\ell)$ outputs a solution $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c})$ where for all $i \in [\ell]$, $A_i \boldsymbol{\pi}_i = \mathbf{t}_i + \mathbf{c}$.
- **Preimage distribution:** We require that the solutions output by SampleMultPre to have a “nice” distribution. Formally, we require that the joint distribution of the solution $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c})$ output by the sampler $\text{SampleMultPre}(td, \mathbf{t}_1, \dots, \mathbf{t}_\ell)$ to be statistically close to sampling $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and $\boldsymbol{\pi}_i \leftarrow A_i^{-1}(\mathbf{t}_i + \mathbf{c})$. In other words, the distribution of each $\boldsymbol{\pi}_i$ should be statistically close to an independent discrete Gaussian $\boldsymbol{\pi}_i$ conditioned on $A_i \boldsymbol{\pi}_i = \mathbf{t}_i + \mathbf{c}$. We use this property to argue hiding for our dual-mode hidden-bits generator ([Theorem 5.11](#)) and for our vector commitment scheme ([Theorem 6.8](#)).
- **Somewhere programmable:** We also require that SIS and LWE are hard with respect to any A_i given $A_1, \dots, A_\ell$ and the trapdoor td . We model this by defining a “somewhere programmable” property which stipulates that there is an auxiliary sampling algorithm GenProg that takes as input an index i and a (random) matrix A_i and outputs a common reference string $\widetilde{\text{crs}}$ that is indistinguishable from the common reference string output by Gen . Moreover, the i^{th} matrix associated with $\widetilde{\text{crs}}$ is precisely the programmed matrix A_i . This property ensures that the marginal distribution of any individual A_i associated with an honestly-generated crs is statistically close to uniform, and in addition, that problems like SIS or LWE remain hard with respect to any individual A_i even given the common reference string. We use this property to argue mode indistinguishability for our dual-mode hidden-bits generator ([Theorem 5.6](#)) and binding for our vector commitment scheme ([Theorem 6.4](#)).

We now give the formal definition and construction.

Definition 4.2 (Shifted Multi-Preimage Trapdoor Sampler). Let λ be a security parameter and ℓ be a dimension. Let n, t, q, s be parameters that are functions of λ and ℓ . An (n, t, q, s) -shifted multi-preimage trapdoor sampler is a tuple of efficient algorithms $\Pi_{\text{samp}} = (\text{Gen}, \text{Expand}, \text{SampleMultPre})$ with the following syntax:

- $\text{Gen}(1^\lambda, 1^\ell) \rightarrow \text{crs}$: On input the security parameter λ and the dimension ℓ , the generator algorithm outputs a common reference string crs .
- $\text{Expand}(1^\lambda, 1^\ell, \text{crs}) \rightarrow (A_1, \dots, A_\ell, td)$: On input the security parameter λ , the dimension ℓ , and the common reference string crs , the expand algorithm outputs matrices $A_1, \dots, A_\ell \in \mathbb{Z}_q^{n \times t}$ and a trapdoor td . This algorithm is *deterministic*.
- $\text{SampleMultPre}(td, \mathbf{t}_1, \dots, \mathbf{t}_\ell) \rightarrow (\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c})$: On input a trapdoor td and a collection of preimages $\mathbf{t}_1, \dots, \mathbf{t}_\ell$, the shifted multi-preimage sampling algorithm outputs a shift $\mathbf{c} \in \mathbb{Z}_q^n$ together with preimages $\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell \in \mathbb{Z}_q^t$.

The shifted multi-preimage trapdoor sampler should satisfy the following properties:

- **Correctness:** For all $\lambda, \ell \in \mathbb{N}$, all crs in the support of $\text{Gen}(1^\lambda, 1^\ell)$, all target vectors $\mathbf{t}_1, \dots, \mathbf{t}_\ell \in \mathbb{Z}_q^n$, and setting $(A_1, \dots, A_\ell, td) = \text{Expand}(1^\lambda, 1^\ell, \text{crs})$, it holds that

$$\Pr[A_i \boldsymbol{\pi}_i = \mathbf{t}_i + \mathbf{c} \text{ for all } i \in [\ell] : (\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c}) \leftarrow \text{SampleMultPre}(td, \mathbf{t}_1, \dots, \mathbf{t}_\ell)] = 1.$$

- **Preimage distribution:** For all polynomials $\ell = \ell(\lambda)$, there exists a negligible function $\text{negl}(\cdot)$ such that with overwhelming probability over the choice of $\text{crs} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$, letting $(A_1, \dots, A_\ell, td) = \text{Expand}(1^\lambda, \text{crs})$, and for all targets $\mathbf{t}_1, \dots, \mathbf{t}_\ell \in \mathbb{Z}_q^n$ and all $\lambda \in \mathbb{N}$, the statistical distance between the following distributions is $\text{negl}(\lambda)$:
 - $\mathcal{D}_0$: Output $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c}) \leftarrow \text{SampleMultPre}(td, \mathbf{t}_1, \dots, \mathbf{t}_\ell)$.
 - $\mathcal{D}_1$: Sample $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and $\boldsymbol{\pi}_i \leftarrow (A_i)_s^{-1}(\mathbf{t}_i + \mathbf{c})$ for each $i \in [\ell]$. Output $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c})$.

- **Somewhere programmable:** There exists an efficient algorithm GenProg such that for all polynomials $\ell = \ell(\lambda)$, the following hold:

- For all $\lambda \in \mathbb{N}$, all indices $i \in [\ell]$, and all matrices $\mathbf{A}_i \in \mathbb{Z}_q^{n \times t}$, it holds that

$$\Pr \left[\mathbf{A}_i = \tilde{\mathbf{A}}_i : \begin{array}{l} \widetilde{\text{crs}} \leftarrow \text{GenProg}(1^\lambda, 1^\ell, i, \mathbf{A}_i) \\ (\tilde{\mathbf{A}}_1, \dots, \tilde{\mathbf{A}}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \widetilde{\text{crs}}) \end{array} \right] = 1.$$

- There exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$ and all $i \in [\ell]$, the statistical distance between the following distributions is $\text{negl}(\lambda)$:

$$\left\{ \text{crs} : \text{crs} \leftarrow \text{Gen}(1^\lambda, 1^\ell) \right\} \quad \text{and} \quad \left\{ \widetilde{\text{crs}} : \begin{array}{l} \mathbf{A}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times t} \\ \widetilde{\text{crs}} \leftarrow \text{GenProg}(1^\lambda, 1^\ell, i, \mathbf{A}_i) \end{array} \right\}.$$

When these distributions are identical, we say the shifted multi-preimage trapdoor sampler satisfies *perfect* somewhere programmability.

Definition 4.3 (Transparent Setup). A shifted multi-preimage trapdoor sampler $(\text{Gen}, \text{Expand}, \text{SampleMultiPre})$ supports transparent setup if the common reference string crs output by Gen just consists of the random coins used to sample crs . Otherwise, we say the common reference string is a *structured* reference string (i.e., sampled using *private* randomness).

Local expansion. The Expand algorithm in Definition 4.2 outputs a collection of ℓ matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ along with a trapdoor $\mathbf{T}$. Consequently, this necessarily takes $\text{poly}(\lambda, \ell)$ time. In some applications (e.g., to vector commitments), we require the shifted multi-preimage trapdoor sampler to support a more fine-grained algorithm ExpandLocal that takes as input a single index $i \in [\ell]$ and outputs $\mathbf{A}_i$. In particular, the local expansion algorithm only needs to read $\text{poly}(\lambda, \log \ell)$ bits of the CRS and runs in time $\text{poly}(\lambda, \log \ell)$. We give the formal definition below:

Definition 4.4 (Local Expansion). A shifted multi-preimage trapdoor sampler $(\text{Gen}, \text{Expand}, \text{SampleMultiPre})$ supports local expansion if there exists an efficient algorithm ExpandLocal that takes as input crs together with an index $i \in [\ell]$ and outputs $\mathbf{A}_i$:

- $\text{ExpandLocal}(1^\lambda, \text{crs}, i)$: On input the security parameter λ , the common reference string crs , and an index i , the local expand algorithm outputs a matrix $\mathbf{A}_i \in \mathbb{Z}_q^{n \times t}$. This algorithm is deterministic.

The requirement for ExpandLocal is that for all $\lambda, \ell \in \mathbb{N}$ and all crs , if $\text{Expand}(1^\lambda, 1^\ell, \text{crs}) = (\mathbf{A}_1, \dots, \mathbf{A}_\ell, \mathbf{T})$, then for all $i \in [\ell]$, $\text{ExpandLocal}(1^\lambda, \text{crs}, i) = \mathbf{A}_i$. In addition, $\text{ExpandLocal}(1^\lambda, \text{crs}, i)$ only needs to read $\text{poly}(\lambda, \log \ell)$ bits of crs , and moreover, runs in time $\text{poly}(\lambda, \log \ell)$ given these bits.

Simulatable openings. In some applications (e.g., to statistically-hiding vector commitments; see Section 6), we require a stronger requirement where we can sample a common reference string crs (that is indistinguishable from a real CRS) together with trapdoors $\mathbf{T}_1, \dots, \mathbf{T}_\ell$ for each of the matrices $\mathbf{A}_1, \dots, \mathbf{A}_\ell$ associated with crs . Notably, the trapdoor $\mathbf{T}_i$ can be used to *efficiently* sample from $\mathbf{A}_i^{-1}(\mathbf{c})$ for any vector $\mathbf{c}$ in the column-space of $\mathbf{A}_i$. We give the formal definition below:

Definition 4.5 (Simulatable Openings). An (n, t, q, s) -shifted multi-preimage trapdoor sampler $\Pi_{\text{samp}} = (\text{Gen}, \text{Expand}, \text{SampleMultiPre})$ has simulatable openings if there exists an efficient algorithm GenTD with the following syntax:

- $\text{GenTD}(1^\lambda, 1^\ell) \rightarrow (\text{crs}, \mathbf{T}_1, \dots, \mathbf{T}_\ell)$: On input the security parameter λ and the dimension ℓ , the trapdoor generator algorithm outputs a common reference string crs together with trapdoors $\mathbf{T}_1, \dots, \mathbf{T}_\ell$.

Moreover, the GenTD algorithm should satisfy the following properties:

- **Mode indistinguishability:** There exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$, the statistical distance between the following distributions is $\text{negl}(\lambda)$:

$$\left\{ \text{crs} : \text{crs} \leftarrow \text{Gen}(1^\lambda, 1^\ell) \right\} \quad \text{and} \quad \left\{ \text{crs} : (\text{crs}, \mathbf{T}_1, \dots, \mathbf{T}_\ell) \leftarrow \text{GenTD}(1^\lambda, 1^\ell) \right\}.$$

- **Trapdoor generation:** For all $\lambda, \ell \in \mathbb{N}$, and all $(\text{crs}, \mathbf{T}_1, \dots, \mathbf{T}_\ell)$ in the support of $\text{GenTD}(1^\lambda, 1^\ell)$, and setting $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs})$, we have that

$$\forall i \in [\ell] : \mathbf{A}_i \mathbf{T}_i = \mathbf{G} \quad \text{and} \quad \|\mathbf{T}_i\| \leq s/(t \log n).$$

Shifted multi-preimage trapdoor sampler construction. We now give our shifted multi-preimage trapdoor sampler and analysis. The construction critically leverages [Lemma 4.1](#). We refer to [Section 2](#) for an overview of our construction.

Construction 4.6 (Shifted Multi-Preimage Trapdoor Sampler). Let λ be a security parameter and ℓ be a dimension. Let $n = n(\lambda, \ell)$, $q = q(\lambda, \ell)$, and $s = s(\lambda, \ell)$ be lattice parameters. Let $m = 3n \lceil \log q \rceil$ and $t = m(\lceil \log \ell \rceil + 1)$. In the following construction, we associate each index $i \in [\ell]$ with a distinct canonical bit-vector $\mathbf{u}_i \in \{0, 1\}^{\lceil \log \ell \rceil}$ (e.g., the bit-vector associated with the binary representation of $i - 1$). We construct an (n, t, q, s) -shifted multi-preimage trapdoor sampler as follows:

- $\text{Gen}(1^\lambda, 1^\ell)$: On input the security parameter λ and the dimension ℓ , the generator algorithm samples $[\mathbf{A} \mid \mathbf{B}] \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times t}$ and outputs $\text{crs} = [\mathbf{A} \mid \mathbf{B}]$.
- $\text{Expand}(1^\lambda, 1^\ell, \text{crs})$: On input the security parameter λ , the dimension ℓ , and the common reference string $\text{crs} = [\mathbf{A} \mid \mathbf{B}]$, where $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and $\mathbf{B} \in \mathbb{Z}_q^{n \times m \lceil \log \ell \rceil}$. Then, the expand algorithm proceeds as follows:

- For each $i \in [\ell]$, let $\mathbf{B}_i = \mathbf{B} - \mathbf{u}_i^\top \otimes \mathbf{G}$. Then, define

$$\mathbf{A}_i = [\mathbf{A} \mid \mathbf{B} - \mathbf{u}_i^\top \otimes \mathbf{G}] = [\mathbf{A} \mid \mathbf{B}_i] \in \mathbb{Z}_q^{n \times t}.$$

Let $\mathbf{D}_\ell = [\text{diag}(\mathbf{A}_1, \dots, \mathbf{A}_\ell) \mid \mathbf{1}^\ell \otimes \mathbf{G}] \in \mathbb{Z}_q^{\ell n \times (\ell t + m)}$.

- Let $\Pi \in \{0, 1\}^{(\ell t + m) \times (\ell t + m)}$ be the permutation matrix where

$$\mathbf{D}_\ell = \left[\begin{array}{cc|cc|cc|c} \mathbf{A} & \mathbf{B}_1 & & & & & \mathbf{G} \\ & & \mathbf{A} & \mathbf{B}_2 & & & \mathbf{G} \\ & & & & \ddots & & \vdots \\ & & & & & \mathbf{A} & \mathbf{B}_\ell \\ & & & & & & \mathbf{G} \end{array} \right] = \left[\begin{array}{c|cc|cc|c} \mathbf{A} & & & & \mathbf{B}_1 & & \mathbf{G} \\ & \ddots & & & & \ddots & \vdots \\ & & \mathbf{A} & & & & \mathbf{B}_\ell \\ & & & & & & \mathbf{G} \end{array} \right] \Pi. \quad (4.3)$$

- Compute $\mathbf{T}' \leftarrow \text{StructTrapGen}(\mathbf{B}, \mathbf{u}_1, \dots, \mathbf{u}_\ell) \in \mathbb{Z}_q^{(\ell m \lceil \log \ell \rceil + m) \times \ell m}$ where StructTrapGen is the algorithm from [Lemma 4.1](#). Define the trapdoor $\mathbf{T} \in \mathbb{Z}_q^{(\ell t + m) \times \ell m}$ where

$$\mathbf{T} = \Pi^{-1} \begin{bmatrix} \mathbf{0}^{\ell m \times \ell m} \\ \mathbf{T}' \end{bmatrix} \in \mathbb{Z}_q^{(\ell t + m) \times \ell m}. \quad (4.4)$$

The algorithm sets $\text{td} = (\mathbf{D}_\ell, \mathbf{T})$ and outputs $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td})$.

- $\text{SampleMultPre}(\text{td}, \mathbf{t}_1, \dots, \mathbf{t}_\ell)$: On input the trapdoor $\text{td} = (\mathbf{D}_\ell, \mathbf{T})$ and target vectors $\mathbf{t}_1, \dots, \mathbf{t}_\ell \in \mathbb{Z}_q^n$, the shifted multi-preimage sampler defines the vector $\mathbf{t} \in \mathbb{Z}_q^{\ell n}$ to be the vertical concatenation of $\mathbf{t}_1, \dots, \mathbf{t}_\ell$ and outputs $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, -\mathbf{G}\hat{\mathbf{c}})$ where

$$\begin{bmatrix} \boldsymbol{\pi}_1 \\ \vdots \\ \boldsymbol{\pi}_\ell \\ \hat{\mathbf{c}} \end{bmatrix} \leftarrow \text{SamplePre}(\mathbf{D}_\ell, \mathbf{T}, \mathbf{t}, s).$$

Theorem 4.7 (Shifted Multi-Preimage Trapdoor Sampler). Let $n = n(\lambda, \ell)$, $q = q(\lambda, \ell)$ be arbitrary non-negative functions where $n \geq \lambda$. Let $m = 3n \lceil \log q \rceil$ and $t = m(\lceil \log \ell \rceil + 1)$. Then for all $s \geq (\ell t + m) \log(\ell n)$, [Construction 4.6](#) is an (n, t, q, s) -shifted multi-preimage trapdoor sampler with perfect somewhere programmability. Moreover, [Construction 4.6](#) is transparent ([Definition 4.3](#)), supports local expansion ([Definition 4.4](#)) and has simulatable openings ([Definition 4.5](#)). The size of the CRS output by $\text{Gen}(1^\lambda, 1^\ell)$ is $nt \log q$.

Proof. Take any polynomial $\ell = \ell(\lambda)$ and any $\lambda \in \mathbb{N}$. Take any $\text{crs} = [\mathbf{A} \mid \mathbf{B}] \in \mathbb{Z}_q^{n \times t}$ where $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and $\mathbf{B} \in \mathbb{Z}_q^{n \times m \lceil \log q \rceil}$. Let $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(\text{crs})$. Then, $\text{td} = (\mathbf{D}_\ell, \mathbf{T})$ where $\mathbf{D}_\ell = [\text{diag}(\mathbf{A}_1, \dots, \mathbf{A}_\ell) \mid \mathbf{1}^\ell \otimes \mathbf{G}]$ and $\mathbf{T}$ is defined as in Eq. (4.4). By construction, $\mathbf{A}_i = [\mathbf{A} \mid \mathbf{B} - \mathbf{u}_i^\top \otimes \mathbf{G}] = [\mathbf{A} \mid \mathbf{B}_i]$. By Eqs. (4.3) and (4.4) and Lemma 4.1, we have

$$\begin{aligned} \mathbf{D}_\ell \mathbf{T} &= \left[\begin{array}{c|c|c} \mathbf{A} & \mathbf{B}_1 & \mathbf{G} \\ & \ddots & \vdots \\ & & \mathbf{B}_\ell & \mathbf{G} \end{array} \right] \cdot \Pi \cdot \Pi^{-1} \cdot \begin{bmatrix} \mathbf{0}^{\ell m \times \ell m} \\ \mathbf{T}' \end{bmatrix} \\ &= \left[\begin{array}{c|c} \mathbf{B} - \mathbf{u}_1^\top \otimes \mathbf{G} & \mathbf{G} \\ & \vdots \\ & \mathbf{B} - \mathbf{u}_\ell^\top \otimes \mathbf{G} & \mathbf{G} \end{array} \right] \mathbf{T}' = \begin{bmatrix} \mathbf{G} & & \\ & \ddots & \\ & & \mathbf{G} \end{bmatrix}. \end{aligned}$$

Thus, $\mathbf{T}$ is a gadget trapdoor for $\mathbf{D}_\ell$. Again by Lemma 4.1, $\|\mathbf{T}'\| = 1$, so $\|\mathbf{T}\| = 1$. We now show Construction 4.6 satisfies each of the required properties.

Correctness. Take any set of target vectors $\mathbf{t}_1, \dots, \mathbf{t}_\ell \in \mathbb{Z}_q^n$ and any $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c})$ in the support of the sampler $\text{SampleMultPre}(\text{td}, \mathbf{t}_1, \dots, \mathbf{t}_\ell)$. By construction, SampleMultPre first constructs $\mathbf{t} \in \mathbb{Z}_q^{\ell n}$ to be the vertical concatenation of $\mathbf{t}_1, \dots, \mathbf{t}_\ell$. Then it samples

$$\begin{bmatrix} \boldsymbol{\pi}_1 \\ \vdots \\ \boldsymbol{\pi}_\ell \\ \hat{\mathbf{c}} \end{bmatrix} \leftarrow \text{SamplePre}(\mathbf{D}_\ell, \mathbf{T}, \mathbf{t}, s)$$

and sets $\mathbf{c} = -\mathbf{G}\hat{\mathbf{c}}$. Since $\mathbf{D}_\ell \mathbf{T} = \mathbf{I}_\ell \otimes \mathbf{G}$, the columns of $\mathbf{D}_\ell$ span $\mathbb{Z}_q^{\ell n}$. By Theorem 3.9, this means that for all $i \in [\ell]$,

$$\mathbf{A}_i \boldsymbol{\pi}_i + \mathbf{G}\hat{\mathbf{c}} = \mathbf{t}_i \implies \mathbf{A}_i \boldsymbol{\pi}_i = \mathbf{t}_i - \mathbf{G}\hat{\mathbf{c}} = \mathbf{t}_i + \mathbf{c}.$$

Preimage distribution. Take any set of target vectors $\mathbf{t}_1, \dots, \mathbf{t}_\ell \in \mathbb{Z}_q^n$ and let $\mathbf{t} \in \mathbb{Z}_q^{\ell n}$ be the vertical concatenation of $\mathbf{t}_1, \dots, \mathbf{t}_\ell$. We now define a sequence of intermediate distributions:

- $\mathcal{D}_0$: Output $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c}) \leftarrow \text{SampleMultPre}(\text{td}, \mathbf{t}_1, \dots, \mathbf{t}_\ell)$. Specifically, sample $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \hat{\mathbf{c}})$ according to $\text{SamplePre}(\mathbf{D}_\ell, \mathbf{T}, \mathbf{t}, s)$ and set $\mathbf{c} = -\mathbf{G}\hat{\mathbf{c}}$.
- $\mathcal{D}_1$: Sample $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \hat{\mathbf{c}})$ from $(\mathbf{D}_\ell)_s^{-1}(\mathbf{t})$. Set $\mathbf{c} = -\mathbf{G}\hat{\mathbf{c}}$ and output $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c})$.
- $\mathcal{D}_2$: Sample $\hat{\mathbf{c}} \leftarrow D_{\mathbb{Z}_S}^m$, set $\mathbf{c} = -\mathbf{G}\hat{\mathbf{c}}$, and sample $\boldsymbol{\pi}_i \leftarrow (\mathbf{A}_i)_s^{-1}(\mathbf{t}_i + \mathbf{c})$ for each $i \in [\ell]$. Output $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c})$.
- $\mathcal{D}_3$: Sample $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and $\boldsymbol{\pi}_i \leftarrow (\mathbf{A}_i)_s^{-1}(\mathbf{t}_i + \mathbf{c})$ for each $i \in [\ell]$. Output $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c})$.

By definition, $\mathcal{D}_0$ and $\mathcal{D}_3$ corresponds to the two distributions in the preimage distribution property. We analyze each adjacent pair of distributions:

- First, $\mathcal{D}_0$ and $\mathcal{D}_1$ are statistically indistinguishable by Theorem 3.9. As argued above, $\mathbf{T}$ is a gadget trapdoor for $\mathbf{D}_\ell$ and $\|\mathbf{T}\| = 1$. Since $t = 3n \lceil \log q \rceil$, $s \geq (\ell t + m) \log(\ell n) = (\ell t + m) \|\mathbf{T}\| \cdot \log(\ell n)$, we appeal to Theorem 3.9 to conclude that the statistical distance between the distribution $\text{SamplePre}(\mathbf{D}_\ell, \mathbf{T}, \mathbf{t}, s)$ and $(\mathbf{D}_\ell)_s^{-1}(\mathbf{t})$ is at most $2^{-\ell n} = \text{negl}(\lambda)$. Thus, $\mathcal{D}_0$ and $\mathcal{D}_1$ are statistically indistinguishable.
- Next, $\mathcal{D}_1$ and $\mathcal{D}_2$ are statistically indistinguishable by Lemma 3.6. To argue this, we show that with overwhelming probability over the choice of crs , all of the matrices $\mathbf{A}_i$ output by $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs})$ are full rank and satisfy $\lambda_1^\infty(\mathbf{A}_i) \geq q/4$. Consider the marginal distribution of each $\mathbf{A}_i$. By construction, $\mathbf{A}_i = [\mathbf{A} \mid \mathbf{B} - \mathbf{u}_i^\top \otimes \mathbf{G}]$ where $[\mathbf{A} \mid \mathbf{B}] \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times t}$. Thus, the marginal distribution of $\mathbf{A}_i$ is uniform over $\mathbb{Z}_q^{n \times t}$. By Lemmas 3.1 and 3.5, all but a negligible fraction of matrices $\mathbf{A} \in \mathbb{Z}_q^{n \times t}$ are full rank and satisfy $\lambda_1^\infty(\mathbf{A}) \geq q/4$. Thus, for each $i \in [\ell]$, with overwhelming probability over the choice of crs , the associated matrix $\mathbf{A}_i$ is full rank. Since $\ell = \text{poly}(\lambda)$, we use a union bound to argue that with overwhelming probability over the choice of crs , for all $i \in [\ell]$, it holds that $\mathbf{A}_i$ is full rank and $\lambda_1^\infty(\mathbf{A}_i) \geq q/4$. As long as $s \geq 4 \log(\ell t)$, the claim holds by Lemma 3.6.

- Finally, $\mathcal{D}_2$ and $\mathcal{D}_3$ are statistically indistinguishable by [Lemma 3.3](#). In particular, when $s \geq \log m$, the distribution of $\mathbf{c} = -\mathbf{G}\hat{\mathbf{c}}$ when $\hat{\mathbf{c}} \leftarrow D_{\mathbb{Z},s}^m$ is statistically close to uniform.

By a hybrid argument, we conclude that with overwhelming probability over the choice of $\text{crs} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$, distributions $\mathcal{D}_0$ and $\mathcal{D}_3$ are statistically indistinguishable, as required.

Somewhere programmable. We define the GenProg algorithm as follows:

- $\text{GenProg}(1^\lambda, 1^\ell, i, \mathbf{A}_i)$: On input the security parameter λ , the dimension ℓ , the index $i \in [\ell]$ and a matrix $\mathbf{A}_i = [\mathbf{A} \mid \mathbf{B}]$ where $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and $\mathbf{B} \in \mathbb{Z}_q^{n \times m \lceil \log \ell \rceil}$, output $\widetilde{\text{crs}} = [\mathbf{A} \mid \mathbf{B} + \mathbf{u}_i^\top \otimes \mathbf{G}]$.

We now show that GenProg satisfies the required properties. Take any polynomial $\ell = \ell(\lambda)$ and index $i \in [\ell]$. We consider each property individually:

- Take any $\lambda \in \mathbb{N}$ and any matrix $\mathbf{A}_i = [\mathbf{A} \mid \mathbf{B}] \in \mathbb{Z}_q^{n \times t}$. Let $\widetilde{\text{crs}} \leftarrow \text{GenProg}(1^\lambda, 1^\ell, i, \mathbf{A}_i)$ and consider $(\widetilde{\mathbf{A}}_1, \dots, \widetilde{\mathbf{A}}_\ell, \text{td}) = \text{Expand}(\widetilde{\text{crs}})$. Then $\widetilde{\text{crs}} = [\mathbf{A} \mid \mathbf{B} + \mathbf{u}_i^\top \otimes \mathbf{G}]$. By definition of Expand, we now have

$$\widetilde{\mathbf{A}}_i = [\mathbf{A} \mid (\mathbf{B} + \mathbf{u}_i^\top \otimes \mathbf{G}) - \mathbf{u}_i^\top \otimes \mathbf{G}] = [\mathbf{A} \mid \mathbf{B}] = \mathbf{A}_i.$$

- Suppose $[\mathbf{A} \mid \mathbf{B}] \leftarrow \text{Gen}(1^\lambda, 1^\ell)$. By construction, the Gen algorithm samples $[\mathbf{A} \mid \mathbf{B}] \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times t}$. We claim that this matches the distribution output by $\text{GenProg}(1^\lambda, 1^\ell, i, \mathbf{A}_i)$ when $\mathbf{A}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times t}$. If $\mathbf{A}_i = [\mathbf{A}' \mid \mathbf{B}']$, then $\text{GenProg}(1^\lambda, 1^\ell, i, \mathbf{A}_i)$ outputs $[\mathbf{A}' \mid \mathbf{B}' + \mathbf{u}_i^\top \otimes \mathbf{G}]$, which is still distributed uniformly over $\mathbb{Z}_q^{n \times t}$. Thus, these two distributions are identical.

We conclude that [Construction 4.6](#) satisfies perfect somewhere programmability.

Transparent setup. The common reference string output by Gen consists of a uniform random matrix $[\mathbf{A} \mid \mathbf{B}] \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times t}$, so the scheme has a transparent setup by construction.

Local expansion. The local expansion property follows by construction of Expand. Namely, we can define $\text{ExpandLocal}(1^\lambda, \text{crs}, i)$ to output $\mathbf{A}_i = [\mathbf{A} \mid \mathbf{B} - \mathbf{u}_i^\top \otimes \mathbf{G}] \in \mathbb{Z}_q^{n \times t}$, where $\text{crs} = [\mathbf{A} \mid \mathbf{B}]$.

Simulatable openings. We define the GenTD algorithm as follows:

- $\text{GenTD}(1^\lambda, 1^\ell)$: On input the security parameter λ and the dimension ℓ , the trapdoor generator algorithm samples $(\mathbf{A}, \mathbf{T}) \leftarrow \text{TrapGen}(1^\lambda, q, m)$ and $\mathbf{B} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times m \lceil \log \ell \rceil}$. It sets $\text{crs} = [\mathbf{A} \mid \mathbf{B}]$ and for each $i \in [\ell]$, it sets $\mathbf{T}_i = \begin{bmatrix} \mathbf{T} \\ 0 \end{bmatrix} \in \mathbb{Z}_q^{t \times m}$. Finally, it outputs $(\text{crs}, \mathbf{T}_1, \dots, \mathbf{T}_\ell)$.

We now show that GenTD satisfies mode indistinguishability and the trapdoor generation properties:

- **Mode indistinguishability:** Consider the distribution of $(\text{crs}, \mathbf{T}_1, \dots, \mathbf{T}_\ell)$ output by $\text{GenTD}(1^\lambda, 1^\ell)$. By construction, $\text{crs} = [\mathbf{A} \mid \mathbf{B}]$ where $(\mathbf{A}, \mathbf{T}) \leftarrow \text{TrapGen}(1^\lambda, q, m)$ and $\mathbf{B} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times m \lceil \log \ell \rceil \times m}$. Since $n \geq \lambda$ and $m = 3n \lceil \log q \rceil$, we appeal to [Theorem 3.9](#) to conclude that the distribution of $\mathbf{A}$ is statistically close to uniform. Correspondingly, this means $[\mathbf{A} \mid \mathbf{B}]$ is statistically close to uniform over $\mathbb{Z}_q^{n \times t}$. This is the distribution output by $\text{Gen}(1^\lambda, 1^\ell)$.
- **Trapdoor generation:** Take any $\lambda, \ell \in \mathbb{N}$ and consider $(\text{crs}, \mathbf{T}_1, \dots, \mathbf{T}_\ell)$ output by $\text{GenTD}(1^\lambda, 1^\ell)$. Let $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs})$. By construction, $\text{crs} = [\mathbf{A} \mid \mathbf{B}]$ where $(\mathbf{A}, \mathbf{T}) \leftarrow \text{TrapGen}(1^\lambda, q, m)$ and $\mathbf{T}_i = \begin{bmatrix} \mathbf{T} \\ 0 \end{bmatrix}$. Since $m \geq 3n \lceil \log q \rceil$, by [Theorem 3.9](#), this means $\mathbf{A}\mathbf{T} = \mathbf{G}$. Moreover, by definition of Expand, $\mathbf{A}_i = [\mathbf{A} \mid \mathbf{B} - \mathbf{u}_i \otimes \mathbf{G}]$. Thus,

$$\mathbf{A}_i \mathbf{T}_i = [\mathbf{A} \mid \mathbf{B} - \mathbf{u}_i \otimes \mathbf{G}] \begin{bmatrix} \mathbf{T} \\ 0 \end{bmatrix} = \mathbf{A}\mathbf{T} = \mathbf{G}.$$

Again by [Theorem 3.9](#), $\|\mathbf{T}\| = 1$ so $\|\mathbf{T}_i\| = 1$. Since $s \geq (\ell t + m) \log(\ell n)$, we have $\|\mathbf{T}_i\| = 1 \leq s/(t \log n)$. $\square$

Marginal distribution of matrices output by the shifted multi-preimage trapdoor sampler. The somewhere programmability requirement of a shifted multi-preimage trapdoor sampler ensures that the marginal distribution of each $\mathbf{A}_i$ obtained by running $\text{crs} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$ and $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs})$ is statistically close to uniform. This will be useful in our applications, so we give the formal statement below:

Lemma 4.8 (Marginal Distribution of $\mathbf{A}_i$). *Let λ be a security parameter and ℓ be a dimension. Suppose $(\text{Gen}, \text{Expand}, \text{SampleMultPre})$ is an (n, t, q, s) -shifted multi-preimage trapdoor sampler. Then, for all polynomials $\ell = \ell(\lambda)$, there exists a negligible function $\text{negl}(\cdot)$ such that for all indices $i \in [\ell]$ and all $\lambda \in \mathbb{N}$, the statistical distance between the following distributions is $\text{negl}(\lambda)$:*

$$\left\{ \mathbf{A}_i : \begin{array}{l} \text{crs} \leftarrow \text{Gen}(1^\lambda, 1^\ell) \\ (\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}) \end{array} \right\} \quad \text{and} \quad \left\{ \mathbf{A}_i : \mathbf{A}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times t} \right\}.$$

Proof. Follows immediately by somewhere programmability. $\square$

5 Dual-Mode Hidden-Bits Model NIZK from LWE

In this section, we show how to use the shifted multi-preimage trapdoor sampler from [Section 4](#) to construct a dual-mode hidden-bits generator. Our construction improves upon the previous dual-mode hidden-bits model NIZK of Waters [[Wat24](#)] in several key dimensions: (1) the size of the CRS in our scheme is linear in the length of the hidden-bits string as opposed to quadratic; (2) in statistically-hiding mode, the CRS in our scheme is a common *random* string as opposed to a structured reference string; (3) security relies on the LWE assumption with a *polynomial* modulus-to-noise ratio as opposed to a sub-exponential one.

Dual-mode hidden-bits generator. We start by recalling the notion of a dual-mode hidden-bits generator [[QRW19](#), [LPWW20](#)]. In the definition, we use the statistical single-bit hiding property from [[Wat24](#)], which is sufficient for a basic application to NIZKs for NP. The single-bit security notion is simpler to analyze, so we focus on it for ease of exposition. In [Remark 5.3](#), we discuss a stronger hiding definition on the hidden-bits generator that allows us to obtain a statistical *multi-theorem* NIZK argument in the *uniform* random string model.

Definition 5.1 (Dual-Mode Hidden-Bits Generator [[QRW19](#), [LPWW20](#), [Wat24](#), adapted]). A dual-mode hidden-bits generator is a tuple of efficient algorithms $\Pi_{\text{HBG}} = (\text{Setup}, \text{GenBits}, \text{Verify})$ with the following syntax:

- $\text{Setup}(1^\lambda, 1^\ell, \text{mode}) \rightarrow \text{crs}$: On input the security parameter λ , the output length ℓ , and $\text{mode} \in \{\text{binding}, \text{hiding}\}$, the setup algorithm outputs a common reference string crs .
- $\text{GenBits}(\text{crs}) \rightarrow (\sigma, \mathbf{r}, (\pi_1, \dots, \pi_\ell))$: On input the common reference string crs , the generator algorithm outputs a commitment σ , a bit-string $\mathbf{r} \in \{0, 1\}^\ell$, and a tuple of proofs $\pi_1, \dots, \pi_\ell$.
- $\text{Verify}(\text{crs}, \sigma, i, \beta, \pi) \rightarrow b$: On input the common reference string crs , a commitment σ , an index i , a bit $\beta \in \{0, 1\}$, and a proof π , the verification algorithm outputs a bit $b \in \{0, 1\}$.

Moreover, we require that Π_{HBG} satisfy the following properties:

- **Correctness:** For all $\lambda, \ell \in \mathbb{N}$, all modes $\text{mode} \in \{\text{binding}, \text{hiding}\}$, and all indices $i \in [\ell]$, we have that

$$\Pr \left[\text{Verify}(\text{crs}, \sigma, i, r_i, \pi_i) = 1 : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell, \text{mode}) \\ (\sigma, \mathbf{r}, (\pi_1, \dots, \pi_\ell)) \leftarrow \text{GenBits}(\text{crs}) \end{array} \right] = 1.$$

- **Mode indistinguishability:** For an adversary $\mathcal{A}$, an output length ℓ , and a bit $b \in \{0, 1\}$, we define the mode-indistinguishability game as follows:

1. If $b = 0$, the challenger sets $\text{mode} = \text{binding}$. If $b = 1$, the challenger sets $\text{mode} = \text{hiding}$. The challenger samples $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell, \text{mode})$ and gives $(1^\lambda, 1^\ell, \text{crs})$ to $\mathcal{A}$.

2. Algorithm $\mathcal{A}$ outputs a bit $b' \in \{0, 1\}$, which is the output of the experiment.

The hidden-bits generator satisfies mode indistinguishability if for all efficient adversaries $\mathcal{A}$ and all polynomials $\ell = \ell(\lambda)$, there exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$,

$$|\Pr[b' = 1 : b = 0] - \Pr[b' = 1 : b = 1]| = \text{negl}(\lambda)$$

in the mode-indistinguishability game.

- **Succinctness:** There exists a fixed polynomial $p(\cdot, \cdot)$ such that for all $\lambda, \ell \in \mathbb{N}$, all $\text{mode} \in \{\text{binding}, \text{hiding}\}$, all crs in the support of $\text{Setup}(1^\lambda, 1^\ell, \text{mode})$, and all commitments σ in the support of $\text{GenBits}(\text{crs})$, we have that $|\sigma| \leq p(\lambda, \log \ell)$.
- **Statistically binding in binding mode:** For all polynomials $\ell = \ell(\lambda)$, there exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$,

$$\Pr \left[\begin{array}{c} \exists(\sigma, i, \pi_0, \pi_1) : \\ \text{Verify}(\text{crs}, \sigma, i, 0, \pi_0) = 1 = \text{Verify}(\text{crs}, \sigma, i, 1, \pi_1) \end{array} : \text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell, \text{binding}) \right] = \text{negl}(\lambda).$$

- **Single-bit statistical hiding in hiding mode:** For an adversary $\mathcal{A}$, an output length ℓ , and a bit $b \in \{0, 1\}$, we define the hiding game as follows:

1. On input the security parameter 1^λ and the length parameter 1^ℓ , algorithm $\mathcal{A}$ outputs an index $i^* \in [\ell]$.
2. The challenger samples $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell, \text{hiding})$ and $(\sigma, \mathbf{r}, (\pi_1, \dots, \pi_\ell)) \leftarrow \text{GenBits}(\text{crs})$. If $b = 0$, the challenger sets $\beta = r_{i^*}$. If $b = 1$, it samples $\beta \xleftarrow{\mathbb{R}} \{0, 1\}$.
3. The challenger gives $(\text{crs}, \sigma, \{(i, r_i, \pi_i)\}_{i \neq i^*}, \beta)$ to $\mathcal{A}$.
4. Algorithm $\mathcal{A}$ outputs a bit $b' \in \{0, 1\}$, which is the output of the experiment.

The hidden-bits generator satisfies statistical hiding in hiding mode if for all (computationally-unbounded) adversaries $\mathcal{A}$ and all polynomials $\ell = \ell(\lambda)$, there exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$,

$$|\Pr[b' = 1 : b = 0] - \Pr[b' = 1 : b = 1]| = \text{negl}(\lambda)$$

in the hiding game.

Theorem 5.2 (NIZKs from Hidden-Bits Generators [FLS90, QRW19, LPWW20, Wat24]). *If there exists a dual-mode hidden-bits generator, then there exists a dual-mode NIZK for NP.*

Remark 5.3 (Multi-Theorem Zero-Knowledge). The works of [FLS90, QRW19, LPWW20, Wat24] show that [Definition 5.1](#) implies a dual-mode NIZK that satisfies single-theorem zero-knowledge (in hiding mode). To achieve *multi-theorem* zero-knowledge where the adversary is allowed to see *multiple* proofs (either real or simulated) on adaptively-chosen statements, the work of [FLS90] describes an elegant and generic transformation using “or-proofs” (which only relies on one-way functions). Thus, the NIZK obtained via [Theorem 5.2](#) also satisfies multi-theorem zero-knowledge. A caveat of the [FLS90] transformation though is that it does *not* preserve statistical zero-knowledge in the common *random* string model. Namely, even if we started with a dual-mode NIZK for NP with a common random string in hiding mode (e.g., [Construction 5.4](#)), the resulting dual-mode multi-theorem NIZK would require a structured reference string in *both* modes.

The concurrent work of [BCD⁺25] show how to obtain a single-theorem to multi-theorem transformation that preserves statistical zero-knowledge in the common random string model. Like the classic [FLS90] framework, their approach also takes an “or-proof” approach and uses the single-theorem NIZK to either prove membership in the original language or knowledge of a “simulation trapdoor.” In this work, we show an alternative approach that directly implies a multi-theorem dual-mode NIZK. Namely, we take the approach from [LPWW20] which showed that a dual-mode hidden-bits generator that satisfies a stronger simulation-based hiding notion immediately implies a multi-theorem dual-mode NIZK with no modification whatsoever. In [Appendix A](#), we show that our dual-mode

hidden-bits generator from [Construction 5.4](#) satisfies this stronger requirement without *any* modification other than assuming that our shifted multi-preimage trapdoor sampler has simulatable openings ([Definition 4.5](#)).⁴ Thus, our work directly gives a dual-mode NIZK that satisfies multi-theorem statistical zero-knowledge in the common random string model and completely avoids the “or-proof” (and resulting non-black-box use of cryptography) needed in other dual-mode NIZK constructions from [LWE \[PS19, BCD⁺25\]](#).

Constructing a dual-mode hidden-bits generator. We now show how to use a shifted multi-preimage trapdoor sampler to construct a dual-mode hidden-bits model generator ([Definition 4.2](#)). We refer to [Section 2.2](#) for an overview of the construction. As discussed in [Section 2.2](#), our construction shares a similar structure with the scheme of Waters [[Wat24](#)], but differs in a few key respects:

- **CRS structure:** The [[Wat24](#)] construction publishes structured preimages in the CRS and the commitment and the openings are derived by computing short linear combinations of the components in the CRS. Because the CRS contains structured preimages (in both modes), the [[Wat24](#)] construction relied on a (quadratic-size) structured CRS in both modes. In our scheme, we replace the structured preimages in the CRS with our shifted multi-preimage trapdoor sampler. This allows us to achieve a linear-size CRS, and moreover in hiding mode, the CRS is a uniform random string (i.e., supports transparent setup).
- **Binding analysis:** In the [[Wat24](#)] scheme, the verification algorithm checks that the opening does not land near a “rounding boundary” (and rejects if the opening is too close). This ensures that at every index, a given commitment can only be opened one way. This is useful for arguing binding. In [[Wat24](#)], the modulus q is super-polynomial so the probability that an opening lands near a rounding boundary is negligible. In our construction, we rely on a polynomial modulus q , so there is an inverse polynomial probability that GenBits samples a commitment and a set of openings where one of the opening lands inside the rounding boundary (and causes the verification algorithm to fail). However, since the verification algorithm is public, the GenBits algorithm can simply resample a commitment and set of openings whenever this happens. By choosing q to be a sufficiently-large polynomial, we ensure that each sampling attempt succeeds with at least constant probability. After λ attempts, the algorithm will sample a commitment and a set of valid openings with overwhelming probability. Thus, we can avoid the super-polynomial modulus q in the binding analysis. We also discuss an alternative approach from [[Wat24](#), Appendix B] and used in the concurrent work [[BCD⁺25](#)] in [Remark 5.21](#).
- **Hiding analysis:** The [[Wat24](#)] also critically relies on a super-polynomial modulus q for the hiding analysis. Specifically, when the CRS is sampled in hiding mode, [[Wat24](#)] first establishes that there exists small perturbations that can be added to a commitment and flip the i^{th} output bit of the hidden-bits string, while leaving all remaining bits unchanged. Then, using a noise smudging argument, [[Wat24](#)] argues that the adversary cannot tell whether a commitment is “normal” or “perturbed.” This suffices to show that the i^{th} bit is statistically hidden from the view of the adversary. In our work, we take a different approach. To argue hiding, we rely on the fact that Gaussian preimages have sufficient min-entropy and use this to extract a string of uniform random bits. This avoids the need for noise smudging and allows us to use a polynomial modulus for the overall construction. This in turn allows us to prove security from LWE with a polynomial modulus-to-noise ratio.

We now describe our construction.

Construction 5.4 (Dual-Mode Hidden-Bits Generator). Let λ be a security parameter and ℓ be a length parameter. Let $\Pi_{\text{samp}} = (\text{Gen}, \text{Expand}, \text{SampleMultPre})$ be a $(n, t, q, s_{\text{samp}})$ -shifted multi-preimage trapdoor sampler. Let $s_{\text{LWE}} = s_{\text{LWE}}(\lambda, \ell)$ be a Gaussian width parameter and $B_{\text{max}} = B_{\text{max}}(\lambda, \ell)$, $B_{\text{round}} = B_{\text{round}}(\lambda, \ell)$ be bounds. Throughout, we assume that $B_{\text{round}} < q/4$. We construct our dual-mode hidden bits generator $\Pi_{\text{HBG}} = (\text{Setup}, \text{GenBits}, \text{Verify})$ as follows:

- **Setup($1^\lambda, 1^\ell$, mode):** On input the security parameter λ , the output length ℓ , and mode $\in \{\text{binding}, \text{hiding}\}$, the setup algorithm samples $\text{crs}_{\text{samp}} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$. Next, it computes $(\mathbf{A}_1, \dots, \mathbf{A}_t, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$, where $\mathbf{A}_i \in \mathbb{Z}_q^{n \times t}$. Next, for each $i \in [\ell]$, it samples a vector $\mathbf{v}_i \in \mathbb{Z}_q^t$ as follows:

⁴In the body, we focus on the weaker notion of single-bit hiding which is easier to argue. We defer the full simulation-based security proof to [Appendix A](#).

- If mode = binding, it samples $\mathbf{s}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$, $\mathbf{e}_i \leftarrow D_{\mathbb{Z}, \text{sLWE}}^t$, and sets $\mathbf{v}_i^\top = \mathbf{s}_i^\top \mathbf{A}_i + \mathbf{e}_i^\top$.
- If mode = hiding, it samples $\mathbf{v}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$.

It outputs $\text{crs} = (1^\lambda, \text{crs}_{\text{samp}}, \mathbf{v}_1, \dots, \mathbf{v}_\ell)$.

- **GenBits(crs)**: On input the common reference string $\text{crs} = (1^\lambda, \text{crs}_{\text{samp}}, \mathbf{v}_1, \dots, \mathbf{v}_\ell)$, the generator algorithm first computes $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$. Then, it repeats the following procedure up to λ times:
 - Sample $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c}) \leftarrow \text{SampleMultPre}(\text{td}, \mathbf{0}^n, \dots, \mathbf{0}^n)$.
 - For each $i \in [\ell]$, if $\|\boldsymbol{\pi}_i\| > B_{\max}$, it sets $r_i = \perp$. Otherwise, compute $u_i = \mathbf{v}_i^\top \boldsymbol{\pi}_i$ and set r_i as follows:

$$r_i = \begin{cases} 0 & u_i \in [-B_{\text{round}}, B_{\text{round}}] \\ 1 & u_i \in [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}] \\ \perp & \text{otherwise.} \end{cases}$$

Note that the intervals $[-B_{\text{round}}, B_{\text{round}}]$ and $[\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}]$ are guaranteed to be disjoint when $B_{\text{round}} < q/4$.

- If $r_i \in \{0, 1\}$ for all $i \in [\ell]$, then it outputs $(\mathbf{c}, \mathbf{r}, (\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell))$. Otherwise, if there exists an index $i \in [\ell]$ where $r_i = \perp$, the generator algorithm restarts the sampling procedure.

If the sampling procedure does not succeed after λ attempts, then the generator algorithm sets $\mathbf{c} = \perp$, $\mathbf{r} = \mathbf{0}^\ell$, and $\boldsymbol{\pi}_i = \perp$ for all $i \in [\ell]$. It outputs $(\mathbf{c}, \mathbf{0}^\ell, (\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell))$.

- **Verify(crs, $\mathbf{c}$, i , β , $\boldsymbol{\pi}$)**: On input the common reference string $\text{crs} = (1^\lambda, \text{crs}_{\text{samp}}, \mathbf{v}_1, \dots, \mathbf{v}_\ell)$, a commitment $\mathbf{c}$, an index $i \in [\ell]$, a bit $\beta \in \{0, 1\}$, and a proof $\boldsymbol{\pi}$, the verification algorithm proceeds as follows:
 - If $\mathbf{c} = \perp$ output 1 if $\beta = 0$ and 0 otherwise.
 - If $\mathbf{c} \in \mathbb{Z}_q^n$ and $\boldsymbol{\pi} \in \mathbb{Z}_q^t$, then compute $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$. Output 1 if

$$\|\boldsymbol{\pi}\| \leq B_{\max} \quad \text{and} \quad \mathbf{A}_i \boldsymbol{\pi} = \mathbf{c} \quad \text{and} \quad \mathbf{v}_i^\top \boldsymbol{\pi}_i \in [\lfloor q/2 \rfloor \beta - B_{\text{round}}, \lfloor q/2 \rfloor \beta + B_{\text{round}}]$$

and 0 otherwise.

In all other cases, output 0.

Theorem 5.5 (Correctness). *If Π_{samp} is correct, then [Construction 5.4](#) is correct.*

Proof. Take any $\lambda, \ell \in \mathbb{N}$ and mode $\in \{\text{binding}, \text{hiding}\}$. Suppose $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell, \text{mode})$ and $(\mathbf{c}, \mathbf{r}, (\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell)) \leftarrow \text{GenBits}(\text{crs})$. Then, we can write $\text{crs} = (1^\lambda, \text{crs}_{\text{samp}}, \mathbf{v}_1, \dots, \mathbf{v}_\ell)$. Let $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$. We consider two possibilities:

- Suppose $\mathbf{c} = \perp$. By construction of GenBits, this means $\mathbf{r} = \mathbf{0}^\ell$. Then **Verify**(crs, $\mathbf{c}$, i , 0, $\boldsymbol{\pi}_i$) outputs 1 by construction.
- Suppose $\mathbf{c} \in \mathbb{Z}_q^n$. This means the GenBits algorithm sampled $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c}) \leftarrow \text{SampleMultPre}(\text{td}, \mathbf{0}^n, \dots, \mathbf{0}^n)$. By correctness of Π_{samp} , this means $\mathbf{A}_i \boldsymbol{\pi}_i = \mathbf{c}$ for all $i \in [\ell]$. Finally, GenBits outputs $\mathbf{c}$ only if

$$\|\boldsymbol{\pi}_i\| \leq B_{\max} \quad \text{and} \quad \mathbf{v}_i^\top \boldsymbol{\pi}_i \in [\lfloor q/2 \rfloor r_i - B_{\text{round}}, \lfloor q/2 \rfloor r_i + B_{\text{round}}].$$

In this case, all of the verification checks pass and **Verify**(crs, $\mathbf{c}$, i , r_i , $\boldsymbol{\pi}_i$) = 1. $\square$

Theorem 5.6 (Mode Indistinguishability). *Suppose Π_{samp} satisfies somewhere programmability. Then, under the $\text{LWE}_{n,t,q,\text{sLWE}}$ assumption, [Construction 5.4](#) satisfies mode indistinguishability.*

Proof. Let $\mathcal{A}$ be an efficient adversary for the mode indistinguishability game. We begin by defining a sequence of hybrid experiments parameterized by an index $i \in \{0, \dots, \ell\}$:

- Hyb_i : In this experiment, the challenger samples $\text{crs}_{\text{samp}} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$ and computes $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$. Then, for each $j \in [\ell]$, it constructs the vector $\mathbf{v}_j \in \mathbb{Z}_q^t$ as follows:
 - If $j > i$, sample $\mathbf{s}_j \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$, $\mathbf{e}_j \leftarrow D_{\mathbb{Z}, s_{\text{LWE}}}^t$, and compute $\mathbf{v}_j^\top = \mathbf{s}_j^\top \mathbf{A}_j + \mathbf{e}_j^\top$.
 - If $j \leq i$, sample $\mathbf{v}_j \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$.

The challenger gives $\text{crs} = (1^\lambda, \text{crs}_{\text{samp}}, \mathbf{v}_1, \dots, \mathbf{v}_\ell)$ to $\mathcal{A}$. The output of the experiment is the output of $\mathcal{A}$.

We write $\text{Hyb}_i(\mathcal{A})$ to denote the output of an execution of Hyb_i with adversary $\mathcal{A}$. By construction, $\text{Hyb}_0(\mathcal{A})$ corresponds to the experiment where the challenger samples $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell, \text{binding})$ while $\text{Hyb}_\ell(\mathcal{A})$ corresponds to the experiment where the challenger samples $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell, \text{hiding})$. To complete the proof, we show that for all $i \in [\ell]$, the output distributions $\text{Hyb}_{i-1}(\mathcal{A})$ and $\text{Hyb}_i(\mathcal{A})$ are computationally indistinguishable. To do so, we introduce two intermediate hybrids:

- $\text{Hyb}_{i,1}$: Same as Hyb_i , except the challenger samples $\mathbf{A}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times t}$ and $\text{crs}_{\text{samp}} \leftarrow \text{GenProg}(1^\lambda, 1^\ell, i, \mathbf{A}_i)$.
- $\text{Hyb}_{i,2}$: Same as $\text{Hyb}_{i,1}$, except the challenger samples $\mathbf{v}_j \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$.

We now show that each adjacent pair of hybrid distributions are indistinguishable.

Lemma 5.7. *If Π_{samp} is somewhere programmable, then $\text{Hyb}_i(\mathcal{A}) \approx \text{Hyb}_{i,1}(\mathcal{A})$.*

Proof. The only difference between Hyb_i and $\text{Hyb}_{i,1}$ is the distribution of crs_{samp} . In Hyb_i , the challenger samples $\text{crs}_{\text{samp}} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$ while in $\text{Hyb}_{i,1}$, the challenger samples $\mathbf{A}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times t}$ and $\text{crs}_{\text{samp}} \leftarrow \text{GenProg}(1^\lambda, 1^\ell, i, \mathbf{A}_i)$. These two distributions are statistically indistinguishable by somewhere programmability. $\square$

Lemma 5.8. *Suppose Π_{samp} is somewhere programmable. Then, under the $\text{LWE}_{n,t,q,s_{\text{LWE}}}$ assumption, for all $i \in [\ell]$, $\text{Hyb}_{i,1}(\mathcal{A}) \approx \text{Hyb}_{i,2}(\mathcal{A})$.*

Proof. Suppose $|\Pr[\text{Hyb}_{i,1}(\mathcal{A}) = 1] - \Pr[\text{Hyb}_{i,2}(\mathcal{A}) = 1]| = \varepsilon(\lambda)$ for some non-negligible ε . We use $\mathcal{A}$ to construct an efficient adversary $\mathcal{B}$ for the $\text{LWE}_{n,t,q,s_{\text{LWE}}}$ assumption:

- On input the LWE challenge $(\mathbf{A}, \mathbf{u}^\top)$ where $\mathbf{A} \in \mathbb{Z}_q^{n \times t}$ and $\mathbf{u} \in \mathbb{Z}_q^t$, algorithm $\mathcal{B}$ computes $\text{crs}_{\text{samp}} \leftarrow \text{GenProg}(1^\lambda, 1^\ell, i, \mathbf{A})$. Then, it computes $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$. For each $j \in [\ell]$, the challenger computes the vector $\mathbf{v}_j$ as follows:
 - If $j < i$, sample $\mathbf{v}_j \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$.
 - If $j = i$, set $\mathbf{v}_j = \mathbf{u}$.
 - If $j > i$, sample $\mathbf{s}_j \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$, $\mathbf{e}_j \leftarrow D_{\mathbb{Z}, s_{\text{LWE}}}^t$, and compute $\mathbf{v}_j^\top = \mathbf{s}_j^\top \mathbf{A}_j + \mathbf{e}_j^\top$.
- Algorithm $\mathcal{B}$ gives $\text{crs} = (1^\lambda, \text{crs}_{\text{samp}}, \mathbf{v}_1, \dots, \mathbf{v}_\ell)$ to $\mathcal{A}$ and outputs whatever $\mathcal{A}$ outputs.

We now analyze the advantage of algorithm $\mathcal{B}$. First, the LWE challenger samples $\mathbf{A} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times t}$, so the distribution of crs_{samp} is identical to the distribution in $\text{Hyb}_{i,1}$ and $\text{Hyb}_{i,2}$. By somewhere programmability, we also have that $\mathbf{A}_i = \mathbf{A}$. Consider now the distribution of $\mathbf{u}$:

- Suppose $\mathbf{u}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ where $\mathbf{s} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and $\mathbf{e} \leftarrow D_{\mathbb{Z}, s_{\text{LWE}}}^t$. Since $\mathbf{A}_i = \mathbf{A}$, we have $\mathbf{v}_i^\top = \mathbf{u}^\top = \mathbf{s}^\top \mathbf{A}_i + \mathbf{e}^\top$, which matches the distribution in $\text{Hyb}_{i,1}$.
- Suppose $\mathbf{u} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$. Then, $\mathbf{v}_i$ is uniform over $\mathbb{Z}_q^t$, which matches the distribution in $\text{Hyb}_{i,2}$.

Algorithm $\mathcal{B}$ breaks $\text{LWE}_{n,t,q,s_{\text{LWE}}}$ with the same non-negligible advantage ε and the lemma follows. $\square$

Lemma 5.9. *If Π_{samp} is somewhere programmable, then $\text{Hyb}_{i,2}(\mathcal{A}) \approx \text{Hyb}_{i+1}(\mathcal{A})$.*

Proof. The only difference between $\text{Hyb}_{i,2}$ and Hyb_{i+1} is the distribution of crs_{samp} . In $\text{Hyb}_{i,2}$, the challenger samples $\mathbf{A}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^{n \times t}$ and $\text{crs}_{\text{samp}} \leftarrow \text{GenProg}(1^\lambda, 1^\ell, i, \mathbf{A}_i)$ whereas in Hyb_i , the challenger samples $\text{crs} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$. These two distributions are statistically indistinguishable by somewhere programmability. $\square$

Mode indistinguishability now follows from [Lemmas 5.7 to 5.9](#). $\square$

Theorem 5.10 (Statistical Binding in Binding Mode). *Suppose $q > 4t\sqrt{\lambda}s_{\text{LWE}}B_{\text{max}} + 4B_{\text{round}}$. Then, [Construction 5.4](#) is statistically binding in binding mode.*

Proof. Take a security parameter $\lambda \in \mathbb{N}$ and any polynomial $\ell = \ell(\lambda)$. Let $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell, \text{binding})$. We parse $\text{crs} = (1^\lambda, \text{crs}_{\text{samp}}, \mathbf{v}_1, \dots, \mathbf{v}_\ell)$ and let $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$. In binding mode, for all $i \in [\ell]$, $\mathbf{v}_i^\top = \mathbf{s}_i^\top \mathbf{A}_i + \mathbf{e}_i^\top$, where $\mathbf{e}_i \leftarrow D_{\mathbb{Z}, s_{\text{LWE}}}^\ell$. By [Lemma 3.2](#), $\|\mathbf{e}_i\| \leq \sqrt{\lambda}s_{\text{LWE}}$ with probability $1 - t \cdot 2^{-\lambda}$. By a union bound, with probability $1 - t\ell \cdot 2^{-\lambda} = 1 - \text{negl}(\lambda)$, the following holds:

$$\forall i \in [\ell] : \|\mathbf{e}_i\| \leq \sqrt{\lambda}s_{\text{LWE}}. \quad (5.1)$$

Suppose now that there exists a tuple $(\mathbf{c}, i, \boldsymbol{\pi}_0, \boldsymbol{\pi}_1)$ where

$$\text{Verify}(\text{crs}, \mathbf{c}, i, 0, \boldsymbol{\pi}_0) = 1 = \text{Verify}(\text{crs}, \mathbf{c}, i, 1, \boldsymbol{\pi}_1). \quad (5.2)$$

We now consider two possibilities:

- Suppose that $\mathbf{c} = \perp$. Then, by construction, $\text{Verify}(\text{crs}, \mathbf{c}, i, 1, \boldsymbol{\pi}_1)$ outputs 0 which contradicts [Eq. \(5.2\)](#).
- Suppose $\mathbf{c} \in \mathbb{Z}_q^n$. By [Eq. \(5.2\)](#) and construction of Verify , the following conditions also hold:

$$\|\boldsymbol{\pi}_0\|, \|\boldsymbol{\pi}_1\| \leq B_{\text{max}} \quad \text{and} \quad \mathbf{A}_i \boldsymbol{\pi}_0 = \mathbf{c} = \mathbf{A}_i \boldsymbol{\pi}_1,$$

and in addition,

$$\begin{aligned} \mathbf{v}_i^\top \boldsymbol{\pi}_0 &\in [-B_{\text{round}}, B_{\text{round}}] \\ \mathbf{v}_i^\top \boldsymbol{\pi}_1 &\in [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}]. \end{aligned}$$

In particular, this means that

$$|\mathbf{v}_i^\top (\boldsymbol{\pi}_0 - \boldsymbol{\pi}_1)| \geq \lfloor q/2 \rfloor - 2B_{\text{round}}. \quad (5.3)$$

Since $\mathbf{A}_i \boldsymbol{\pi}_0 = \mathbf{A}_i \boldsymbol{\pi}_1$, we have

$$\mathbf{v}_i^\top (\boldsymbol{\pi}_0 - \boldsymbol{\pi}_1) = \mathbf{s}_i^\top (\mathbf{A}_i \boldsymbol{\pi}_0 - \mathbf{A}_i \boldsymbol{\pi}_1) + \mathbf{e}_i^\top (\boldsymbol{\pi}_0 - \boldsymbol{\pi}_1) = \mathbf{e}_i^\top (\boldsymbol{\pi}_0 - \boldsymbol{\pi}_1).$$

Since $\|\boldsymbol{\pi}_0\|, \|\boldsymbol{\pi}_1\| \leq B_{\text{max}}$, and $\|\mathbf{e}_i\| \leq \sqrt{\lambda}s_{\text{LWE}}$ from [Eq. \(5.1\)](#), we conclude that

$$|\mathbf{v}_i^\top (\boldsymbol{\pi}_0 - \boldsymbol{\pi}_1)| = |\mathbf{e}_i^\top (\boldsymbol{\pi}_0 - \boldsymbol{\pi}_1)| \leq 2t\sqrt{\lambda}s_{\text{LWE}}B_{\text{max}}.$$

However, this contradicts [Eq. \(5.3\)](#) whenever $q > 4t\sqrt{\lambda}s_{\text{LWE}}B_{\text{max}} + 4B_{\text{round}}$.

We conclude that no such tuple $(\mathbf{c}, i, \boldsymbol{\pi}_0, \boldsymbol{\pi}_1)$ can exist when [Eq. \(5.1\)](#) holds. Since [Eq. \(5.1\)](#) holds with $1 - \text{negl}(\lambda)$ over the randomness of Setup , the theorem follows. $\square$

Theorem 5.11 (Single-Bit Statistical Hiding). *Suppose Π_{samp} satisfies the preimage distribution property and $n \geq 4\lambda + 2\log q$, $t \geq 3n \lceil \log q \rceil$, q is prime, $q \geq 4B_{\text{round}} + 2$, $s_{\text{samp}} \geq \log t$, $B_{\text{max}} \geq \sqrt{t}s_{\text{samp}}$, and $B_{\text{round}} \geq q/4 - q/(8\ell) + 1/2$. Then, [Construction 5.4](#) satisfies single-bit statistical hiding in hiding mode.*

Proof. Let $\mathcal{A}$ be a distinguisher for the hiding game. We start by defining a sequence of hybrid experiments:

- $\text{Hyb}_0^{(b)}$: This is the hiding experiment with the bit b . Specifically, the adversary (on input 1^λ and 1^ℓ) starts by outputting an index $i^* \in [\ell]$. The challenger then samples $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell, \text{hiding})$. Namely, the challenger samples $\text{crs}_{\text{samp}} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$ and for each $i \in [\ell]$, $\mathbf{v}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$. It sets $\text{crs} = (1^\lambda, \text{crs}_{\text{samp}}, \mathbf{v}_1, \dots, \mathbf{v}_\ell)$. Next, for each $d \in [\lambda]$, the challenger proceeds as follows:⁵

- Sample $(\boldsymbol{\pi}_{d,1}, \dots, \boldsymbol{\pi}_{d,\ell}, \mathbf{c}_d) \leftarrow \text{SampleMultPre}(\text{td}, \mathbf{0}^n, \dots, \mathbf{0}^n)$.
- For each $i \in [\ell]$, the challenger computes $u_{d,i} = \mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}$. If $\|\boldsymbol{\pi}_{d,i}\| > B_{\max}$, set $r_{d,i} = \perp$. Otherwise, set

$$r_{d,i} = \begin{cases} 0 & u_{d,i} \in [-B_{\text{round}}, B_{\text{round}}] \\ 1 & u_{d,i} \in [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}] \\ \perp & \text{otherwise.} \end{cases} \quad (5.4)$$

The challenger then constructs the challenge as follows:

- Suppose for all $d \in [\lambda]$, there exists an index $i \in [\ell]$ where $r_{d,i} = \perp$. Then, the challenger sets $\mathbf{c} = \perp$ and for all $i \in [\ell]$, $r_i = 0$ and $\boldsymbol{\pi}_i = \perp$.
- Otherwise, let $d^* \in [\lambda]$ be the first index where $r_{d^*,i} \in \{0, 1\}$ for all $i \in [\ell]$. Then the challenger sets $\mathbf{c} = \mathbf{c}_{d^*}$ and for all $i \in [\ell]$, $r_i = r_{d^*,i}$ and $\boldsymbol{\pi}_i = \boldsymbol{\pi}_{d^*,i}$.

Finally, if $b = 0$, then the challenger sets $\beta = r_{i^*}$. If $b = 1$, the challenger samples $\beta \xleftarrow{\mathbb{R}} \{0, 1\}$. The challenger gives $(\text{crs}, \mathbf{c}, \{(i, r_i, \boldsymbol{\pi}_i)\}_{i \in [\ell]}, \beta)$ to $\mathcal{A}$. At the end of the game, algorithm $\mathcal{A}$ outputs a bit $b' \in \{0, 1\}$, which is the output of the experiment.

- $\text{Hyb}_1^{(b)}$: Same as $\text{Hyb}_0^{(b)}$, except for all $d \in [\lambda]$, the challenger changes how it samples $\mathbf{c}_d$ and $\boldsymbol{\pi}_{d,i}$. Specifically, after sampling crs as in $\text{Hyb}_0^{(b)}$, the challenger proceeds as follows for each $d \in [\lambda]$:

- Sample $\mathbf{c}_d \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and for each $i \in [\ell]$, sample $\boldsymbol{\pi}_{d,i} \leftarrow (\mathbf{A}_i)_{\text{samp}}^{-1}(\mathbf{c}_d)$.
- For each $i \in [\ell]$, the challenger computes $u_{d,i} = \mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}$. If $\|\boldsymbol{\pi}_{d,i}\| > B_{\max}$, set $r_{d,i} = \perp$. Otherwise, it sets $r_{d,i}$ according to Eq. (5.4).

The rest of the experiment proceeds as in $\text{Hyb}_0^{(b)}$.

- $\text{Hyb}_2^{(b)}$: Same as $\text{Hyb}_1^{(b)}$, except the challenger **no longer checks the norm constraint on $\boldsymbol{\pi}_{d,i}$ when computing $r_{d,i}$** . Specifically, after sampling crs as in $\text{Hyb}_0^{(b)}$, the challenger proceeds as follows for each $d \in [\lambda]$:

- Sample $\mathbf{c}_d \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and for each $i \in [\ell]$, sample $\boldsymbol{\pi}_{d,i} \leftarrow (\mathbf{A}_i)_{\text{samp}}^{-1}(\mathbf{c}_d)$.
- For each $i \in [\ell]$, the challenger computes $u_{d,i} = \mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}$ and sets $r_{d,i}$ according to Eq. (5.4). **In particular, the challenger no longer checks if $\|\boldsymbol{\pi}_{d,i}\| \leq B_{\max}$.**

The rest of the experiment proceeds as in $\text{Hyb}_0^{(b)}$.

- $\text{Hyb}_3^{(b)}$: Same as $\text{Hyb}_2^{(b)}$, except for all $d \in [\lambda]$, the challenger samples $u_{d,i^*} \xleftarrow{\mathbb{R}} \mathbb{Z}_q$. Specifically, after sampling crs as in $\text{Hyb}_0^{(b)}$, the challenger proceeds as follows for each $d \in [\lambda]$:

- Sample $\mathbf{c}_d \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and for each $i \in [\ell]$, sample $\boldsymbol{\pi}_{d,i} \leftarrow (\mathbf{A}_i)_{\text{samp}}^{-1}(\mathbf{c}_d)$.
- For each $i \in [\ell] \setminus \{i^*\}$, the challenger computes $u_{d,i} = \mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}$. It then samples $u_{d,i^*} \xleftarrow{\mathbb{R}} \mathbb{Z}_q$. Finally, the challenger sets $r_{d,i}$ according to Eq. (5.4).

The rest of the experiment proceeds as in $\text{Hyb}_0^{(b)}$.

⁵In this description, we explicitly unroll the (up to) λ iterations of rejection sampling that the GenBits algorithm performs. Specifically, the challenger samples λ commitments and openings, and the output is defined to be the first instance that is successful (i.e., the first instance that GenBits would have accepted). As such, the description here is identical to the procedure in GenBits, but is more convenient to analyze.

- $\text{Hyb}_4^{(b)}$: Same as $\text{Hyb}_3^{(b)}$, except for all $d \in [\lambda]$, the challenger changes how it samples r_{d,i^*} . Specifically, after sampling crs as in $\text{Hyb}_0^{(b)}$, the challenger proceeds as follows for each $d \in [\lambda]$:
 - Sample $\mathbf{c}_d \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and for each $i \in [\ell]$, sample $\boldsymbol{\pi}_{d,i} \leftarrow (\mathbf{A}_i)_{s_{\text{samp}}}^{-1}(\mathbf{c}_d)$.
 - For each $i \in [\ell] \setminus \{i^*\}$, the challenger computes $u_{d,i} = \mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}$ and sets $r_{d,i}$ according to Eq. (5.4).
 - With probability $(4B_{\text{round}} + 2)/q$, the challenger samples $r_{d,i^*} \xleftarrow{\mathbb{R}} \{0, 1\}$, and with probability $1 - (4B_{\text{round}} + 2)/q$, the challenger sets $r_{d,i^*} = \perp$.

The rest of the experiment proceeds as in $\text{Hyb}_0^{(b)}$.

- $\text{Hyb}_5^{(b)}$: Same as $\text{Hyb}_4^{(b)}$, except the challenger samples $r_{i^*} \xleftarrow{\mathbb{R}} \{0, 1\}$ in the case for all $d \in [\lambda]$, there exists an index $i \in [\ell]$ where $r_{d,i} = \perp$. Specifically, after sampling crs as in $\text{Hyb}_0^{(b)}$, the challenger proceeds as follows for each $d \in [\lambda]$:
 - Sample $\mathbf{c}_d \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and for each $i \in [\ell]$, sample $\boldsymbol{\pi}_{d,i} \leftarrow (\mathbf{A}_i)_{s_{\text{samp}}}^{-1}(\mathbf{c}_d)$.
 - For each $i \in [\ell] \setminus \{i^*\}$, the challenger computes $u_{d,i} = \mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}$ and sets $r_{d,i}$ according to Eq. (5.4).
 - With probability $(4B_{\text{round}} + 2)/q$, the challenger samples $r_{d,i^*} \xleftarrow{\mathbb{R}} \{0, 1\}$, and with probability $1 - (4B_{\text{round}} + 2)/q$, the challenger sets $r_{d,i^*} = \perp$.

The challenger then constructs the challenge as follows:

- Suppose for all $d \in [\lambda]$, there exists an index $i \in [\ell]$ where $r_{d,i} = \perp$. Then, the challenger sets $\mathbf{c} = \perp$ and for all $i \in [\ell] \setminus \{i^*\}$, it sets $r_i = 0$ and $\boldsymbol{\pi}_i = \perp$. The challenger samples $r_{i^*} \xleftarrow{\mathbb{R}} \{0, 1\}$.
- Otherwise, let $d^* \in [\lambda]$ be the first index where $r_{d^*,i} \in \{0, 1\}$ for all $i \in [\ell]$. Then the challenger sets $\mathbf{c} = \mathbf{c}_{d^*}$ and for all $i \in [\ell]$, $r_i = r_{d^*,i}$ and $\boldsymbol{\pi}_i = \boldsymbol{\pi}_{d^*,i}$.

Finally, if $b = 0$, then the challenger sets $\beta = r_{i^*}$. If $b = 1$, the challenger samples $\beta \xleftarrow{\mathbb{R}} \{0, 1\}$. The challenger gives $(\text{crs}, \mathbf{c}, \{(i, r_i, \boldsymbol{\pi}_i)\}_{i \neq i^*}, \beta)$ to $\mathcal{A}$. At the end of the game, algorithm $\mathcal{A}$ outputs a bit $b' \in \{0, 1\}$, which is the output of the experiment.

We write $\text{Hyb}_i^{(b)}(\mathcal{A})$ to denote the random variable corresponding to the output of an execution of hybrid $\text{Hyb}_i^{(b)}$ with adversary $\mathcal{A}$. We now analyze each adjacent pair of distributions.

Lemma 5.12. *Suppose Π_{samp} satisfies the preimage distribution property. Then, for all $b \in \{0, 1\}$, $\text{Hyb}_0^{(b)}(\mathcal{A}) \stackrel{s}{\approx} \text{Hyb}_1^{(b)}(\mathcal{A})$.*

Proof. For each $j \in \{0, \dots, \lambda\}$, we define an intermediate hybrid as follows:

- $\text{Hyb}_{0,j}^{(b)}$: Same as $\text{Hyb}_0^{(b)}$, except for all $d \leq j$, the challenger samples $(\boldsymbol{\pi}_{d,1}, \dots, \boldsymbol{\pi}_{d,\ell}, \mathbf{c}_d)$ according to the procedure in $\text{Hyb}_1^{(b)}$. For all $d > j$, the challenger samples $(\boldsymbol{\pi}_{d,1}, \dots, \boldsymbol{\pi}_{d,\ell}, \mathbf{c}_d)$ according to the procedure in $\text{Hyb}_0^{(b)}$.

By construction, $\text{Hyb}_{0,0}^{(b)}(\mathcal{A}) \equiv \text{Hyb}_0^{(b)}(\mathcal{A})$ and $\text{Hyb}_{0,\lambda}^{(b)}(\mathcal{A}) \equiv \text{Hyb}_1^{(b)}(\mathcal{A})$. We now argue that for every $j \in [\lambda]$, the statistical distance between $\text{Hyb}_{0,j-1}^{(b)}(\mathcal{A})$ and $\text{Hyb}_{0,j}^{(b)}(\mathcal{A})$ is $\text{negl}(\lambda)$. The only difference between these two distributions is the distribution of $(\boldsymbol{\pi}_{j,1}, \dots, \boldsymbol{\pi}_{j,\ell}, \mathbf{c}_j)$. With overwhelming probability over the choice of crs_{samp} , these two distributions are statistically indistinguishable by the preimage distribution property of Π_{samp} . $\square$

Lemma 5.13. *Suppose $n \geq \lambda$, $t \geq 2n \log q$, $s_{\text{samp}} \geq \log t$, and $B_{\text{max}} \geq \sqrt{t} s_{\text{samp}}$. Then, for all $b \in \{0, 1\}$, $\text{Hyb}_1^{(b)}(\mathcal{A}) \stackrel{s}{\approx} \text{Hyb}_2^{(b)}(\mathcal{A})$.*

Proof. These experiments are identical unless in an execution of $\text{Hyb}_1^{(b)}$, there exists an index $d \in [\lambda]$ and $i \in [\ell]$ where $\|\pi_{d,i}\| > B_{\max}$. In $\text{Hyb}_1^{(b)}$, the challenger samples $\pi_{d,i} \leftarrow (\mathbf{A}_i)_{s_{\text{samp}}}^{-1}(\mathbf{c}_d)$. By Lemma 4.8, the marginal distribution of $\mathbf{A}_i$ in $\text{Hyb}_1^{(b)}$ is statistically close to uniform. Since $t \geq 2n \log q$, $s_{\text{samp}} \geq \log t$, and $B_{\max} \geq \sqrt{t} s_{\text{samp}}$, by Lemma 3.2, with overwhelming probability over the choice of $\pi_{d,i}$, it holds that $\|\pi_{d,i}\| \leq B_{\max}$. By a union bound over all $d \in [\lambda]$ and $i \in [\ell]$, we conclude that with overwhelming probability, in an execution of $\text{Hyb}_1^{(b)}$, it holds that $\|\pi_{d,i}\| \leq B_{\max}$ for all $d \in [\lambda]$ and $i \in [\ell]$. In this case, the output of $\text{Hyb}_1^{(b)}$ and $\text{Hyb}_2^{(b)}$ is the same. $\square$

Lemma 5.14. *Suppose $n \geq 4\lambda + 2 \log q$, $t \geq 2n \log q$, q is prime, and $s_{\text{samp}} \geq \log t$. Then, for all $b \in \{0, 1\}$, $\text{Hyb}_2^{(b)}(\mathcal{A}) \approx \text{Hyb}_3^{(b)}(\mathcal{A})$.*

Proof. For each $j \in \{0, \dots, \lambda\}$, we define an intermediate hybrid as follows:

- $\text{Hyb}_{2,j}^{(b)}$: Same as $\text{Hyb}_2^{(b)}$, except for all $d \leq j$, the challenger samples $u_{d,i^*} \xleftarrow{\mathbb{R}} \mathbb{Z}_q$. For all $d > j$, the challenger sets $u_{d,i^*} = \mathbf{v}_{i^*}^\top \pi_{d,i^*}$ as in $\text{Hyb}_1^{(b)}$.

By construction, $\text{Hyb}_{2,0}^{(b)}(\mathcal{A}) \equiv \text{Hyb}_2^{(b)}(\mathcal{A})$ and $\text{Hyb}_{2,\lambda}^{(b)}(\mathcal{A}) \equiv \text{Hyb}_3^{(b)}(\mathcal{A})$. We argue that for all $j \in [\lambda]$, the statistical distance between $\text{Hyb}_{2,j-1}^{(b)}$ and $\text{Hyb}_{2,j}^{(b)}$ is $\text{negl}(\lambda)$. The only difference between these two distributions is the distribution of u_{j,i^*} . We show that these two distributions are statistically indistinguishable.

- By Lemma 4.8, the marginal distribution of $\mathbf{A}_{i^*}$ in $\text{Hyb}_{2,j-1}^{(b)}$ and $\text{Hyb}_{2,j}^{(b)}$ is statistically close to uniform over $\mathbb{Z}_q^{n \times t}$.
- In $\text{Hyb}_{2,j-1}^{(b)}$ and $\text{Hyb}_{2,j}^{(b)}$, the challenger samples $\pi_{j,i^*} \leftarrow (\mathbf{A}_{i^*})_{s_{\text{samp}}}^{-1}(\mathbf{c}_d)$. Since $t \geq 2n \log q$ and $s_{\text{samp}} \geq \log t$, we appeal to Lemma 3.8 to conclude that with overwhelming probability over the choice of $\mathbf{A}_{i^*}$,

$$\mathbf{H}_\infty(\pi_{j,i^*}) \geq n/2 \geq 2\lambda + \log q.$$

- In $\text{Hyb}_{2,j-1}^{(b)}$ and $\text{Hyb}_{2,j}^{(b)}$, the challenger samples $\mathbf{v}_{i^*} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$. By the leftover hash lemma (Lemma 3.7), the statistical distance between the distributions

$$\left\{ (\mathbf{v}_{i^*}, \mathbf{v}_{i^*}^\top \pi_{j,i^*}) : \mathbf{v}_{i^*} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t \right\} \quad \text{and} \quad \left\{ (\mathbf{v}_{i^*}, u_{j,i^*}) : \mathbf{v}_{i^*} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t, u_{j,i^*} \xleftarrow{\mathbb{R}} \mathbb{Z}_q \right\}$$

is at most $2^{-\lambda} = \text{negl}(\lambda)$. The distribution in the left-hand side corresponds to $\text{Hyb}_{2,j-1}^{(b)}$ while the one on the right-hand side corresponds to $\text{Hyb}_{2,j}^{(b)}$.

We conclude that for all $j \in [\lambda]$, the distributions $\text{Hyb}_{2,j-1}^{(b)}(\mathcal{A})$ and $\text{Hyb}_{2,j}^{(b)}(\mathcal{A})$ are statistically indistinguishable. The lemma now follows by a hybrid argument. $\square$

Lemma 5.15. *If $q > 4B_{\text{round}} + 2$, then for all $b \in \{0, 1\}$, $\text{Hyb}_3^{(b)}(\mathcal{A}) \equiv \text{Hyb}_4^{(b)}(\mathcal{A})$.*

Proof. The distributions $\text{Hyb}_3^{(b)}(\mathcal{A})$ and $\text{Hyb}_4^{(b)}(\mathcal{A})$ are identically distributed as long as $q > 4B_{\text{round}} + 2$. In this case, the intervals $[-B_{\text{round}}, B_{\text{round}}]$ and $[\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}]$ are disjoint and each has size $2B_{\text{round}} + 1$. The two experiments only differ in how they compute r_{d,i^*} for $d \in [\lambda]$. We show that these two procedures are distributed identically for each $d \in [\lambda]$.

- In $\text{Hyb}_3^{(b)}$, the challenger samples $u_{d,i^*} \xleftarrow{\mathbb{R}} \mathbb{Z}_q$ and then sets $r_{d,i^*} = 0$ if $u_{d,i^*} \in [-B_{\text{round}}, B_{\text{round}}]$. Over the randomness of u_{d,i^*} , this happens with probability $(2B_{\text{round}} + 1)/q$. Alternatively, it sets $r_{d,i^*} = 1$ if $u_{d,i^*} \in [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}]$. This also happens with probability $(2B_{\text{round}} + 1)/q$ over the randomness of u_{d,i^*} . Finally, if neither event holds, which occurs with probability $1 - (4B_{\text{round}} + 2)/q$, then the challenger sets $r_{d,i^*} = \perp$.
- In $\text{Hyb}_4^{(b)}$, $r_{d,i^*} = 0$ with probability $(1/2) \cdot (4B_{\text{round}} + 2)/q = (2B_{\text{round}} + 1)/q$, which matches the probability in $\text{Hyb}_3^{(b)}$. The challenger sets $r_{d,i^*} = 1$ with the same probability. Finally, the challenger sets $r_{d,i^*} = \perp$ with probability $1 - (4B_{\text{round}} + 2)/q$, which is identical to the behavior in $\text{Hyb}_3^{(b)}$.

We conclude that the distribution of r_{d,i^*} is identical in the two experiments for all $d \in [\lambda]$. Correspondingly, the outputs of these two experiments are identically distributed. $\square$

Lemma 5.16. *Suppose $n \geq 4\lambda + 2 \log q$, $t \geq 2n \log q$, q is prime, $s_{\text{samp}} > \log t$, and $B_{\text{round}} \geq q/4 - q/(8\ell) - 1/2$. Then, for all $b \in \{0, 1\}$, $\text{Hyb}_4^{(b)}(\mathcal{A}) \approx \text{Hyb}_5^{(b)}(\mathcal{A})$.*

Proof. $\text{Hyb}_4^{(b)}$ and $\text{Hyb}_5^{(b)}$ are identical experiments unless for all $d \in [\lambda]$, there exists an index $i \in [\ell]$ where $r_{d,i} = \perp$. We show that this event happens with negligible probability. To analyze the probability of this event, we first define the following sequence of distributions and argue that each adjacent pair is statistically indistinguishable:

- $\mathcal{D}_0$: This is the distribution of $r_{d,i}$ in $\text{Hyb}_4^{(b)}$ and $\text{Hyb}_5^{(b)}$. Namely, the distribution samples $\text{crs}_{\text{samp}} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$, $(A_1, \dots, A_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$, and $\mathbf{v}_1, \dots, \mathbf{v}_\ell \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$. Then for each $d \in [\lambda]$, it samples $\mathbf{c}_d \leftarrow \mathbb{Z}_q^n$. For each $d \in [\lambda]$ and $i \in [\ell] \setminus \{i^*\}$, it samples $\boldsymbol{\pi}_{d,i} \leftarrow (A_i)_{s_{\text{samp}}}^{-1}(\mathbf{c}_d)$, sets $u_{d,i} = \mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}$, and sets $r_{d,i}$ according to Eq. (5.4). Finally, with probability $(4B_{\text{round}} + 2)/q$, sample $r_{d,i^*} \xleftarrow{\mathbb{R}} \{0, 1\}$ and with probability $1 - (4B_{\text{round}} + 2)/q$, set $r_{d,i^*} = \perp$. The output is $(r_{1,1}, \dots, r_{1,\ell}, \dots, r_{\lambda,1}, \dots, r_{\lambda,\ell})$.
- $\mathcal{D}_1$: In this distribution, for all $d \in [\lambda]$ and $i \in [\ell] \setminus \{i^*\}$, sample $u_{d,i} \xleftarrow{\mathbb{R}} \mathbb{Z}_q$ and set $r_{d,i}$ according to Eq. (5.4). Then, with probability $(4B_{\text{round}} + 2)/q$, sample $r_{d,i^*} \xleftarrow{\mathbb{R}} \{0, 1\}$ and with probability $1 - (4B_{\text{round}} + 2)/q$, set $r_{d,i^*} = \perp$. The output is $(r_{1,1}, \dots, r_{1,\ell}, \dots, r_{\lambda,1}, \dots, r_{\lambda,\ell})$.
- $\mathcal{D}_2$: In this distribution, for all $d \in [\lambda]$ and $i \in [\ell]$, sample $r_{d,i} \xleftarrow{\mathbb{R}} \{0, 1\}$ and with probability $1 - (4B_{\text{round}} + 2)/q$, set $r_{d,i} = \perp$. The output is $(r_{1,1}, \dots, r_{1,\ell}, \dots, r_{\lambda,1}, \dots, r_{\lambda,\ell})$.

We start by showing that distributions $\mathcal{D}_0$ and $\mathcal{D}_1$ are statistically indistinguishable. Formally, we analyze the distribution $\mathcal{D}_0$:

- Since $\text{crs}_{\text{samp}} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$ and $(A_1, \dots, A_\ell, \text{td}) = \text{Expand}(\text{crs}_{\text{samp}})$, we appeal to Lemma 4.8 to conclude that the marginal distribution of each A_i is statistically close to uniform over $\mathbb{Z}_q^{n \times t}$. We declare $A_i \in \mathbb{Z}_q^{n \times t}$ to be “good” if the property in Lemma 3.8 holds: namely, A_i is good if for all $\mathbf{y} \in \mathbb{Z}_q^n$, the distribution $(A_i)_{s_{\text{samp}}}^{-1}(\mathbf{y})$ has min-entropy at least $n/2$. Since $t \geq 2n \log q$ and $s_{\text{samp}} \geq \log t$, all but a $\text{negl}(n)$ fraction of A_i ’s are “good”. Since the marginal distribution of each A_i is statistically close to uniform over $\mathbb{Z}_q^{n \times t}$, it follows that each A_i is good with probability $1 - \text{negl}(n)$. By a union bound (and since $\ell = \text{poly}(\lambda)$), we conclude that with overwhelming probability, all of the A_i are good.
- If A_i is good for all $i \in [\ell]$, this means that $H_\infty(\boldsymbol{\pi}_{d,i}) \geq n/2 \geq 2\lambda + \log q$ for all $d \in [\lambda]$ and $i \in [\ell]$, where $\boldsymbol{\pi}_{d,i} \leftarrow (A_i)_{s_{\text{samp}}}^{-1}(\mathbf{c}_d)$.
- By Lemma 3.7, the following pair of distributions are statistically indistinguishable for all $i \in [\ell]$:
 - Sample $\mathbf{v}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$. For each $d \in [\lambda]$, sample $\mathbf{c}_d \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and $\boldsymbol{\pi}_{d,i} \leftarrow (A_i)_{s_{\text{samp}}}^{-1}(\mathbf{c}_d)$. Output $(\mathbf{v}_i, \mathbf{v}_i^\top \boldsymbol{\pi}_{1,i}, \dots, \mathbf{v}_i^\top \boldsymbol{\pi}_{\lambda,i})$.
 - Sample $\mathbf{v}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$. For each $d \in [\lambda]$, sample $u_{d,i} \xleftarrow{\mathbb{R}} \mathbb{Z}_q$. Output $(\mathbf{v}_i, u_{1,i}, \dots, u_{\lambda,i})$.

Since this holds for all $i \in [\ell]$ and $\ell = \text{poly}(\lambda)$, we conclude that the joint distribution of $(u_{d,i})_{(d \in [\lambda], i \in [\ell])}$ in $\mathcal{D}_0$ and $\mathcal{D}_1$ is statistically indistinguishable. Since both distributions derive $r_{d,i}$ from $u_{d,i}$ using the same procedure, we conclude that $\mathcal{D}_0$ and $\mathcal{D}_1$ are statistically indistinguishable.

Next, $\mathcal{D}_1$ and $\mathcal{D}_2$ are identical distributions. Namely, if $u_{d,i} \xleftarrow{\mathbb{R}} \mathbb{Z}_q$, then $r_{d,i} = 0$ with probability $(2B_{\text{round}} + 1)/q$, $r_{d,i} = 1$ with probability $(2B_{\text{round}} + 1)/q$, and $r_{d,i} = \perp$ with probability $1 - (4B_{\text{round}} + 2)/q$. By a hybrid argument, we conclude that $\mathcal{D}_0$ and $\mathcal{D}_2$ are statistically indistinguishable.

Consider now the probability that in an execution of $\text{Hyb}_4^{(b)}$ and $\text{Hyb}_5^{(b)}$, it happens that for all $d \in [\lambda]$, there exists an index $i \in [\ell]$ where $r_{d,i} = \perp$. For a tuple $(r_{1,1}, \dots, r_{1,\ell}, \dots, r_{\lambda,1}, \dots, r_{\lambda,\ell})$ and an index $d \in [\lambda]$, we define the event Bad_d to be the event that there exists $i \in [\ell]$ where $r_{d,i} = \perp$. By a union bound, we have for all $d \in [\lambda]$,

$$\Pr[\text{Bad}_d : (r_{d,i})_{d \in [\lambda], i \in [\ell]} \leftarrow \mathcal{D}_2] \leq \sum_{i \in [\ell]} \frac{q - (4B_{\text{round}} + 2)}{q} \leq \sum_{i \in [\ell]} \frac{1}{2\ell} = \frac{1}{2},$$

when $B_{\text{round}} \geq q/4 - q/(8\ell) - 1/2$. By definition of $\mathcal{D}_2$, we moreover have that

$$\Pr \left[\bigwedge_{d \in [\lambda]} \text{Bad}_d : (u_{d,i})_{d \in [\lambda], i \in [\ell]} \leftarrow \mathcal{D}_1 \right] = \prod_{d \in [\lambda]} \Pr[\text{Bad}_d : (u_{d,i})_{d \in [\lambda], i \in [\ell]} \leftarrow \mathcal{D}_2] = \frac{1}{2^\lambda}.$$

Finally, since $\mathcal{D}_0$ and $\mathcal{D}_1$ are statistically indistinguishable, there exists a negligible function $\text{negl}(\cdot)$ where

$$\Pr \left[\bigwedge_{d \in [\lambda]} \text{Bad}_d : (u_{d,i})_{d \in [\lambda], i \in [\ell]} \leftarrow \mathcal{D}_0 \right] \leq \Pr \left[\bigwedge_{d \in [\lambda]} \text{Bad}_d : (u_{d,i})_{d \in [\lambda], i \in [\ell]} \leftarrow \mathcal{D}_2 \right] + \text{negl}(\lambda) = \frac{1}{2^\lambda} + \text{negl}(\lambda).$$

Thus, in an execution of $\text{Hyb}_4^{(b)}$ and $\text{Hyb}_5^{(b)}$, the probability that for all $d \in [\lambda]$, there exists an index $i \in [\ell]$ where $r_{d,i} = \perp$ is at most $2^{-\lambda} + \text{negl}(\lambda)$. Thus, with overwhelming probability, the adversary's view in $\text{Hyb}_4^{(b)}$ and $\text{Hyb}_5^{(b)}$ is identical. The lemma follows. $\square$

Lemma 5.17. *It holds that $\text{Hyb}_5^{(0)}(\mathcal{A}) \equiv \text{Hyb}_5^{(1)}(\mathcal{A})$.*

Proof. In $\text{Hyb}_5^{(0)}$, the challenger sets $\beta = r_{i^*}$ whereas in $\text{Hyb}_5^{(1)}$, the challenger samples $\beta \xleftarrow{\mathcal{R}} \{0, 1\}$. We argue that the distribution of r_{i^*} in $\text{Hyb}_5^{(0)}$ is uniformly random (and independent of all other quantities in the adversary's view). Consider an execution of $\text{Hyb}_5^{(0)}$. We consider two cases:

- Suppose for all $d \in [\lambda]$, there exists an index $i \in [\ell]$ where $r_{d,i} = \perp$. In this case, the challenger samples $r_{i^*} \xleftarrow{\mathcal{R}} \{0, 1\}$.
- Otherwise, let $d^* \in [\lambda]$ be the smallest index where $r_{d^*,i} \in \{0, 1\}$ for all $i \in [\ell]$. Then, the challenger sets $r_{i^*} = r_{d^*,i^*}$. In $\text{Hyb}_5^{(0)}$, the challenger either sets $r_{d^*,i^*} = \perp$ or samples $r_{d^*,i^*} \xleftarrow{\mathcal{R}} \{0, 1\}$. Since $r_{d^*,i^*} \neq \perp$, this means the challenger must have sampled $r_{d^*,i^*} \xleftarrow{\mathcal{R}} \{0, 1\}$.

Finally, none of the other components in the adversary's view depend on the value of r_{d^*,i^*} . As such, we conclude that the distribution of r_{i^*} in $\text{Hyb}_5^{(0)}$ is uniform and independent of all other quantities in the adversary's view. Thus, the distribution of β in the two experiments is identical. $\square$

Combining [Lemmas 5.12 to 5.17](#), we conclude that $\text{Hyb}_0^{(0)} \approx \text{Hyb}_0^{(1)}$, which completes the proof. $\square$

Theorem 5.18 (Succinctness). *If $n \log q = \text{poly}(\lambda, \log \ell)$, then [Construction 5.4](#) is succinct.*

Proof. The size of the commitment $\mathbf{c}$ output by GenBits in [Construction 5.4](#) is either an element of $\mathbb{Z}_q^n$ or $\perp$. Thus, we can describe $\mathbf{c}$ by a string of length $n \log q + 1$. If $n \log q = \text{poly}(\lambda, \log \ell)$, then succinctness holds. $\square$

Parameter instantiation. Let λ be a security parameter and let ℓ be the length of the hidden-bits string. We now provide one possible instantiation of the parameters in [Construction 5.4](#) to satisfy [Theorems 5.5, 5.6, 5.10 and 5.11](#). In the following, we assume that $\ell \leq 2^\lambda$, so $\log \ell \leq \lambda$.

- When setting parameters, we work under the assumption that $q \leq 2^{O(\lambda)}$. Our final parameter instantiations will satisfy this property. In this case, $\log q = O(\lambda)$.
- We require that $n \geq 4\lambda + 2 \log q$, so we can take $n = 4\lambda + O(\lambda) = O(\lambda)$.
- We set $t = 3n \lceil \log q \rceil \cdot (\lceil \log \ell \rceil + 1) = O(\lambda^3)$.
- We set $s_{\text{LWE}} = \lambda^\delta$ for some constant $\delta > 0$, which we will set later.
- We set $s_{\text{samp}} = (\ell t + 3n \lceil \log q \rceil) \log(\ell n) = O(\lambda^4 \ell)$. We take $B_{\text{max}} > \sqrt{t s_{\text{samp}}} = O(\lambda^{11/2} \ell)$.

- We choose $q = \text{poly}(\lambda, \ell)$ and δ such that $q > 8\ell t \sqrt{\lambda} s_{\text{LWE}} B_{\text{max}} + 4\ell = O(\lambda^{9+\delta} \ell^2)$ and the $\text{LWE}_{n,t,q,s_{\text{LWE}}}$ assumption holds. In particular, since $q = \text{poly}(\lambda, \ell)$ and $\ell < 2^\lambda$, this means $q \leq 2^{O(\lambda)}$, which satisfies our initial assumption.
- We set $B_{\text{round}} = q/4 - q/(8\ell) - 1/2$.
- Finally, we instantiate (Gen, Expand, SampleMultPre) with the $(n, t, q, s_{\text{samp}})$ -shifted multi-preimage trapdoor sampler from [Theorem 4.7](#) ([Construction 4.6](#)).

We briefly verify that these parameters satisfy the necessary requirements:

- [Theorem 5.10](#) requires that $q > 4t \sqrt{\lambda} s_{\text{LWE}} B_{\text{max}} + 4B_{\text{round}}$. Since $B_{\text{round}} = q/4 - q/(8\ell) - 1/2$, this is equivalent to requiring that $q/(2\ell) > 4t \sqrt{\lambda} s_{\text{LWE}} B_{\text{max}} - 2$, or equivalently, that $q > 8\ell t \sqrt{\lambda} s_{\text{LWE}} B_{\text{max}} - 4\ell$.
- All of the conditions of [Theorem 5.11](#) are satisfied by construction. In particular, the requirement $q > 4B_{\text{round}} + 2$ is always satisfied when $B_{\text{round}} = q/4 - q/(8\ell) - 1/2$.

With this setting of parameters, we obtain a dual-mode hidden-bits generator with the following properties:

- **CRS size:** By [Theorem 4.7](#), the size of the CRS is $\lambda + nt \log q + \ell t \log q = \ell \cdot \text{poly}(\lambda, \log \ell)$. Moreover, in hiding mode, the CRS sampling algorithm is *transparent*.
- **Commitment and opening size:** The size of a commitment $\mathbf{c} \in \mathbb{Z}_q^n$ is $n \log q = \text{poly}(\lambda, \log \ell)$ bits. The size of an opening $\boldsymbol{\pi} \in \mathbb{Z}_q^t$ is $t \log q = \text{poly}(\lambda, \log \ell)$ bits.

Finally, for all $\ell = \text{poly}(\lambda)$, security relies on the $\text{LWE}_{n,t,q,s_{\text{LWE}}}$ assumption with a *polynomial* modulus-to-noise ratio (in this case, $q = \text{poly}(\lambda, \ell) = \text{poly}(\lambda)$ and $s_{\text{LWE}} = \lambda^\delta$ for constant $\delta > 0$). We summarize our instantiation in the following corollaries:

Corollary 5.19 (Dual-Mode Hidden-Bits Generator from LWE). *Let λ be a security parameter. Then, for all polynomials $\ell = \ell(\lambda)$, under the LWE assumption with a polynomial modulus-to-noise ratio, there exists a dual-mode hidden-bits generator with a CRS of size $\ell \cdot \text{poly}(\lambda, \log \ell)$. Moreover, in (statistically) hiding mode, the common reference string can be sampled using a transparent setup algorithm.*

Dual-mode NIZKs for NP. [Corollary 5.19](#) suffices to obtain a dual-mode NIZK for NP from the plain LWE assumption with a polynomial modulus-to-noise ratio. Specifically, by [Theorem 5.2](#), it gives a computational single-theorem NIZK proof in the structured reference string model and a statistical single-theorem NIZK argument in the uniform random string model. Using the or-proof transformations from [\[FLS90, BCD⁺25\]](#), this can be upgraded to satisfy multi-theorem zero-knowledge in the structured reference string model. As discussed in [Remark 5.3](#), we can also directly achieve multi-theorem zero-knowledge by showing that the hidden-bits generator in [Construction 5.4](#) satisfies a stronger simulation-based notion of security (see [Appendix A](#)). This yields an (adaptive) multi-theorem zero-knowledge argument without any additional transformation (or using the NIZK to prove membership in a cryptographic language).

Corollary 5.20 (Dual-Mode NIZK for NP). *Under the plain LWE assumption with a polynomial modulus-to-noise ratio, there exists a dual-mode multi-theorem NIZK for NP. Specifically, there exists a computational multi-theorem NIZK proof in the structured reference string model and a statistical multi-theorem NIZK argument in the uniform random string model.*

Remark 5.21 (Handling Values Inside the Rounding Boundary). As noted at the beginning of [Section 5](#), the verification algorithm of our dual-mode NIZK checks that the opening is not close to a rounding boundary. This property is used in the binding analysis to argue that each commitment can only be opened to (at most) one possible hidden-bits string. When working with a polynomial modulus, there is a small, but noticeable, probability that a randomly-sampled set of commitment and openings contains at least one opening that lands inside a rounding boundary. In [Construction 5.4](#), we handle this by having GenBits resample when this happens. By choosing the modulus q accordingly, we can show that after λ retries, the algorithm is successful with overwhelming probability. The previous work of [\[Wat24\]](#) used a super-polynomial modulus, in which case the resampling is unnecessary because the probability of landing inside the rounding boundary is negligible.

The work of [Wat24, Appendix B] also proposes an alternative approach where instead of resampling the commitment and openings whenever an opening lands in the rounding boundary, the verification algorithm simply allows the bit in question to be opened to an *arbitrary* value. This requires a more delicate binding analysis where instead of showing that each commitment can be opened to exactly one hidden-bits string, one shows that each commitment can only be opened to a small number of hidden-bits strings. The recent concurrent work of [BCD⁺25] takes this approach in their lattice-based hidden-bits model NIZK construction.

6 Statistically-Hiding Vector Commitments from SIS

In this section, we show how to use our shifted multi-preimage trapdoor sampler to construct a statistically-hiding vector commitment from the SIS assumption. Our vector commitment scheme supports transparent setup. Moreover, the size of the CRS, the commitment, and the opening all scale polylogarithmically with the input dimension. This improves upon the earlier constructions of de Castro and Peikert [dCP23], which does not support statistically-hiding openings as well as the construction of Wee and Wu [WW23b], which required a structured common reference string (with size *quadratic* in the input dimension). We start by recalling the definition of a vector commitment and then provide our construction and analysis. We refer to Section 2.1 for a high-level overview of the construction. Our definitions are adapted from [WW23b]:

Definition 6.1 (Vector Commitment). Let λ be a security parameter and ℓ be a dimension. A vector commitment scheme with succinct local openings over a message space $\mathcal{M} = \{\mathcal{M}_{\lambda, \ell}\}_{\lambda, \ell \in \mathbb{N}}$ consists of a tuple of efficient algorithms $\Pi_{VC} = (\text{Setup}, \text{Commit}, \text{Open}, \text{Verify})$ with the following properties:

- $\text{Setup}(1^\lambda, 1^\ell) \rightarrow \text{crs}$: On input the security parameter λ and the vector length ℓ , the setup algorithm outputs a common reference string crs .
- $\text{Commit}(\text{crs}, \mathbf{x}) \rightarrow (\sigma, \text{st})$: On input the common reference string crs and a vector $\mathbf{x}$, the commit algorithm outputs a commitment σ and a state st .
- $\text{Open}(\text{st}, i) \rightarrow \pi$: On input a commitment state st and an index i , the open algorithm outputs an opening π . Note that the opening algorithm could be randomized.
- $\text{Verify}(\text{crs}, \sigma, i, x, \pi) \rightarrow b$: On input the common reference string crs , a commitment σ , an index i , a message x , and an opening π , the verification algorithm outputs a bit $b \in \{0, 1\}$.

We now define several standard properties on vector commitment schemes:

- **Correctness:** For all polynomials $\ell = \ell(\lambda)$, there exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$ and all inputs $\mathbf{x} = (x_1, \dots, x_\ell) \in \mathcal{M}_{\lambda, \ell}^\ell$,

$$\Pr \left[\begin{array}{c} \text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell) \\ \text{Verify}(\text{crs}, \sigma, i, x_i, \pi) = 1 : \begin{array}{c} (\sigma, \text{st}) \leftarrow \text{Commit}(\text{crs}, \mathbf{x}) \\ \pi \leftarrow \text{Open}(\text{st}, i) \end{array} \end{array} \right] = 1 - \text{negl}(\lambda).$$

- **Succinctness:** The vector commitment scheme is succinct if there exist fixed polynomials p_1, p_2 such that for all $\lambda, \ell \in \mathbb{N}$, all crs in the support of $\text{Setup}(1^\lambda, 1^\ell)$, all vectors $\mathbf{x} \in \mathcal{M}_{\lambda, \ell}^\ell$, all (σ, st) in the support of $\text{Commit}(\text{crs}, \mathbf{x})$, and all π in the support of $\text{Open}(\text{st}, i)$, we have that $|\sigma| = p_1(\lambda, \log \ell)$ and $|\pi| = p_2(\lambda, \log \ell)$.
- **Computational binding:** We say the commitment scheme is computationally binding if for all polynomials $\ell = \ell(\lambda)$ and all efficient adversaries $\mathcal{A}$, there exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$,

$$\Pr \left[\begin{array}{c} \text{Verify}(\text{crs}, \sigma, i, x, \pi) = 1 \\ \text{and } x \neq x' \text{ and} \\ \text{Verify}(\text{crs}, \sigma, i, x', \pi') = 1 \end{array} : \begin{array}{c} \text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell); \\ (\sigma, i, (x, \pi), (x', \pi')) \leftarrow \mathcal{A}(1^\lambda, 1^\ell, \text{crs}) \end{array} \right] = \text{negl}(\lambda).$$

- **Statistical hiding:** For a vector dimension ℓ , an adversary $\mathcal{A}$, and a simulator $\mathcal{S} = (\mathcal{S}_0, \mathcal{S}_1)$, we define two distributions $\text{Real}_{\mathcal{A}}(\lambda, \ell)$ and $\text{Ideal}_{\mathcal{A}, \mathcal{S}}(\lambda, \ell)$ as follows:

$\text{Real}_{\mathcal{A}}(\lambda, \ell)$: 1. Sample $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell)$ and give crs to $\mathcal{A}$. 2. Algorithm $\mathcal{A}$ outputs an input $\mathbf{x} \in \mathcal{M}_{\lambda, \ell}^\ell$. 3. Compute $(\sigma, \text{st}) \leftarrow \text{Commit}(\text{crs}, \mathbf{x})$ and give σ to $\mathcal{A}$. 4. Algorithm $\mathcal{A}$ can adaptively query for openings. On each query, it provides an index $i \in [\ell]$, and the challenger replies with $\pi_i \leftarrow \text{Open}(\text{st}, i)$. 5. Algorithm $\mathcal{A}$ outputs a bit $b \in \{0, 1\}$ which is the output of the experiment.	$\text{Ideal}_{\mathcal{A}, \mathcal{S}}(\lambda, \ell)$: 1. Sample $(\text{crs}, \sigma, \text{st}) \leftarrow \mathcal{S}_0(1^\lambda, 1^\ell)$ and give crs to $\mathcal{A}$. 2. Algorithm $\mathcal{A}$ outputs an input $\mathbf{x} \in \mathcal{M}_{\lambda, \ell}^\ell$. 3. Give σ to $\mathcal{A}$. 4. Algorithm $\mathcal{A}$ can adaptively query for openings. On each query, it provides an index $i \in [\ell]$, and the challenger computes $(\pi_i, \text{st}) \leftarrow \mathcal{S}_1(\text{st}, i, x_i)$. It replies to $\mathcal{A}$ with π_i. 5. Algorithm $\mathcal{A}$ outputs a bit $b \in \{0, 1\}$ which is the output of the experiment.
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We say that the vector commitment scheme is statistically hiding if there exists an efficient simulator $\mathcal{S} = (\mathcal{S}_0, \mathcal{S}_1)$ and such that for all polynomials $\ell = \ell(\lambda)$ and all (possibly unbounded) adversaries $\mathcal{A}$, there exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$,

$$|\Pr[\text{Real}_{\mathcal{A}}(\lambda, \ell) = 1] - \Pr[\text{Ideal}_{\mathcal{A}, \mathcal{S}}(\lambda, \ell) = 1]| = \text{negl}(\lambda).$$

Vector commitment scheme. We now describe our vector commitment scheme. As described in [Section 2.1](#), our construction can be viewed as replacing the CRS in the Wee-Wu vector commitment scheme [[WW23b](#), Construction 3.9] with the CRS for our shifted multi-preimage trapdoor sampler. Our analysis follows via a similar structure as the analysis in [[WW23b](#)], except we now appeal to the properties of the shifted multi-preimage trapdoor sampler. We give the full description and analysis below:

Construction 6.2 (Vector Commitment). Let λ be a security parameter and ℓ be an input length parameter. Let $\Pi_{\text{samp}} = (\text{Gen}, \text{GenTD}, \text{Expand}, \text{ExpandLocal})$ be a (n, t, q, s) -shifted multi-preimage trapdoor sampler that supports local expansion ([Definition 4.4](#)). Let $B = B(\lambda, \ell)$ be a bound. We construct a vector commitment $\Pi_{\text{VC}} = (\text{Setup}, \text{Commit}, \text{Open}, \text{Verify})$ scheme over the message space $\mathcal{M} = \mathbb{Z}_q = \{\mathbb{Z}_{q(\lambda, \ell)}\}_{\lambda, \ell \in \mathbb{N}}$ as follows:

- **Setup** $(1^\lambda, 1^\ell)$: On input the security parameter λ and the vector dimension ℓ , the setup algorithm samples $\text{crs}_{\text{samp}} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$. It outputs the common reference string $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$.
- **Commit** $(\text{crs}, \mathbf{x})$: On input the common reference string $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$ and a vector $\mathbf{x} \in \mathbb{Z}_q^\ell$, the commit algorithm computes $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$. Then it samples $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c}) \leftarrow \text{SampleMultPre}(\text{td}, x_1 \mathbf{e}_1, \dots, x_\ell \mathbf{e}_1)$ where $\mathbf{e}_1 = [1, 0, \dots, 0]^\top \in \mathbb{Z}_q^n$ is the first standard basis vector. It outputs the commitment $\mathbf{c} \in \mathbb{Z}_q^n$ and the state $\text{st} = (\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell)$.
- **Open** (st, i) : On input the state $\text{st} = (\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell)$ and the index $i \in [\ell]$, the opening algorithm outputs $\pi_i \in \mathbb{Z}_q^t$.
- **Verify** $(\text{crs}, \mathbf{c}, i, x, \boldsymbol{\pi})$: On input the common reference string $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$, a commitment $\mathbf{c} \in \mathbb{Z}_q^n$, an index $i \in [\ell]$, a message $x \in \mathbb{Z}_q$, and an opening $\boldsymbol{\pi} \in \mathbb{Z}_q^t$, the verification algorithm computes $\mathbf{A}_i = \text{ExpandLocal}(1^\lambda, \text{crs}_{\text{samp}}, i)$ and outputs 1 if $\|\boldsymbol{\pi}\| \leq B$ and $\mathbf{A}_i \boldsymbol{\pi} = \mathbf{c} + x \mathbf{e}_1$.

Theorem 6.3 (Correctness). Suppose Π_{samp} satisfies correctness and the preimage distribution property. If $s \geq \log t$ and $B \geq \sqrt{ts}$, then [Construction 6.2](#) is correct.

Proof. Take any polynomial $\ell = \ell(\lambda)$ and any $\mathbf{x} \in \mathbb{Z}_q^\ell$. Let $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell)$, $(\mathbf{c}, \text{st}) \leftarrow \text{Commit}(\text{crs}, \mathbf{x})$, and $\pi_i \leftarrow \text{Open}(\text{st}, i)$. Parse $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$. Let $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$. By construction, the commit algorithm samples $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c}) \leftarrow \text{SampleMultPre}(\text{td}, x_1 \mathbf{e}_1, \dots, x_\ell \mathbf{e}_1)$. Consider the value of $\text{Verify}(\text{crs}, \mathbf{c}, i, x_i, \boldsymbol{\pi}_i)$.

- By correctness of Π_{samp} , $\mathbf{A}_i \boldsymbol{\pi}_i = \mathbf{c} + x_i \mathbf{e}_1$. It suffices to argue that $\|\boldsymbol{\pi}_i\| \leq B$.
- Since Π_{samp} satisfies the preimage distribution property, the distribution of $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c})$ is statistically close to the distribution obtained by sampling $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and $\boldsymbol{\pi}_i \leftarrow (\mathbf{A}_i)_s^{-1}(x_i \mathbf{e}_1 + \mathbf{c})$ for all $i \in [\ell]$.

- By Lemma 4.8, the marginal distribution of $\mathbf{A}_i$ is statistically close to uniform. Since $s \geq \log t$, by Lemma 3.2, it holds that $\|\pi_i\| \leq \sqrt{ts} \leq B$ with overwhelming probability.

Finally, the local expansion property (Definition 4.4) ensures that $\text{ExpandLocal}(1^\lambda, \text{crs}_{\text{samp}}, i) = \mathbf{A}_i$. The above analysis shows that $\mathbf{A}_i \pi_i = \mathbf{c} + x_i \mathbf{e}_1$ and $\|\pi_i\| \leq B$, so Verify outputs 1 with overwhelming probability. $\square$

Theorem 6.4 (Computational Binding). *Suppose Π_{samp} satisfies somewhere programmability. Then, under the $\text{SIS}_{n-1,t,q,2B}$ assumption, Construction 6.2 is computationally binding.*

Proof. Take any polynomial $\ell = \ell(\lambda)$ and any efficient adversary $\mathcal{A}$ for the computational binding game. We begin by defining a sequence of hybrid experiments:

- Hyb_0 : This is the real binding experiment:
 - The challenger begins by sampling $\text{crs}_{\text{samp}} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$ and gives $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$ to $\mathcal{A}$.
 - Algorithm $\mathcal{A}$ outputs a commitment $\mathbf{c} \in \mathbb{Z}_q^n$, an index $i \in [\ell]$ and two pairs (x, π) and (x', π') , where $x, x' \in \mathcal{M}_{\lambda,\ell}$ and $\pi, \pi' \in \mathbb{Z}_q^t$.
 - The challenger then computes $\mathbf{A}_i = \text{ExpandLocal}(1^\lambda, \text{crs}_{\text{samp}}, i)$ and outputs 1 if $x \neq x'$, $\|\pi\|, \|\pi'\| \leq B$, and $\mathbf{c} = \mathbf{A}_i \pi - x \mathbf{e}_1 = \mathbf{A}_i \pi' - x' \mathbf{e}_1$. Otherwise, the challenger outputs 0.
- Hyb_1 : Same as Hyb_0 , except at the beginning of the game, the challenger samples an index $i^* \xleftarrow{\mathcal{R}} [\ell]$. The output of the experiment is 1 if the conditions in Hyb_0 hold and $i = i^*$.
- Hyb_2 : Same as Hyb_1 , except the challenger samples $\mathbf{A}_{i^*} \xleftarrow{\mathcal{R}} \mathbb{Z}_q^{n \times t}$ and $\text{crs}_{\text{samp}} \leftarrow \text{GenProg}(1^\lambda, 1^\ell, i^*, \mathbf{A}_{i^*})$.

We write $\text{Hyb}_i(\mathcal{A})$ to denote the random variable corresponding to the output of an execution of Hyb_i with adversary $\mathcal{A}$. We now analyze each pair of adjacent experiments:

Lemma 6.5. *It holds that $\Pr[\text{Hyb}_0(\mathcal{A}) = 1] = \ell \cdot \Pr[\text{Hyb}_1(\mathcal{A})]$.*

Proof. The only difference between these two experiments is the additional condition that $i^* = i$ in Hyb_1 . Since the adversary's view in Hyb_1 is independent of i^* (and in fact, the challenger can sample i^* after the adversary outputs i), the probability that $\text{Hyb}_1(\mathcal{A}) = 1$ is exactly $1/\ell \cdot \Pr[\text{Hyb}_0(\mathcal{A}) = 1]$. The claim follows. $\square$

Lemma 6.6. *If Π_{samp} is somewhere programmable, then $\text{Hyb}_1(\mathcal{A}) \stackrel{s}{\approx} \text{Hyb}_2(\mathcal{A})$.*

Proof. The only difference between these two experiments is the distribution of crs . These two distributions are statistically indistinguishable by somewhere programmability of Π_{samp} . $\square$

Lemma 6.7. *Suppose Π_{samp} satisfies somewhere programmability and local expansion. Then, under the $\text{SIS}_{n-1,t,q,2B}$ assumption, there exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$, $\Pr[\text{Hyb}_2(\mathcal{A}) = 1] = \text{negl}(\lambda)$.*

Proof. Suppose $\Pr[\text{Hyb}_2 = 1] \geq \varepsilon(\lambda)$ for some non-negligible ε . We use $\mathcal{A}$ to construct an adversary $\mathcal{B}$ for the SIS assumption:

- On input an SIS challenge matrix $\mathbf{A} \in \mathbb{Z}_q^{(n-1) \times t}$, algorithm $\mathcal{B}$ samples $i^* \xleftarrow{\mathcal{R}} [\ell]$ and $\mathbf{a} \xleftarrow{\mathcal{R}} \mathbb{Z}_q^t$. It sets $\mathbf{A}_{i^*} = \begin{bmatrix} \mathbf{a}^\top \\ \mathbf{A} \end{bmatrix} \in \mathbb{Z}_q^{n \times t}$. Next, algorithm $\mathcal{B}$ samples $\text{crs}_{\text{samp}} \leftarrow \text{GenProg}(1^\lambda, 1^\ell, i^*, \mathbf{A}_{i^*})$ and gives $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$ to $\mathcal{A}$.
- Algorithm $\mathcal{A}$ outputs a commitment $\mathbf{c}$, an index i , and two pairs (x, π) and (x', π') . Algorithm $\mathcal{B}$ outputs $\pi - \pi'$.

Since the SIS challenger samples $\mathbf{A} \xleftarrow{\mathcal{R}} \mathbb{Z}_q^{(n-1) \times t}$, the distribution of $\mathbf{A}_{i^*}$ is uniform over $\mathbb{Z}_q^{n \times t}$. Thus, algorithm $\mathcal{B}$ perfectly simulates an execution of Hyb_2 for $\mathcal{A}$. Thus, with probability ε , algorithm $\mathcal{A}$ outputs $\mathbf{c} \in \mathbb{Z}_q^n$, $i = i^*$, $x, x' \in \mathcal{M}_{\lambda,\ell}$, and $\pi, \pi' \in \mathbb{Z}_q^t$ where

$$x \neq x' \quad \text{and} \quad \|\pi\|, \|\pi'\| \leq B \quad \text{and} \quad \mathbf{c} = \mathbf{A}_{i^*} \pi - x \mathbf{e}_1 = \mathbf{A}_{i^*} \pi' - x' \mathbf{e}_1.$$

Here, we have implicitly used the fact that the $(i^*)^{\text{th}}$ matrix output by $\text{ExpandLocal}(1^\lambda, i, \text{crs})$ is $\mathbf{A}_{i^*}$, which is guaranteed by somewhere programmability of Π_{samp} and the local expansion property (Definition 4.4). This means

$$\mathbf{A}_{i^*}(\boldsymbol{\pi} - \boldsymbol{\pi}') = (x - x')\mathbf{e}_1 = \begin{bmatrix} x - x' \\ \mathbf{0}^{n-1} \end{bmatrix}. \quad (6.1)$$

Since $x \neq x'$, we conclude that $\boldsymbol{\pi} - \boldsymbol{\pi}' \neq \mathbf{0}$. Since $\mathbf{A}_{i^*} = \begin{bmatrix} \mathbf{a}^\top \\ \mathbf{A} \end{bmatrix}$, Eq. (6.1) now implies that $\mathbf{A}(\boldsymbol{\pi} - \boldsymbol{\pi}') = \mathbf{0}^{n-1}$. Thus $\boldsymbol{\pi} - \boldsymbol{\pi}'$ is a non-trivial SIS solution. Finally, $\|\boldsymbol{\pi}\|, \|\boldsymbol{\pi}'\| \leq B$, so $\|\boldsymbol{\pi} - \boldsymbol{\pi}'\| \leq 2B$ and algorithm $\mathcal{B}$ succeeds in breaking SIS with the same advantage ε . $\square$

Combining Lemmas 6.5 to 6.7, we conclude that $\Pr[\text{Hyb}_0(\mathcal{A}) = 1] \leq \ell \cdot \text{negl}(\lambda)$. Since $\ell = \text{poly}(\lambda)$, computational binding holds. $\square$

Theorem 6.8 (Statistical Hiding). *Suppose Π_{samp} supports simulatable openings (Definition 4.5) and moreover, that $n \geq \lambda$ and q is prime. Then Construction 6.2 satisfies statistical hiding.*

Proof. Since Π_{samp} supports simulatable openings, let GenTD be the trapdoor generator algorithm. We construct an efficient simulator $\mathcal{S} = (\mathcal{S}_0, \mathcal{S}_1)$ as follows:

- $\mathcal{S}_0(1^\lambda, 1^\ell)$: On input the security parameter λ and the vector dimension ℓ , the simulator setup algorithm first samples $(\text{crs}_{\text{samp}}, \mathbf{T}_1, \dots, \mathbf{T}_\ell) \leftarrow \text{GenTD}(1^\lambda, 1^\ell)$. Next, it samples $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$. The simulator also initializes an empty dictionary $\mathbf{D}$ to keep track of indices and openings. Finally, it outputs the common reference string $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$, the commitment $\mathbf{c}$, and the simulation state $\text{st} = (1^\lambda, 1^\ell, \text{crs}_{\text{samp}}, \mathbf{c}, \mathbf{T}_1, \dots, \mathbf{T}_\ell, \mathbf{D})$.
- $\mathcal{S}(\text{st}, i, x_i)$: On input the simulation state $\text{st} = (1^\lambda, 1^\ell, \text{crs}_{\text{samp}}, \mathbf{c}, \mathbf{T}_1, \dots, \mathbf{T}_\ell, \mathbf{D})$, an index $i \in [\ell]$, and an input $x_i \in \mathcal{M}_{\lambda, \ell}$, the simulator first checks if there is a mapping $(i \mapsto \boldsymbol{\pi}_i)$ in $\mathbf{D}$. If so, it replies with $\boldsymbol{\pi}_i$. Otherwise, the simulator computes $\mathbf{A}_i \leftarrow \text{ExpandLocal}(1^\lambda, \text{crs}_{\text{samp}}, i)$ and samples $\boldsymbol{\pi}_i \leftarrow \text{SamplePre}(\mathbf{A}_i, \mathbf{T}_i, x_i \mathbf{e}_1 + \mathbf{c})$. It adds the mapping $(i \mapsto \boldsymbol{\pi}_i)$ to $\mathbf{D}$ and outputs the opening $\boldsymbol{\pi}_i$ together with the updated state $\text{st} = (1^\lambda, 1^\ell, \text{crs}_{\text{samp}}, \mathbf{c}, \mathbf{T}_1, \dots, \mathbf{T}_\ell, \mathbf{D})$.

We now show that this simulator $\mathcal{S}$ satisfies the statistical hiding definition. Take any (possibly unbounded) adversary $\mathcal{A}$. We proceed via a hybrid argument:

- Hyb_0 : This is the distribution $\text{Real}_{\mathcal{A}}(\lambda, \ell)$. Specifically, in this experiment, the challenger proceeds as follows:
 - The challenger samples $\text{crs}_{\text{samp}} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$ and gives $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$ to $\mathcal{A}$.
 - Algorithm $\mathcal{A}$ outputs a vector $\mathbf{x}$ and the challenger computes $(\mathbf{c}, \text{st}) \leftarrow \text{Commit}(\text{crs}, \mathbf{x})$. Specifically, the challenger first computes $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$. Then it samples $(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_\ell, \mathbf{c}) \leftarrow \text{SampleMultPre}(\text{td}, x_1 \mathbf{e}_1, \dots, x_\ell \mathbf{e}_\ell)$. The challenger responds to $\mathcal{A}$ with the commitment $\mathbf{c}$.
 - Whenever the adversary requests an opening on an index $i \in [\ell]$, the challenger replies with $\boldsymbol{\pi}_i$.
 - At the end of the game, algorithm $\mathcal{A}$ outputs a bit $b \in \{0, 1\}$ which is the output of the experiment.
- Hyb_1 : Same as Hyb_0 , except the challenger changes how it samples the commitment and the openings:
 - The challenger samples $\text{crs}_{\text{samp}} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$ and gives $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$ to $\mathcal{A}$.
 - Algorithm $\mathcal{A}$ outputs a vector $\mathbf{x}$. The challenger responds to $\mathcal{A}$ with the commitment $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$. Then, the challenger computes $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$ and for each $i \in [\ell]$, $\boldsymbol{\pi}_i \leftarrow (\mathbf{A}_i)_s^{-1}(x_i \mathbf{e}_1 + \mathbf{c})$.
 - The rest of the experiment proceeds as in Hyb_0 .
- Hyb_2 : Same as Hyb_1 , except the challenger changes how it samples the common reference string:
 - The challenger samples $(\text{crs}_{\text{samp}}, \mathbf{T}_1, \dots, \mathbf{T}_\ell) \leftarrow \text{GenTD}(1^\lambda, 1^\ell)$ and gives $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$ to $\mathcal{A}$.
 - Algorithm $\mathcal{A}$ outputs a vector $\mathbf{x}$. The challenger responds to $\mathcal{A}$ with the commitment $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$. Then, the challenger computes $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$ and for each $i \in [\ell]$, $\boldsymbol{\pi}_i \leftarrow (\mathbf{A}_i)_s^{-1}(x_i \mathbf{e}_1 + \mathbf{c})$.

- The rest of the experiment proceeds as in Hyb_0 .
- Hyb_3 : Same as Hyb_2 , except the challenger changes how it constructs the openings:
 - The challenger samples $(\text{crs}_{\text{samp}}, \mathbf{T}_1, \dots, \mathbf{T}_\ell) \leftarrow \text{GenTD}(1^\lambda, 1^\ell)$ and gives $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$ to $\mathcal{A}$.
 - Algorithm $\mathcal{A}$ outputs a vector $\mathbf{x}$. The challenger responds to $\mathcal{A}$ with the commitment $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$. Then, the challenger computes $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) = \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$ and for each $i \in [\ell]$, it samples $\pi_i \leftarrow \text{SamplePre}(\mathbf{A}_i, \mathbf{T}_i, x_i \mathbf{e}_1 + \mathbf{c}, s)$.
 - The rest of the experiment proceeds as in Hyb_0 .
- Hyb_4 : Same as Hyb_3 , except the challenger changes how it computes $\mathbf{A}_i$:
 - The challenger samples $(\text{crs}_{\text{samp}}, \mathbf{T}_1, \dots, \mathbf{T}_\ell) \leftarrow \text{GenTD}(1^\lambda, 1^\ell)$ and gives $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$ to $\mathcal{A}$.
 - Algorithm $\mathcal{A}$ outputs a vector $\mathbf{x}$. The challenger responds to $\mathcal{A}$ with the commitment $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$. For each $i \in [\ell]$, the challenger computes $\mathbf{A}_i = \text{ExpandLocal}(1^\lambda, \text{crs}_{\text{samp}}, i)$ and $\pi_i \leftarrow \text{SamplePre}(\mathbf{A}_i, \mathbf{T}_i, x_i \mathbf{e}_1 + \mathbf{c}, s)$.
 - The rest of the experiment proceeds as in Hyb_0 .
- Hyb_5 : Same as Hyb_4 , except the challenger samples π_i only when the adversary requests an opening on index $i \in [\ell]$:
 - The challenger samples $(\text{crs}_{\text{samp}}, \mathbf{T}_1, \dots, \mathbf{T}_\ell) \leftarrow \text{GenTD}(1^\lambda, 1^\ell)$ and gives $\text{crs} = (1^\lambda, \ell, \text{crs}_{\text{samp}})$ to $\mathcal{A}$. **The challenger also initializes an (empty) dictionary $\mathbf{D}$ to keep track of indices and openings.**
 - Algorithm $\mathcal{A}$ outputs a vector $\mathbf{x}$. The challenger responds to $\mathcal{A}$ with the commitment $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$.
 - **Whenever the adversary requests an opening on an index $i \in [\ell]$, the challenger first checks if there is a mapping $(i \mapsto \pi_i)$ in $\mathbf{D}$. If so, it replies with π_i . Otherwise, the challenger computes $\mathbf{A}_i = \text{ExpandLocal}(1^\lambda, \text{crs}_{\text{samp}}, i)$ and samples $\pi_i \leftarrow \text{SamplePre}(\mathbf{A}_i, \mathbf{T}_i, x_i \mathbf{e}_1 + \mathbf{c}, s)$. It adds the mapping $(i \mapsto \pi_i)$ to $\mathbf{D}$ and gives π_i to $\mathcal{A}$.**
 - At the end of the game, algorithm $\mathcal{A}$ outputs a bit $b \in \{0, 1\}$ which is the output of the experiment.

This is the experiment $\text{Ideal}_{\mathcal{A}, \mathcal{S}}(\lambda, \ell)$.

We write $\text{Hyb}_i(\mathcal{A}, \mathcal{S})$ to denote the random variable (indexed implicitly by the security parameter λ) corresponding to the output of Hyb_i with adversary $\mathcal{A}$ and simulator $\mathcal{S}$. We now analyze each pair of hybrid experiments:

Lemma 6.9. *Suppose Π_{samp} satisfies the preimage distribution property. Then $\text{Hyb}_0(\mathcal{A}, \mathcal{S}) \stackrel{\approx}{\sim} \text{Hyb}_1(\mathcal{A}, \mathcal{S})$.*

Proof. By the preimage sampling property, with overwhelming probability over the choice of crs_{samp} , the distribution of $(\pi_1, \dots, \pi_\ell, \mathbf{c})$ output by $\text{SampleMultPre}(\text{td}, x_1 \mathbf{e}_1, \dots, x_\ell \mathbf{e}_1)$ is statistically close to the distribution obtained by sampling $\mathbf{c} \leftarrow \mathbb{Z}_q^n$ and $\pi_i \leftarrow (\mathbf{A}_i)_s^{-1}(x_i \mathbf{e}_1 + \mathbf{c})$ for all $i \in [\ell]$. Thus, $\text{Hyb}_0(\mathcal{A}, \mathcal{S})$ and $\text{Hyb}_1(\mathcal{A}, \mathcal{S})$ are statistically indistinguishable. $\square$

Lemma 6.10. *Suppose Π_{samp} has simulatable openings. Then, $\text{Hyb}_1(\mathcal{A}, \mathcal{S}) \stackrel{\approx}{\sim} \text{Hyb}_2(\mathcal{A}, \mathcal{S})$.*

Proof. By definition of mode indistinguishability (Definition 4.5), the distributions of crs_{samp} output by $\text{Gen}(1^\lambda, 1^\ell)$ and $\text{GenTD}(1^\lambda, 1^\ell)$ are statistically indistinguishable. $\square$

Lemma 6.11. *Suppose Π_{samp} has simulatable openings. Then, $\text{Hyb}_2(\mathcal{A}, \mathcal{S}) \stackrel{\approx}{\sim} \text{Hyb}_3(\mathcal{A}, \mathcal{S})$.*

Proof. By the trapdoor generation property (Definition 4.5), we have that $\mathbf{A}_i \mathbf{T}_i = \mathbf{G}$ and moreover, that $\|\mathbf{T}_i\| \leq s/(t \log n)$. By Theorem 3.9, this means the distribution of $\pi_i \leftarrow \text{SamplePre}(\mathbf{A}_i, \mathbf{T}_i, x_i \mathbf{e}_1 + \mathbf{c}, s)$ is statistically close to sampling $\pi_i \leftarrow (\mathbf{A}_i)_s^{-1}(x_i \mathbf{e}_1 + \mathbf{c})$. This holds for all $i \in [\ell]$. Since $\ell = \text{poly}(\lambda)$, we conclude that $\text{Hyb}_2(\mathcal{A}, \mathcal{S})$ and $\text{Hyb}_3(\mathcal{A}, \mathcal{S})$ are statistically indistinguishable by a hybrid argument. $\square$

Lemma 6.12. *Suppose ExpandLocal is a correct local-expansion procedure. Then, $\text{Hyb}_3(\mathcal{A}, \mathcal{S}) \equiv \text{Hyb}_4(\mathcal{A}, \mathcal{S})$.*

Proof. Immediate by correctness of the local expansion procedure (Definition 4.4). $\square$

Lemma 6.13. *It holds that $\text{Hyb}_4(\mathcal{A}, \mathcal{S}) \equiv \text{Hyb}_5(\mathcal{A}, \mathcal{S})$.*

Proof. The only difference between these experiments is the order in which the challenger samples different components. As such, the output of these two experiments is identically distributed. $\square$

Statistical hiding now follows by combining Lemmas 6.9 to 6.13. $\square$

Theorem 6.14 (Succinctness). *Suppose $n \log q \leq \text{poly}(\lambda, \log \ell)$ and $t \log q \leq \text{poly}(\lambda, \log \ell)$. Then Construction 6.2 is succinct.*

Proof. This is immediate from the assumptions. Namely, each commitment in Construction 6.2 is an element $\mathbf{c} \in \mathbb{Z}_q^n$, which has size $|\mathbf{c}| = n \log q \leq \text{poly}(\lambda, \log \ell)$. Similarly, each opening $\pi_i \in \mathbb{Z}_q^t$ has length at most $t \log q \leq \text{poly}(\lambda, \log \ell)$. $\square$

Parameter instantiations. Let λ be a security parameter and ℓ be the vector dimension. We provide one possible instantiation of the parameters in Construction 6.2 to satisfy Theorems 6.3, 6.4, 6.8 and 6.14. In the following, we will assume that ℓ is polynomially-bounded in λ (i.e., $\ell \leq \lambda^c$ for some constant $c \in \mathbb{N}$).

- We set the lattice dimension to be $n = \lambda$.
- We set $t = 3n \lceil \log q \rceil \cdot (\lceil \log \ell \rceil + 1)$. When setting parameters, we work under the assumption that $\log q \leq k \log \lambda$ for some constant $k > 0$. It is easy to check that this is satisfied by our final instantiation. In this case, $t = O(\lambda \log \ell \log \lambda) = O(\lambda \log^2 \lambda)$ since $\log \ell \leq c \log \lambda$.
- We set $s = (\ell t + 3n \lceil \log q \rceil) \log(\ell n) = O(\ell \lambda \log^3 \lambda)$.
- We set $B = \sqrt{t} s = O(\ell \lambda^{3/2} \log^4 \lambda)$.
- We choose a prime $q = 2B \cdot \text{poly}(n)$ where the $\text{SIS}_{n-1, t, q, 2B}$ assumption holds. In this case, $\log q = O(\log \lambda + \log \ell)$. Since $\log \ell \leq c \log \lambda$, we can bound $(\log q)$ by $(k \log \lambda)$ for some sufficiently-large constant $k > 0$, as required.
- We instantiate (Gen, GenTD, Expand, ExpandLocal) with the (n, t, q, s) -shifted multi-preimage trapdoor sampler from Theorem 4.7 (Construction 4.6).

With this setting of parameters, we obtain a vector commitment scheme over $\mathbb{Z}_q^\ell$ with the following properties:

- **CRS size:** From Theorem 4.7, Construction 6.2 supports a transparent setup and the size of the CRS is $\lambda + \log \ell + nt \log q = O(\lambda^2 \log^3 \lambda)$.
- **Commitment size:** A commitment in Construction 6.2 consists of a vector $\mathbf{c} \in \mathbb{Z}_q^n$ which has size $n \log q = O(\lambda \log \lambda)$.
- **Opening size:** An opening in Construction 6.2 consists of a vector $\pi_i \in \mathbb{Z}_q^t$, which has size $t \log q = O(\lambda \log^3 \lambda)$.

We summarize the instantiation in the following corollary:

Corollary 6.15 (Vector Commitments from SIS). *Let λ be a security parameter. Then, for all polynomials $\ell = \ell(\lambda)$, under the SIS assumption with a polynomial noise bound $\beta = \text{poly}(\lambda, \ell)$ and a polynomial modulus $q = \text{poly}(\lambda, \ell)$, there exists a vector commitment scheme over $\mathbb{Z}_q^\ell$ with a transparent CRS of size $O(\lambda^2 \log^3 \lambda) = \text{poly}(\lambda)$, commitments of size $O(\lambda \log \lambda)$ and openings of size $O(\lambda \log^3 \lambda)$. The vector commitment is computationally binding and statistically hiding.*

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A Simulation-Based Statistical Hiding

In this section, we recall the *stronger* simulation-based statistical hiding property on dual-mode hidden-bits generators from [LPWW20]. In this setting, we require that there exist an efficient simulator that can simulate a commitment together with the openings to *any* random string. While this definition is more complex than the single-bit statistical hiding definition from Definition 5.1, the work of [LPWW20] shows that it directly implies adaptive multi-theorem statistical zero-knowledge for the resulting NIZK (as opposed to single-theorem statistical zero-knowledge). This allows us to achieve multi-theorem security without needing the “or-proof” approach from [FLS90, BCD⁺25] (which requires using the underlying single-theorem NIZK to prove membership in a cryptographic language). We refer to

[Remark 5.3](#) for more discussion. We recall the formal definition below and then show that [Construction 5.4](#) satisfies this stronger property (with the same parameter requirements as in [Theorem 5.11](#)).

Definition A.1 (Statistical Simulation in Hiding Mode [[LPWW20](#), Definition 3.1, adapted]). Let $\Pi_{\text{HBG}} = (\text{Setup}, \text{GenBits}, \text{Verify})$ be a hidden-bits generator. For an adversary $\mathcal{A}$, a simulator $\mathcal{S} = (\mathcal{S}_0, \mathcal{S}_1)$, an output length $\ell \in \mathbb{N}$, and a bit $b \in \{0, 1\}$, we define the simulation-based statistical hiding game as follows:

- If $b = 0$, the challenger samples $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^\ell, \text{hiding})$. If $b = 1$, the challenger samples $(\text{td}_{\mathcal{S}}, \text{crs}) \leftarrow \mathcal{S}_0(1^\lambda, 1^\ell)$. The challenger gives crs to $\mathcal{A}$.
- Algorithm $\mathcal{A}$ can now issue challenge queries. On each query, the challenger proceeds as follows:
 - If $b = 0$, the challenger computes $(\sigma, \mathbf{r}, (\pi_1, \dots, \pi_\ell)) \leftarrow \text{GenBits}(\text{crs})$ and gives $\mathbf{r}$ to $\mathcal{A}$. If $b = 1$, the challenger responds with $\mathbf{r} \xleftarrow{\mathbb{R}} \{0, 1\}^\ell$.
 - Algorithm $\mathcal{A}$ then specifies a set $I \subseteq [\rho]$.
 - If $b = 0$, the challenger responds with $(\sigma, \{(i, \pi_i)\}_{i \in I})$. If $b = 1$, it responds with $\mathcal{S}_1(\text{td}_{\mathcal{S}}, \{(i, r_i)\}_{i \in I})$.
- At the end of the game, algorithm $\mathcal{A}$ outputs a bit $b' \in \{0, 1\}$ which is the output of the experiment.

We say that Π_{HBG} satisfies statistical simulation in hiding mode if for all polynomials $\ell = \ell(\lambda)$ and $Q = Q(\lambda)$, all unbounded adversaries $\mathcal{A}$ making at most Q challenge queries, there exists an efficient simulator $\mathcal{S} = (\mathcal{S}_0, \mathcal{S}_1)$ and a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$,

$$|\Pr[b' = 1 : b = 0] - \Pr[b' = 1 : b = 1]| = \text{negl}(\lambda)$$

in the simulation-based statistical hiding game.

Theorem A.2 (Dual-Mode Multi-Theorem NIZK from Hidden-Bits Generators [[LPWW20](#)]). *If there exists a dual-mode hidden-bits generator that satisfies statistical simulation in hiding mode, then there exists a dual-mode multi-theorem NIZK for NP. Moreover, if the CRS for the binding (resp., hiding) mode of the hidden-bits generator is a uniform random string, then the same holds for the statistical soundness (resp., statistical multi-theorem zero-knowledge) mode of the resulting NIZK construction.*

Theorem A.3 (Statistical Simulation in Hiding Mode). *Suppose Π_{samp} satisfies the preimage distribution property and that it supports simulatable openings. In addition, suppose $n \geq 4\lambda + 2 \log q$, $t \geq 2n \log q$, $q > 4B_{\text{round}} + 2$ is prime, $s_{\text{samp}} \geq \log t$, $B_{\text{max}} \geq \sqrt{t s_{\text{samp}}}$, and $B_{\text{round}} \geq q/4 - q/(8\ell) - 1/2$. Then, [Construction 5.4](#) satisfies statistical simulation in hiding mode.*

Proof. Since Π_{samp} supports simulatable openings, let GenTD be the trapdoor generator algorithm. We construct an efficient simulator $\mathcal{S} = (\mathcal{S}_0, \mathcal{S}_1)$ as follows:

- $\mathcal{S}_0(1^\lambda, 1^\ell)$: On input the security parameter λ and the output length ℓ , the simulator setup algorithm samples $(\text{crs}_{\text{samp}}, \mathbf{T}_1, \dots, \mathbf{T}_\ell) \leftarrow \text{GenTD}(1^\lambda, 1^\ell)$. Then, for each $i \in [\ell]$, it samples $\mathbf{v}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$ and outputs the simulated common reference string $\text{crs} = (1^\lambda, \text{crs}_{\text{samp}}, \mathbf{v}_1, \dots, \mathbf{v}_\ell)$ along with the simulator trapdoor $\text{td}_{\mathcal{S}} = (\text{crs}_{\text{samp}}, \mathbf{T}_1, \dots, \mathbf{T}_\ell)$.
- $\mathcal{S}_1(\text{td}_{\mathcal{S}}, \{(i, r_i)\}_{i \in I})$: On input the simulation trapdoor $\text{td}_{\mathcal{S}} = (\mathbf{T}_1, \dots, \mathbf{T}_\ell)$ and a collection of bits $r_i \in \{0, 1\}$ for $i \in I$, the simulator algorithm proceeds as follows:
 1. Compute $(\mathbf{A}_1, \dots, \mathbf{A}_\ell, \text{td}) \leftarrow \text{Expand}(1^\lambda, 1^\ell, \text{crs}_{\text{samp}})$.
 2. Sample $\mathbf{c} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$.
 3. For each $i \in I$, and $d \in [\lambda]$, sample $\pi_{d,i} \leftarrow \text{SamplePre}(\mathbf{A}_i, \mathbf{T}_i, \mathbf{c})$. The simulator now sets $\pi_i = \pi_{d^*,i}$ where $d^* \in [\lambda]$ is the smallest index where $\|\pi_{d^*,i}\| \leq B_{\text{max}}$ and

$$\begin{aligned} \mathbf{v}_i^\top \pi_{d^*,i} &\in [-B_{\text{round}}, B_{\text{round}}] && \text{if } r_i = 0 \\ \mathbf{v}_i^\top \pi_{d^*,i} &\in [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}] && \text{if } r_i = 1. \end{aligned}$$

If there does not exist an index d^* with this property, then the simulator outputs $\perp$.

4. Output $(c, \{(i, \pi_i)\}_{i \in I})$.

We now show that the simulator $\mathcal{S}$ satisfies the required property. Take any polynomial $\ell = \ell(\lambda)$ and $Q = Q(\lambda)$ and let $\mathcal{A}$ be an (unbounded) adversary that makes at most Q challenge queries. We define a sequence of hybrid experiments:

- Hyb₀: This is the statistical simulation experiment where $b = 0$. Specifically, the experiment proceeds as follows:

- First, the challenger samples $\text{crs}_{\text{samp}} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$ and for each $i \in [\ell]$, the challenger samples $\mathbf{v}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^t$. It gives the common reference string $\text{crs} = (1^\lambda, \text{crs}_{\text{samp}}, \mathbf{v}_1, \dots, \mathbf{v}_\ell)$.
- When algorithm $\mathcal{A}$ makes its k^{th} challenge query (where $k \in [Q]$), the challenger proceeds as follows for each $d \in [\lambda]$:⁶
 - * Sample $(\pi_{d,1}^{(k)}, \dots, \pi_{d,\ell}^{(k)}, \mathbf{c}_d^{(k)}) \leftarrow \text{SampleMultPre}(\text{td}, \mathbf{0}^n, \dots, \mathbf{0}^n)$.
 - * For each $i \in [\ell]$, the challenger computes $u_{d,i}^{(k)} = \mathbf{v}_i^\top \pi_{d,i}^{(k)}$. If $\|\pi_{d,i}^{(k)}\| > B_{\max}$, the challenger sets $r_{d,i}^{(k)} = \perp$. Otherwise, it sets

$$r_{d,i}^{(k)} = \begin{cases} 0 & u_{d,i}^{(k)} \in [-B_{\text{round}}, B_{\text{round}}] \\ 1 & u_{d,i}^{(k)} \in [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}] \\ \perp & \text{otherwise.} \end{cases} \quad (\text{A.1})$$

- * Suppose for all $d \in [\lambda]$, there exists an index $i \in [\ell]$ where $r_{d,i}^{(k)} = \perp$. Then, the challenger sets

$$\mathbf{c}^{(k)} = \perp \quad \text{and} \quad \forall i \in [\ell] : r_i^{(k)} = 0 \quad \text{and} \quad \pi_i^{(k)} = \perp.$$

Otherwise, let $d^* \in [\lambda]$ be the first index where $r_{d^*,i}^{(k)} \in \{0, 1\}$ for all $i \in [\ell]$. Then the challenger sets

$$\mathbf{c}^{(k)} = \mathbf{c}_{d^*}^{(k)} \quad \text{and} \quad \forall i \in [\ell] : r_i^{(k)} = r_{d^*,i}^{(k)} \quad \text{and} \quad \pi_i^{(k)} = \pi_{d^*,i}^{(k)}.$$

- * The challenger responds to $\mathcal{A}$ with $\mathbf{r}^{(k)} = (r_1^{(k)}, \dots, r_\ell^{(k)})$.
- * After algorithm $\mathcal{A}$ specifies a set $I_k \subseteq [\rho]$, the challenger responds with $\mathbf{c}^{(k)}$ and $\{(i, \pi_i^{(k)})\}_{i \in I_k}$.
- At the end of the game, algorithm $\mathcal{A}$ outputs a bit $b' \in \{0, 1\}$, which is the output of the experiment.
- Hyb₁: Same as Hyb₀ except for all $k \in [Q]$ and $d \in [\lambda]$, the challenger instead samples $\mathbf{c}_d^{(k)} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and for all $i \in [\ell]$, the challenger samples $\pi_{d,i}^{(k)} \leftarrow (\mathbf{A}_i)_{\text{samp}}^{-1}(\mathbf{c}_d^{(k)})$.
- Hyb₂: Same as Hyb₁ except the challenger uses the following procedure to respond to challenge queries:

- Sample $\mathbf{c}^{(k)} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and $\mathbf{r}^{(k)} \xleftarrow{\mathbb{R}} \{0, 1\}^\ell$.
- Then, for each $i \in [\ell]$, sample $\pi_i^{(k)} \leftarrow (\mathbf{A}_i)_{\text{samp}}^{-1}(\mathbf{c}^{(k)})$ conditioned on $\|\pi_i^{(k)}\| \leq B_{\max}$ and

$$\begin{aligned} \mathbf{v}_i^\top \pi_i^{(k)} &\in [-B_{\text{round}}, B_{\text{round}}] && \text{if } r_i^{(k)} = 0 \\ \mathbf{v}_i^\top \pi_i^{(k)} &\in [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}] && \text{if } r_i^{(k)} = 1. \end{aligned}$$

If there does not exist $\pi_i^{(k)}$ in the support of $(\mathbf{A}_i)_{\text{samp}}^{-1}(\mathbf{c}^{(k)})$ that satisfies the requisite condition, then the challenger halts the experiment and outputs 0.

- Hyb₃: Same as Hyb₂ except the challenger uses the following procedure to respond to challenge queries:
- Sample $\mathbf{c}^{(k)} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and $\mathbf{r}^{(k)} \xleftarrow{\mathbb{R}} \{0, 1\}^\ell$.

⁶Similar to the proof of [Theorem 5.11](#), we explicitly unroll the λ iterations of rejection sampling that the GenBits algorithm performs.

- Then, for each $i \in [\ell]$ and $d \in [\lambda]$, sample $\pi_{d,i}^{(k)} \leftarrow (\mathbf{A}_i)_{s_{\text{samp}}}^{-1}(\mathbf{c}^{(k)})$. Then, set $\pi_i^{(k)} = \pi_{d^*,i}^{(k)}$ where $d^* \in [\lambda]$ is the smallest index where $\|\pi_{d^*,i}^{(k)}\| \leq B_{\text{max}}$ and

$$\begin{aligned} \mathbf{v}_i^\top \pi_{d^*,i}^{(k)} &\in [-B_{\text{round}}, B_{\text{round}}] & \text{if } r_i^{(k)} = 0 \\ \mathbf{v}_i^\top \pi_{d^*,i}^{(k)} &\in [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}] & \text{if } r_i^{(k)} = 1. \end{aligned}$$

If there does not exist an index d^* with this property, then after the adversary responds with I , the challenger responds with $\perp$.

- Hyb_4 : Same as Hyb_3 except the challenger uses GenTD to sample crs_{samp} . Specifically, at the beginning of the experiment, the challenger samples $(\text{crs}_{\text{samp}}, \mathbf{T}_1, \dots, \mathbf{T}_\ell) \leftarrow \text{GenTD}(1^\lambda, 1^\ell)$.
- Hyb_5 : Same as Hyb_4 except when responding to challenge queries, the challenger now samples $\pi_{d,i}^{(k)} \leftarrow \text{SamplePre}(\mathbf{A}_i, \mathbf{T}_i, \mathbf{c}^{(k)}, s_{\text{samp}})$ for all $k \in [Q]$, $d \in [\lambda]$, and $i \in [\ell]$. This is the statistical simulation experiment where $b = 1$.

We write $\text{Hyb}_i(\mathcal{A})$ to denote the random variable corresponding to the output of an execution of Hyb_i with adversary $\mathcal{A}$. We now analyze each adjacent pair of distributions.

Lemma A.4. *Suppose Π_{samp} satisfies the preimage distribution property. Then, $\text{Hyb}_0(\mathcal{A}) \stackrel{s}{\approx} \text{Hyb}_1(\mathcal{A})$.*

Proof. This follows by the same argument as that used in the proof of Lemma 5.12. Specifically, by the preimage distribution property, the distributions $(\pi_{d,1}^{(k)}, \dots, \pi_{d,\ell}^{(k)}, \mathbf{c}_d^{(k)}) \leftarrow \text{SampleMultPre}(\text{td}, \mathbf{0}^n, \dots, \mathbf{0}^n)$ from Hyb_0 and the distribution $\mathbf{c}_d^{(k)} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and $\pi_{d,i}^{(k)} \leftarrow (\mathbf{A}_i)_{s_{\text{samp}}}^{-1}(\mathbf{c}_d^{(k)})$ from Hyb_1 are statistically close. Since $d \in [\lambda]$, $k \in [Q]$, and $Q = \text{poly}(\lambda)$, the claim follows by a standard hybrid argument. $\square$

Lemma A.5. *Suppose $n \geq 4\lambda + 2 \log q$, $t \geq 2n \log q$, $q > 4B_{\text{round}} + 2$ is prime, $s_{\text{samp}} \geq \log t$, $B_{\text{max}} \geq \sqrt{t}s_{\text{samp}}$, and $B_{\text{round}} \geq q/4 - q/(8\ell) - 1/2$. Then $\text{Hyb}_1(\mathcal{A}) \stackrel{s}{\approx} \text{Hyb}_2(\mathcal{A})$.*

Proof. Fix any challenge query index $k \in [Q]$. We analyze the joint distribution of $(\mathbf{c}^{(k)}, \pi_1^{(k)}, \dots, \pi_\ell^{(k)}, r_1^{(k)}, \dots, r_\ell^{(k)})$ sampled using the procedure in Hyb_1 and the distribution sampled using the procedure in Hyb_2 . We will show that these distributions are statistically indistinguishable. The claim then follows by a standard hybrid argument (since $Q = \text{poly}(\lambda)$). Consider the distribution in Hyb_1 . For each $d \in [\lambda]$, we say an event E_d occurs if for all indices $i \in [\ell]$, it holds that $r_{d,i}^{(k)} \neq \perp$. We now show the following claims:

Claim A.6. *In Hyb_1 , for all $d \in [\lambda]$, there exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$, $\Pr[E_d] \geq 1/2 - \text{negl}(\lambda)$.*

Proof. We will use a union bound. Fix any index $i \in [\ell]$. Consider the probability that $r_{d,i}^{(k)} = \perp$. There are two cases where this might happen:

- $\|\pi_{d,i}^{(k)}\| > B_{\text{max}}$; or
- $u_{d,i}^{(k)} \notin [-B_{\text{round}}, B_{\text{round}}] \cup [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}]$.

We analyze the probability of each of these events:

- In Hyb_1 , the challenger samples $\pi_{d,i}^{(k)} \leftarrow (\mathbf{A}_i)_{s_{\text{samp}}}^{-1}(\mathbf{c}_d^{(k)})$. By Lemma 4.8, the marginal distribution of $\mathbf{A}_i$ in Hyb_1 is statistically close to uniform. Since $t \geq 2n \log q$, $s_{\text{samp}} \geq \log t$, and $B_{\text{max}} \geq \sqrt{t}s_{\text{samp}}$, by Lemma 3.2, with overwhelming probability over the choice of $\pi_{d,i}^{(k)}$, it holds that $\|\pi_{d,i}^{(k)}\| \leq B_{\text{max}}$. Thus, the first event occurs with negligible probability.

- For the second property, we appeal to [Lemma 3.8](#) to conclude that with overwhelming probability over the choice of $\mathbf{A}_i$,

$$\mathbf{H}_\infty(\boldsymbol{\pi}_{d,i}^{(k)}) \geq n/2 \geq 2\lambda + \log q.$$

Since the challenger samples $\mathbf{v}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$, by the leftover hash lemma ([Lemma 3.7](#)), the marginal distribution of $\mathbf{u}_{d,i}^{(k)} = \mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}^{(k)}$ is statistically close to uniform over $\mathbb{Z}_q$. Thus,

$$\begin{aligned} \Pr \left[\mathbf{u}_{d,i}^{(k)} \notin [-B_{\text{round}}, B_{\text{round}}] \cup [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}] \right] &\leq \frac{q - (4B_{\text{round}} + 2)}{q} + \text{negl}(\lambda) \\ &\leq \frac{1}{2\ell} + \text{negl}(\lambda), \end{aligned}$$

since $4B_{\text{round}} + 2 \geq q - q/(2\ell)$. Note that when $q > 4B_{\text{round}} + 2$, the intervals $[-B_{\text{round}}, B_{\text{round}}]$ and $[\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}]$ are disjoint and each has size $2B_{\text{round}} + 1$.

By a union bound, for any fixed $i \in [\ell]$, the probability that $r_{d,i}^{(k)} = \perp$ is $1/(2\ell) + \text{negl}(\lambda)$. Since $\ell = \text{poly}(\lambda)$, we can appeal to a union bound to conclude that the probability that there *exists* an index $i \in [\ell]$ where $r_{d,i}^{(k)} = \perp$ is at most $1/2 + \text{negl}(\lambda)$. Correspondingly, the probability that E_d occurs is at least $1 - (1/2 + \text{negl}(\lambda)) = 1/2 - \text{negl}(\lambda)$. $\square$

Claim A.7. For $d \in [\lambda]$, let $\mathcal{X}_d$ be the joint distribution of $(\mathbf{c}_d^{(k)}, \boldsymbol{\pi}_{d,1}^{(k)}, \dots, \boldsymbol{\pi}_{d,\ell}^{(k)}, r_{d,1}^{(k)}, \dots, r_{d,\ell}^{(k)})$ in an execution in Hyb_1 (which is a function of crs_{samp} and $\mathbf{v}_1, \dots, \mathbf{v}_\ell$). Then for all $d \in [\lambda]$, the conditional distribution of $\mathcal{X}_d$ given E_d is statistically close to the following distribution $\mathcal{Y}_d$:

- Sample $\mathbf{c}_d^{(k)} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$. For each $i \in [\ell]$, sample $r_{d,i}^{(k)} \xleftarrow{\mathbb{R}} \{0, 1\}$.
- For each $i \in [\ell]$, sample $\boldsymbol{\pi}_{d,i}^{(k)} \leftarrow (\mathbf{A}_i)_{\text{samp}}^{-1}(\mathbf{c}_d^{(k)})$ conditioned on $\|\boldsymbol{\pi}_{d,i}^{(k)}\| \leq B_{\text{max}}$ and

$$\begin{aligned} \mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}^{(k)} &\in [-B_{\text{round}}, B_{\text{round}}] && \text{if } r_{d,i}^{(k)} = 0 \\ \mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}^{(k)} &\in [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}] && \text{if } r_{d,i}^{(k)} = 1 \end{aligned}$$

If there does not exist $\boldsymbol{\pi}_i^{(k)}$ in the support of $(\mathbf{A}_i)_{\text{samp}}^{-1}(\mathbf{c}^{(k)})$ that satisfies the condition, then the output is $\perp$.

- Output $(\mathbf{c}_d^{(k)}, \boldsymbol{\pi}_{d,1}^{(k)}, \dots, \boldsymbol{\pi}_{d,\ell}^{(k)}, r_{d,1}^{(k)}, \dots, r_{d,\ell}^{(k)})$.

Proof. Take any $d \in [\lambda]$. We proceed via a hybrid argument. Specifically, we define the following distributions:

- $\mathcal{D}_0$: This is the conditional distribution $\mathcal{X}_d$ given E_d . Namely, for each $i \in [\ell]$, the distribution first samples $\mathbf{c}_d^{(k)} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$. Then it samples $\boldsymbol{\pi}_{d,i}^{(k)} \leftarrow (\mathbf{A}_i)_{\text{samp}}^{-1}(\mathbf{c}_d^{(k)})$ conditioned on $\|\boldsymbol{\pi}_{d,i}^{(k)}\| \leq B_{\text{max}}$ and $\mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}^{(k)} \in [-B_{\text{round}}, B_{\text{round}}] \cup [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}]$. Then, it sets $r_{d,i}^{(k)}$ according to [Eq. \(A.1\)](#).
- $\mathcal{D}_1$: This distribution first samples $\mathbf{c}_d^{(k)} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$. Then, let $\mathcal{R}_d$ be the conditional distribution of $(r_{d,1}^{(k)}, \dots, r_{d,\ell}^{(k)})$ given E_d . This distribution samples $(r_{d,1}^{(k)}, \dots, r_{d,\ell}^{(k)}) \leftarrow \mathcal{R}_d$ conditioned on the value $\mathbf{c}_d^{(k)}$. Then for each $i \in [\ell]$, it samples $\boldsymbol{\pi}_{d,i}^{(k)} \leftarrow (\mathbf{A}_i)_{\text{samp}}^{-1}(\mathbf{c}_d^{(k)})$ conditioned on $\|\boldsymbol{\pi}_{d,i}^{(k)}\| \leq B_{\text{max}}$ and

$$\begin{aligned} \mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}^{(k)} &\in [-B_{\text{round}}, B_{\text{round}}] && \text{if } r_{d,i}^{(k)} = 0 \\ \mathbf{v}_i^\top \boldsymbol{\pi}_{d,i}^{(k)} &\in [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}] && \text{if } r_{d,i}^{(k)} = 1. \end{aligned}$$

The output is still $(\mathbf{c}_d^{(k)}, \boldsymbol{\pi}_{d,1}^{(k)}, \dots, \boldsymbol{\pi}_{d,\ell}^{(k)}, r_{d,1}^{(k)}, \dots, r_{d,\ell}^{(k)})$.

- $\mathcal{D}_2$: Same as $\mathcal{D}_1$ except this distribution samples $r_{d,i}^{(k)} \xleftarrow{\mathbb{R}} \{0, 1\}$ for all $i \in [\ell]$. In this distribution, if there does not exist $\boldsymbol{\pi}_{d,i}^{(k)}$ in the support of $(\mathbf{A}_i)_{\text{samp}}^{-1}(\mathbf{c}^{(k)})$ that satisfies the requisite condition, then the output is $\perp$.

First, we claim that by construction, distributions $\mathcal{D}_0$ and $\mathcal{D}_1$ are identical. In particular, let Π_d be the conditional distribution of $(\pi_{d,1}^{(k)}, \dots, \pi_{d,\ell}^{(k)})$ given E_d . Then, the distributions can be described as follows:

- Distribution $\mathcal{D}_0$ corresponds to sampling $\mathbf{c}_d^{(k)} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and then sampling $(\pi_{d,1}^{(k)}, \dots, \pi_{d,\ell}^{(k)}) \leftarrow \Pi_d$ conditioned on the value $\mathbf{c}_d^{(k)}$. Finally, the distribution samples $(r_{d,1}^{(k)}, \dots, r_{d,\ell}^{(k)}) \leftarrow \mathcal{R}_d$ conditioned on the values of $\mathbf{c}_d^{(k)}$ and $(\pi_{d,1}^{(k)}, \dots, \pi_{d,\ell}^{(k)})$. Note that given $\pi_{d,i}^{(k)}$, the value of $r_{d,i}^{(k)}$ is fully determined.
- Distribution $\mathcal{D}_1$ correspond to sampling $\mathbf{c}_d^{(k)} \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$ and then sampling $(r_{d,1}^{(k)}, \dots, r_{d,\ell}^{(k)}) \leftarrow \mathcal{R}_d$ conditioned on the value $\mathbf{c}_d^{(k)}$. Finally, the distribution samples $(\pi_{d,1}^{(k)}, \dots, \pi_{d,\ell}^{(k)}) \leftarrow \Pi_d$ conditioned on the values of $\mathbf{c}_d^{(k)}$ and $(r_{d,1}^{(k)}, \dots, r_{d,\ell}^{(k)})$.

By definition, these are equivalent ways to sampling from the conditional distribution of $\mathcal{X}_d$ given E_d . Thus, $\mathcal{D}_0$ and $\mathcal{D}_1$ are identical distributions. It suffices to argue that distributions $\mathcal{D}_1$ and $\mathcal{D}_2$ are statistically indistinguishable. To do so, we analyze the distribution $\mathcal{R}_d$ given $\mathbf{c}_d^{(k)}$ (for an arbitrary vector $\mathbf{c}_d^{(k)} \in \mathbb{Z}_q^n$). Fix any $i \in [\ell]$. We consider the probability that $r_{d,i}^{(k)} = 0$ conditioned on the event E_d and the choice of $\mathbf{c}_d^{(k)}$. First, let $E_{d,i}$ be the event that $r_{d,i} \neq \perp$. In this case, $E_d = \bigwedge_{i \in [\ell]} E_{d,i}$. Moreover, since each $\pi_{d,i}^{(k)}$ is sampled independently, it follows that

$$\begin{aligned} \Pr \left[r_{d,i}^{(k)} = 0 \mid E_d, \mathbf{c}_d^{(k)} \right] &= \Pr \left[r_{d,i}^{(k)} = 0 \mid E_{d,i}, \mathbf{c}_d^{(k)} \right] = \Pr \left[\mathbf{v}_i^\top \pi_{d,i}^{(k)} \in [-B_{\text{round}}, B_{\text{round}}] \mid E_{d,i}, \mathbf{c}_d^{(k)} \right] \\ &= \frac{\Pr \left[\mathbf{v}_i^\top \pi_{d,i}^{(k)} \in [-B_{\text{round}}, B_{\text{round}}] \wedge E_{d,i} \mid \mathbf{c}_d^{(k)} \right]}{\Pr \left[E_{d,i} \mid \mathbf{c}_d^{(k)} \right]} \end{aligned} \quad (\text{A.2})$$

where the probability in each expression is taken over the choice of $\pi_{d,i}^{(k)} \leftarrow (\mathbf{A}_i)_{\text{samp}}^{-1}(\mathbf{c}_d^{(k)})$. We now analyze the probability of these events:

- As in the proof of [Claim A.6](#), the marginal distribution of $\mathbf{A}_i$ in Hyb_1 is statistically close to uniform, so with overwhelming probability over the choice of $\pi_{d,i}^{(k)}$, it holds that $\|\pi_{d,i}^{(k)}\| \leq B_{\text{max}}$.
- By the same argument as in the proof of [Claim A.6](#) (specifically, by [Lemmas 3.7](#) and [3.8](#)), over the randomness of $\mathbf{v}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$, the marginal distribution of $u_{d,i}^{(k)} = \mathbf{v}_i^\top \pi_{d,i}^{(k)}$ is statistically close to uniform over $\mathbb{Z}_q$. This means

$$\Pr \left[\mathbf{v}_i^\top \pi_{d,i}^{(k)} \in [-B_{\text{round}}, B_{\text{round}}] \wedge E_{d,i} \mid \mathbf{c}_d^{(k)} \right] \in \left[\frac{2B_{\text{round}} + 1}{q} - \delta_1, \frac{2B_{\text{round}} + 1}{q} + \delta_1 \right],$$

where $\delta_1(\lambda) = \text{negl}(\lambda)$ is a negligible function. Similarly, we can write

$$\Pr \left[E_{d,i} \mid \mathbf{c}_d^{(k)} \right] \in \left[\frac{4B_{\text{round}} + 2}{q} - \delta_2, \frac{4B_{\text{round}} + 2}{q} + \delta_2 \right],$$

where $\delta_2(\lambda) = \text{negl}(\lambda)$. Thus, [Eq. \(A.2\)](#) implies that

$$\Pr \left[r_{d,i}^{(k)} = 0 \mid E_d, \mathbf{c}_d^{(k)} \right] = \frac{\Pr \left[\mathbf{v}_i^\top \pi_{d,i}^{(k)} \in [-B_{\text{round}}, B_{\text{round}}] \wedge E_{d,i} \mid \mathbf{c}_d^{(k)} \right]}{\Pr \left[E_{d,i} \mid \mathbf{c}_d^{(k)} \right]} \in \left[\frac{1}{2} - \delta, \frac{1}{2} + \delta \right],$$

for some negligible function $\delta(\lambda) = \text{negl}(\lambda)$. Moreover, conditioned on E_d , we have that $r_{d,i}^{(k)} \in \{0, 1\}$, so the probability that $r_{d,i}^{(k)} = 1$ is also negligibly close to $1/2$.

- Finally, since $\pi_{d,i}^{(k)}$ is sampled independently for all $i \in [\ell]$, the distribution of each $r_{d,i}^{(k)}$ is independent.

We conclude that in distribution $\mathcal{D}_1$, each $r_{d,i}^{(k)}$ is sampled independently from a distribution that is statistically close to the uniform distribution over $\{0, 1\}$. Since $\ell = \text{poly}(\lambda)$, we conclude that distributions $\mathcal{D}_1$ and $\mathcal{D}_2$ are statistically close by a standard hybrid argument. $\square$

Completing the proof of Lemma A.5. We now return to the proof of Lemma A.5. By Claim A.6, we conclude that with overwhelming probability, there exists some $d \in [\lambda]$ such that event E_d occurs. Let $d^* \in [\lambda]$ be the first such index where E_{d^*} occurs. In this case, the challenger in Hyb_1 sets $\mathbf{c}^{(k)} = \mathbf{c}_{d^*}^{(k)}$ and for all $i \in [\ell]$, it sets $r_i^{(k)} = r_{d^*,i}^{(k)}$ and $\pi_i^{(k)} = \pi_{d^*,i}^{(k)}$. By Claim A.7, the conditional distribution of $(\mathbf{c}^{(k)}, \pi_1^{(k)}, \dots, \pi_\ell^{(k)}, r_1^{(k)}, \dots, r_\ell^{(k)})$ when event E_{d^*} occurs is statistically close to the distribution $\mathcal{Y}_{d^*}$. This is precisely the distribution in Hyb_2 . $\square$

Lemma A.8. Suppose $n \geq 4\lambda + 2 \log q$, $t \geq 2n \log q$, $q > 4B_{\text{round}} + 2$ is prime, $s_{\text{samp}} \geq \log t$, $B_{\text{max}} \geq \sqrt{t}s_{\text{samp}}$, and $B_{\text{round}} \geq q/4 - q/(8\ell) - 1/2$. Then $\text{Hyb}_2(\mathcal{A}) \stackrel{s}{\approx} \text{Hyb}_3(\mathcal{A})$.

Proof. By construction, the two distributions are identical unless in an execution of Hyb_3 , there exists indices $k \in [Q]$ and $i \in [\ell]$ where for all $d \in [\lambda]$, it holds that either $\|\pi_{d,i}^{(k)}\| > B_{\text{max}}$ or

$$\begin{aligned} \mathbf{v}_i^\top \pi_{d,i}^{(k)} &\notin [-B_{\text{round}}, B_{\text{round}}] && \text{if } r_i^{(k)} = 0 \\ \mathbf{v}_i^\top \pi_{d,i}^{(k)} &\notin [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}] && \text{if } r_i^{(k)} = 1. \end{aligned}$$

Let $E_{k,i}$ be the event that this happens. We show that for all $k \in [Q]$ and $i \in [\ell]$, the event $E_{k,i}$ happens with negligible probability. We follow a similar analysis as in the proof of Claim A.6:

- In Hyb_3 , the challenger samples $\pi_{d,i}^{(k)} \leftarrow (\mathbf{A}_i)_{s_{\text{samp}}}^{-1}(\mathbf{c}_d^{(k)})$. By Lemma 4.8, the marginal distribution of $\mathbf{A}_i$ in Hyb_3 is statistically close to uniform. Since $t \geq 2n \log q$, $s_{\text{samp}} \geq \log t$, and $B_{\text{max}} \geq \sqrt{t}s_{\text{samp}}$, by Lemma 3.2, with overwhelming probability over the choice of $\pi_{d,i}^{(k)}$, it holds that $\|\pi_{d,i}^{(k)}\| \leq B_{\text{max}}$.
- Next, by Lemma 3.8, with overwhelming probability over the choice of $\mathbf{A}_i$,

$$\mathbf{H}_\infty(\pi_{d,i}^{(k)}) \geq n/2 \geq 2\lambda + \log q.$$

Since the challenger samples $\mathbf{v}_i \xleftarrow{\mathbb{R}} \mathbb{Z}_q^n$, by the leftover hash lemma (Lemma 3.7), the marginal distribution of $\mathbf{u}_{d,i}^{(k)} = \mathbf{v}_i^\top \pi_{d,i}^{(k)}$ is statistically close to uniform over $\mathbb{Z}_q$. Consider now two cases:

- If $r_i^{(k)} = 0$, then

$$\begin{aligned} \Pr \left[\mathbf{u}_{d,i}^{(k)} \notin [-B_{\text{round}}, B_{\text{round}}] \right] &\leq \frac{q - (2B_{\text{round}} + 1)}{q} + \text{negl}(\lambda) \\ &\leq \frac{1}{2} + \frac{1}{4\ell} + \text{negl}(\lambda) \\ &\leq \frac{3}{4} + \text{negl}(\lambda), \end{aligned}$$

since $2B_{\text{round}} + 1 \geq q/2 - q/(4\ell)$ and $\ell \geq 1$.

- Conversely, if $r_i^{(k)} = 1$, then by the same calculation,

$$\begin{aligned} \Pr \left[\mathbf{u}_{d,i}^{(k)} \notin [\lfloor q/2 \rfloor - B_{\text{round}}, \lfloor q/2 \rfloor + B_{\text{round}}] \right] &\leq \frac{q - (2B_{\text{round}} + 1)}{q} + \text{negl}(\lambda) \\ &\leq \frac{3}{4} + \text{negl}(\lambda). \end{aligned}$$

By a union bound, the probability that $\pi_{d,i}^{(k)}$ does not satisfy the required conditions is at most $3/4 + \text{negl}(\lambda)$. Since each $\pi_{d,i}^{(k)}$ is sampled independently, the probability that for all $d \in [\lambda]$, the vector $\pi_{d,i}^{(k)}$ does not satisfy the required conditions is then at most

$$\Pr[E_{k,i}] \leq \left(\frac{3}{4} + \text{negl}(\lambda) \right)^\lambda = \text{negl}(\lambda).$$

Since $Q = \text{poly}(\lambda)$ and $\ell = \text{poly}(\lambda)$, by a union bound, the probability that there exists $k \in [Q]$ and $i \in [\ell]$ where event $E_{k,i}$ happens in an execution of Hyb_3 is also negligible. Correspondingly, the outputs of Hyb_2 and Hyb_3 are statistically indistinguishable. $\square$

Lemma A.9. *Suppose Π_{samp} supports simulatable openings. Then $\text{Hyb}_3(\mathcal{A}) \stackrel{s}{\approx} \text{Hyb}_4(\mathcal{A})$.*

Proof. By mode indistinguishability (Definition 4.5) the distribution of crs_{samp} output by $\text{crs}_{\text{samp}} \leftarrow \text{Gen}(1^\lambda, 1^\ell)$ and $(\text{crs}_{\text{samp}}, \mathbf{T}_1, \dots, \mathbf{T}_\ell) \leftarrow \text{GenTD}(1^\lambda, 1^\ell)$ are statistically indistinguishable. The former corresponds to the distribution of crs_{samp} in Hyb_3 while the latter corresponds to its distribution in Hyb_4 . The claim follows. $\square$

Lemma A.10. *Suppose Π_{samp} supports simulatable openings. Then, $\text{Hyb}_4(\mathcal{A}) \stackrel{s}{\approx} \text{Hyb}_5(\mathcal{A})$.*

Proof. By trapdoor generation (Definition 4.5), we have that $\mathbf{A}_i \mathbf{T}_i = \mathbf{G}$ and moreover, that $\|\mathbf{T}_i\| \leq s_{\text{samp}}/(t \log n)$. By Theorem 3.9, this means the distribution of $\pi_{d,i}^{(k)} \leftarrow \text{SamplePre}(\mathbf{A}_i, \mathbf{T}_i, \mathbf{c}^{(k)}, s_{\text{samp}})$ is statistically close to sampling $\pi_{d,i}^{(k)} \leftarrow (\mathbf{A}_i)_{s_{\text{samp}}}^{-1}(\mathbf{c}^{(k)})$. Since $Q = \text{poly}(\lambda)$ and $\ell = \text{poly}(\lambda)$, $\text{Hyb}_4(\mathcal{A})$ and $\text{Hyb}_5(\mathcal{A})$ are statistically indistinguishable by a hybrid argument. $\square$

Statistical simulation in hiding mode now follows by Lemmas A.4, A.5 and A.8 to A.10. $\square$